

503B Compounding - Regulatory Basis and Industry Good Practices for Outsourcing Facilities

**Current Good
Manufacturing Practices**

Quality Systems

Production Controls

Beyond-Use Dating

Media Fills

Inspection Readiness





GUIDE:

503B Compounding – Regulatory Basis and Industry Good Practices for Outsourcing Facilities

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Preface

As part of ISPE's commitment to our members, we monitor the regulatory environment and have noted the need for a dedicated focus on the pharmaceutical compounding community. This community, primarily 503B outsourcing facilities, has expressed the need for guidance that is more granular than what is currently available for aseptic facilities and that focuses on practices scaled to meet their needs. Although not typically viewed as traditional drug manufacturers, 503B compounders should operate similarly. Current Good Manufacturing Practices (CGMPs) for finished drug products are applicable to 503B outsourcing facilities and the FDA has cited 21 CFR 211 accordingly when outsourcing facilities do not comply.

To assist 503B outsourcing facilities, the FDA has issued numerous guidance documents on their current thinking regarding the regulations and laws. This ISPE Guide aims to help 503B facilities by combining information provided in the applicable regulations and guidances with good industry practices into one document. It covers basic CGMP concepts along with topics specific to 503B outsourcing facilities, such as default beyond-use dates and FDA's regulatory discretion. The references to original sources from the FDA and relevant organizations (such as ISO, PDA, and USP-NF) included in this ISPE Guide provide the reader with the rationale for implementing effective processes.

As of this ISPE Guide publication, the FDA's guidance document, Current Good Manufacturing Practice – Guidance for Human Drug Compounding Outsourcing Facilities Under Section 503B of the FD&C Act Guidance for Industry, is in draft form and subject to change.

The FDA states the following in their draft guidances:

"This draft guidance, when finalized, will represent the current thinking of the Food and Drug Administration (FDA or Agency) on this topic. It does not establish any rights for any person and is not binding on FDA or the public. You can use an alternative approach if it satisfies the requirements of the applicable statutes and regulations. To discuss an alternative approach, contact the FDA staff responsible for this guidance as listed on the title page."

It is recognized that the 503B regulatory environment is evolving, and this ISPE Guide reflects an understanding of it as of the publication date. The ISPE Pharmaceutical Compounding Community of Practice Steering Committee will continue to monitor the 503B regulatory environment and address changes as needed.

Acknowledgements

Recognizing the need for a dedicated focus on the pharmaceutical compounding community, ISPE engaged with multiple organizations with an interest in 503B outsourcing facilities resulting in this ISPE Guide.

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1 Introduction

The US FDA describes drug compounding as “the process of combining, mixing, or altering ingredients to create a medication tailored to the needs of an individual patient.” [1] Compounding may include the combining of two or more commercially available drugs already approved by the FDA. Compounded drugs, however, are not FDA-approved, which means the FDA does not verify the safety or effectiveness of compounded drugs through the traditional drug approval process via IND, ANDA, or NDA¹ submission.

1.1 Background

In 2012, the US Centers for Disease Control and Prevention (CDC), in collaboration with state and local health departments and the FDA, began investigating a multistate outbreak of fungal meningitis among patients that received contaminated preservative-free Methylprednisolone Acetate steroid injections from a compounding facility in Massachusetts. [2] These contaminated compounded injectable drug products were shipped across the country and, due to poor compounding practices, caused a fungal meningitis outbreak that resulted in more than 60 deaths and 750 cases of infection, affecting patients in 20 states.

As a result of this crisis and the inspection findings, the need for federal oversight of compounding pharmacies was revealed. In 2013, the US Congress enacted Title I of the Drug Quality and Security Act (DQSA) [3], also known as the Compounding Quality Act, making important updates to the Federal Food, Drug, and Cosmetic Act (FD&C Act) [4] with respect to drug compounding. The DQSA added a new Section 503B [5] to the FD&C Act, establishing a new voluntary category of compounders called outsourcing facilities.

1.2 Current Situation

The Compounding Quality Act has had a positive impact on public health. Many compounders have implemented corrective actions to improve the quality of drugs produced. [6] A 2021 Pharmacy Times article concluded [7]: “503B compliance has raised the bar on quality and safety for patients and providers alike in the United States.... The benefit of the DQSA to the public is being realized.” However, more work is needed.

As noted in a May 2023 Pharmacy Times article [8]: “Pharmacies evaluating a 503B outsourcing service continue to face challenges. In 2023, supplier and supply chain issues emerged at...one of the country's largest facilities, furloughing workers and facing possible closure after it received cGMP violations from the FDA. In addition, [another facility]...has shut down all operations.”

To assist outsourcing facilities with meeting the improved standards set forth in Section 503B [5], the FDA has clarified regulations by issuing several Guidance for Industry (GFI) documents specifically for 503B outsourcing facilities. The most notable GFI is the Current Good Manufacturing Practice – Guidance for Human Drug Compounding Outsourcing Facilities Under Section 503B of the FD&C Act Guidance for Industry² [9], which is in draft form at the time of this ISPE Guide's publication.

Note: GFIs provide information on the agency's current thinking on a topic. While not legally binding, GFIs play a critical role in helping firms understand the FDA's expectations.

¹ IND = Investigational New Drugs, ANDA = Abbreviated New Drug Application, NDA = New Drug Application

² Referred to as draft GFI – CGMPs for 503B Facilities throughout this ISPE Guide.

There are also times when FDA issues guidances for a specific product or situation, such as a drug shortage, wherein CGMP requirements may be reduced to a certain extent to meet the public health demand. Examples of FDA guidances issued to assist compounding facilities in meeting the public health demand include:

- Compounding Certain Ibuprofen Oral Suspension Products Under Section 503B of the Federal Food, Drug, and Cosmetic Act (Issued April 2023) [10]
- Temporary Policy on Repackaging or Combining Propofol Drug Products During the COVID-19 Public Health Emergency (Issued April 2020, Withdrawn March 2023) [11]
- Policy for Temporary Compounding of Certain Alcohol-Based Hand Sanitizer Products During the Public Health Emergency (Issued March 2020, Withdrawn December 2021) [12]
- Temporary Policy Regarding Non-Standard PPE Practices for Sterile Compounding by Pharmacy Compounders not Registered as Outsourcing Facilities During the COVID-19 Public Health Emergency (Issued April 2020, Withdrawn May 2023) [13]

503B outsourcing facilities are encouraged to monitor FDA activity, as the regulatory environment continues to evolve. For example, in May 2023 the FDA issued the GFI Testing of Glycerin, Propylene Glycol, Maltitol Solution, Hydrogenated Starch Hydrolysate, Sorbitol solution, and other High-Risk Drug Components for Diethylene Glycol and Ethylene Glycol [14]. This GFI was finalized for immediate implementation to *“alert pharmaceutical manufacturers, compounders...to the potential public health hazard of glycerin and other high-risk drug components contaminated with diethylene glycol (DEG) or ethylene glycol (EG).”* This guidance dictates additional testing requirements for products using the referenced excipients.

Another FDA initiative is the establishment of the Office of Compounding Quality and Compliance (OCQC) [15], whose aim is to protect patients from unsafe, ineffective, and poor-quality compounded drugs by evaluating incidents, compliance, regulatory and enforcement actions related to compounded drugs. The OCQC has developed and implemented the Compounding Quality Center of Excellence to assist outsourcing facilities and related stakeholders in their efforts to deliver high-quality drugs to patients by providing relevant and up-to-date information on:

- Laws and policies, oversight, and compliance actions
- Risk alerts
- Inspections, recalls, and other actions
- Compounding research
- Compounding information for states
- Registered outsourcing facilities
- 503B Bulks List [16]
- Trainings – both on-demand and live, instructor led
- Annual conferences
- Compounding Outsourcing Facilities Annual Study

1.3 Purpose

The purpose of this ISPE Guide is to provide detailed guidance on the regulatory considerations for 503B compounding outsourcing facilities and the current industry practices that should be followed. Discussions include the relevant portions of Section 503B [5] and CGMP regulations, focusing on the Quality System (QS) and its application throughout the process from incoming materials to facility and equipment, production controls, laboratory testing, and packaging and labeling operations. This ISPE Guide presents recommendations as described in the FDA's draft GFI – CGMPs for 503B Facilities [9]. This ISPE Guide also provides information on industry good practices and points readers to additional standards and guidances from other technical organizations such as the International Standard for Standardization (ISO) [17], Parenteral Drug Association (PDA) [18], United States Pharmacopeia or National Formulary (USP-NF) [19].

This ISPE Guide is intended for use by 503B compounding outsourcing facilities.

1.4 Scope

The scope of this ISPE Guide is 503B compounding outsourcing facilities. This Guide presents FDA policies and expectations as well as pharmaceutical industry standards to help 503B facilities compound and deliver safe, effective drug products. Pharmacy compounding under Section 503A of the FD&C Act [4] is out of scope.

Given the multitude of individual state regulations, the details on the rules of the State Board of Pharmacy (SBOP) and other equivalent state regulatory groups are out of scope.

While Drug Enforcement Administration (DEA) information on controlled substances is mentioned where relevant, not all DEA rules and regulations are discussed.

2 Regulatory Basis

2.1 Federal and State Oversight

Drugs compounded under Section 503B of the FD&C Act [5] are NOT exempt from following CGMPs, meaning 503B outsourcing facilities are subject to CGMP requirements in Title 21 of the Code of Federal Regulations (CFR) – Part 210 [20] and Part 211 [21]. Oversight and controls for the manufacture of drugs need to be established to ensure the quality of the compounded product as well as the protection of patients. The FDA recommends that 503B outsourcing facilities follow the guidance given in the draft GFI – CGMPs for 503B Facilities [9].

For 503B entities, the FDA is authorized to conduct inspections of the systems used by the facility and take appropriate regulatory actions (i.e., untitled letters, warning letters, recalls, or consent decree) when warranted.³ It is important to note that the FDA generally does not publish FDA-483 observations of drug manufacturers upon closure of inspections. However, for compounding pharmacies, particularly 503B facilities, FDA-483 observations may be posted on the FDA website after the inspection closes. Other regulatory actions such as consent decrees and risk alerts may also be posted [22].

Although 503B outsourcing facilities are primarily under FDA jurisdiction, the FDA's compounding program closely collaborates with state officials. 503B outsourcing facilities should also comply with the SBOP or other state regulatory governing group rules and regulations in the states where they manufacture and the states to which the finished drug product is shipped.

Most states have their own set of regulations and regulating groups for pharmacies and/or pharmaceutical manufacturing facilities. The National Association of Boards of Pharmacy (NABP) provides model language for state laws or board rules, known as The Model State Pharmacy Act and Model Rules of the National Association of Boards of Pharmacy (Model Act) [23]. Each state passes laws or statutes that govern pharmacy practice, generally referred to as a state Pharmacy Practice Act (PPA). The PPA is the basis upon which the regulatory authority is established, usually known as the SBOP. The SBOP may elect to inspect 503B outsourcing facilities. However, each state has different laws/statutes, and the compounding pharmacy should be familiar with and follow the regulations of the SBOP for each state in which they manufacture and/or distribute the compounded drugs. See Section 2.4 for an overview of state pharmacy rules.

2.2 Establishment of Outsourcing Facilities

2.2.1 Definition

Section 503B(d)(4) [5] defines an outsourcing facility as:

“a facility at one geographic location or address that—

(i) is engaged in the compounding of sterile drugs;

(ii) has elected to register as an outsourcing facility; and

(iii) complies with all of the requirements of this section [Section 503B].”

³ FDA also has authority to inspect 503A pharmacies, especially in cases where there is suspicion that they are not functioning within the definition of a 503A pharmacy.

503B compounders are:

- Inspected by the FDA according to a risk-based schedule
- Required to report Adverse Drug Experiences (ADEs) that are categorized as both serious and unexpected per 21 CFR 310.305 [24] (see additional discussion in Section 3.10.2 of this ISPE Guide)
- Required to submit to the FDA twice yearly reports on the drugs compounded by the facility in the previous 6 months

Under Section 503B, compounding must be by or under the direct supervision of a licensed pharmacist. 503B outsourcing facilities are not required to be a licensed pharmacy. They are also not required to compound drugs only for an identified individual patient based on the receipt of a valid prescription. As such, they may distribute compounded drugs either based on a patient-specific prescription or in response to an order from a healthcare provider, such as a hospital. [5]

The FDA guidance, Facility Definition Under Section 503B of the Federal Food, Drug, and Cosmetic Act Guidance for Industry⁴ [25], defines the conditions that a facility must meet in order to be considered as a 503B outsourcing facility. These conditions are outlined below to provide a clear definition of what would qualify a facility as a 503B outsourcing facility:

- The facility must be at one geographic location.
- The compounding must be done by or under the direct supervision of a licensed pharmacist.
- The facility must be engaged in the compounding of sterile drugs.
- The facility elects to register as an outsourcing facility.
- The organization cannot compound as a 503A pharmacy in the same facility.

The FDA has issued the following GFIs regarding the activities that an outsourcing facility may conduct:

- Repackaging of Certain Human Drug Products by Pharmacies and Outsourcing Facilities [26]
- Mixing, Diluting, or Repackaging Biological Products Outside the Scope of an Approved Biologics License Application [27]

2.2.2 Registration

Although registration as an outsourcing facility is voluntary, firms that make products without patient-specific prescriptions and compound sterile drug products may be inspected by the FDA, bringing them under the oversight of the agency. Some reasons a facility may choose to register as a 503B outsourcing facility are:

- Outsourcing facilities may want to compound approved drugs without obtaining ANDA or NDA approval for the compounded product (provided certain criteria are met, such as being on the Drug Shortage list [28] and meeting the requirements listed in Section 2.2.3).
- Outsourcing facilities may want to distribute compounded drugs to healthcare facilities and practitioners without first receiving a patient-specific prescription.

⁴ Referred to as GFI – 503B Facility Definition throughout this ISPE Guide.

- Healthcare facilities and practitioners often prefer to purchase compounded drugs from registered outsourcing facilities because the FD&C Act [4] subjects outsourcing facilities to CGMP requirements and federal oversight, including inspections on a risk-based schedule and specific adverse event reporting requirements.

The FDA issued two GFIs to help firms determine when registration is advised:

- Registration of Human Drug Compounding Outsourcing Facilities Under Section 503B of the FD&C Act [29]
- Guidance for Entities Considering Whether to Register as Outsourcing Facilities Under Section 503B of the Federal Food, Drug, and Cosmetic Act [30]

2.2.3 Requirements

Section 503B [5] provides the set of conditions that 503B outsourcing facilities must meet in order to be exempt from the following:

- FD&C Act Section 502(f)(1) [4], which requires adequate directions for use in labeling.
- FD&C Act Section 505 [4], which states that no person shall introduce or deliver for introduction into interstate commerce any new drug, unless an approval of an application filed pursuant to subsection 505(b) (NDA) or 505(j) (ANDA) is effective with respect to such drug.
- Section 582 of the Drug Supply Chain Security Act [31], which outlines the requirements of the Enhanced Drug Distribution Security.

This means that outsourcing facilities not compliant with Section 503B conditions are subject to the requirements in these three sections. Thus, stating that an outsourcing facility is not required to file an ANDA or NDA is not exactly correct; however, if an outsourcing facility meets the conditions outlined in Section 503B to qualify as an outsourcing facility, the facility can be considered exempt from filing ANDAs and NDAs.

Section 503B [5] of the FD&C Act [4] states “Sections 502(f)(1), 505, and 582 shall not apply to a drug compounded by or under the direct supervision of a licensed pharmacist in a facility that elects to register as an outsourcing facility if each of the following conditions is met:”

1. Registration and reporting – The drug is compounded in a 503B facility that complies with the requirements outlined in the section “Registration of Outsourcing Facilities and Reporting of Drugs”.
2. Bulk drug substances/API⁵ – The drug is compounded from a bulk drug substance that appears on the 503B Bulks List [16] (designated for compounding in a 503B facility and published periodically in the Federal Register) or the drug product appears on the FDA Drug Shortages list [28] in effect under Section 506E at the time of compounding, distribution, and dispensing. In addition:
 - If an applicable monograph exists, the bulk drug substance must comply with the applicable monograph in the USP-NF [19] or another compendium or pharmacopeia recognized by the FDA.
 - The bulk drug substance must be manufactured by an establishment that is registered under Section 510 (including a foreign establishment that is registered under Section 510(i)).
 - The bulk drug substance must be accompanied by a valid Certificate of Analysis (COA).

⁵ Active Pharmaceutical Ingredient (API) and drug substance can be used interchangeably.

3. Ingredients (other than bulk drug substances) – If any ingredients (other than bulk drug substances) are used in compounding the drug, they should comply with the standards of the applicable USP or NF monograph, if such monograph exists, or of another compendia or pharmacopeia recognized by the FDA.
4. Drugs withdrawn or removed because they are unsafe or not effective – The drug cannot be compounded if it appears on the FDA-published list of drug products that have been withdrawn or removed from the market for reasons of safety or effectiveness (21 CFR 216.24 [24]).
5. Essentially a copy of an approved drug – The drug cannot be compounded if it is essentially a copy of an approved drug.
6. Drugs presenting demonstrable difficulties for compounding – The drug does not meet the definition in Section 503B(a)(6)(A) and (B).
7. Elements to assure safe use – If the drug is compounded from a drug or bulk drug substance that is a component of a Risk Evaluation and Mitigation Strategy (REMS) with elements to assure safe use pursuant to Section 505-1, the 503B facility must follow the requirements in Section 503B(a)(7).
8. Prohibition on wholesaling – The drug will not be sold or transferred by an entity other than the outsourcing facility that compounded the drug, in accordance with Section 503B(a)(8).
9. Fees – The drug is compounded in an outsourcing facility that has paid all fees pursuant to Section 744K.
10. Labeling of drugs – The requirements for labeling compounded drugs and container, as outlined in Section 503B(10), are followed.
11. Outsourcing facility requirements – The drug is compounded in an outsourcing facility in which the compounding of drugs occurs only in accordance with Section 503B.

Section 503B(d) [5] provides the following definitions:

- The term **compounding** includes *“the combining, admixing, mixing, diluting, pooling, reconstituting, or otherwise altering of a drug or bulk drug substance to create a drug.”*
- The term **essentially a copy of an approved drug** is defined in Section 503B(d)(2) as:
 - A drug that is identical/nearly identical to an approved/ marketed drug not subject to Section 503(b) and not subject to approval in an application submitted under Section 505, unless, in the case of an approved drug, the drug appears on the drug shortage list in effect under Section 506E at the time of compounding, distribution, and dispensing (Section 503B(d)(2)(A)); or
 - A commercially available drug product or a component of which is a bulk drug substance that is a component of an approved drug or a marketed drug that is not subject to Section 503(b) and is not subject to approval in an application submitted under Section 505 that has a change needed for an individual patient, e.g., due to an allergy, difference in delivery (from solid dosage to liquid), or different dose (>10% difference from a commercially available dose).

Appendix A of the FDA GFI, Compounded Drug Products That Are Essentially Copies of Approved Drug Products Under Section 503B of the Federal Food, Drug, and Cosmetic Act [32], can help in determining whether the compounded drug product fits the definition.

- The term **approved drug** means a drug that is approved under Section 505 and does not appear on the list of drug products that have been withdrawn or removed from the market for reasons of safety or effectiveness (21 CFR 216.24 [24]).

- The term **sterile drug** means “a drug that is required to be sterile under federal or state law. a drug that is intended for parenteral administration, an ophthalmic or oral inhalation drug in aqueous format, or a drug that is required to be sterile under Federal or State law.”

2.3 FDA Regulation of Outsourcing Facilities

2.3.1 Insanitary Conditions

Under Section 501(a)(2)(A) of the FD&C Act, a drug is deemed to have been adulterated “if it has been prepared, packed, or held under insanitary conditions whereby it may have been contaminated with filth, or whereby it may have been rendered injurious to health.” [4] Insanitary conditions are a critical area of focus for the FDA, as this was a large contributor to the 2012 fungal meningitis outbreak discussed in Section 1.1 of this ISPE Guide. As such, the FDA issued the Guidance for Industry – Insanitary Conditions at Compounding Facilities⁶ [33], finalized in November 2020. The GFI – Insanitary Conditions gives specific examples of the insanitary conditions of concern from FDA for different areas of compounding, based on the types of products being compounded and the areas of the process or manufacturing. This topic is discussed in more detail in Section 5.4 of this ISPE Guide.

It is critical that compounding facilities are vigilant about insanitary conditions, as both “the facility and responsible individuals may be subject to several regulatory actions, including, but not limited to, a warning letter, seizure of product, or injunction. FDA may also recommend that the facility initiate a recall of some or all of its drugs and cease operations until the insanitary conditions have been adequately addressed.” (GFI – Insanitary Conditions [33])

2.3.2 Conforming to CGMPs

The draft GFI – CGMPs for 503B Facilities [9] provides the following background:

- Under Section 501(a)(2)(B) [4], a drug is deemed to be adulterated if “the methods used in, or the facilities or controls used for, its manufacture, processing, packing, or holding do not conform to ... [CGMPs] to assure that such drug meets the requirement of this [Act] as to safety and has the identity and strength, and meets the quality and purity characteristics, which it purports or is represented to possess.”
- Under Section 501 [4], the term CGMP “includes the implementation of oversight and controls over the manufacture of drugs to ensure quality, including managing the risk of and establishing the safety of raw materials, materials used in the manufacturing of drugs, and finished drug products.”

Note that even though 503B references CGMP requirements established in 21 CFR 211 [21], not all regulations under Part 211 are mentioned in the draft GFI – CGMPs for 503B Facilities [9]. Adherence to all CGMPs, even those not listed in the draft GFI, are required for 503B facilities; however, there are areas of enforcement discretion that may be granted if the facility follows the guidance.

A principle of CGMP that the draft GFI – CGMPs for 503B Facilities [9] states is that “quality is best assured by implementing appropriate controls throughout the manufacturing process, with end-product testing providing additional assurance.” End-product testing confirms product quality, but is not the only measure. This principle is known as building-in quality to the process/product, as opposed to testing quality into the product. In order to establish the appropriate level of controls, a risk-based quality approach may be utilized, with focus on the critical aspects of compounding, such as “control of raw materials, facility design and maintenance, production techniques and control, and personnel practices” and training. Evaluation of risk should also consider the compounding risk level/compounded drug type as well as sterile/non-sterile compounded products.

⁶ Referred to as GFI – Insanitary Conditions throughout this ISPE Guide.

Expectations of CGMPs in any manufacturing or compounding facility include an oversight of quality, materials, facilities and equipment, production, packaging and labeling, and laboratory operations. These may not be specifically discussed in the draft GFI – CGMPs for 503B Facilities [9]; however, they should be followed as core concepts of CGMPs. They are further discussed throughout this ISPE Guide and include:

- Risk assessment procedures and processes, if a risk-based approach is used
- Defined/established responsibilities and procedures for quality oversight along with evidence of oversight (Chapter 3)
- Training requirements for all employees, consultants, and temporary employees of the 503B outsourcing facility (Section 3.3)
- Procedures related to manufacturing processes and controls, including batch production records, environmental monitoring records, cleaning, room clearance, aseptic processing/gowning/qualification process/training, compounding methods for sterile and non-sterile products, room access control, and room segregation (when applicable) (Chapters 3, 5, 6, and 7)
- Procedures related to Good Documentation Practices (GDP) (e.g., no sticky notes, no pencils, “if it is not written down, it didn’t happen”) (Section 3.4)
- Procedures related to training (Section 3.3)
- Procedures for, and evidence of, equipment cleaning, maintenance, and qualification (Section 5.3)
- Procedures for and records of programs for change management, complaint monitoring, and adverse events (Chapter 3)
- Established Quality Control (QC) program, including processes for assessment of Raw Materials (RMs), packaging, and labeling used for drug compounding (Chapter 8)
- Compliance with the defined QS as outlined in Chapter 3

2.3.3 Regulatory Discretion

The GFIs help compounders understand the applicable regulations for 503B facilities; however, they can also define instances where FDA will take regulatory discretion.

Appendix A of the draft GFI – CGMPs for 503B Facilities [9] outlines conditions under which FDA generally does not intend to take regulatory action regarding certain release testing requirements. Within this appendix, Table A outlines these conditions for non-sterile drug products.

For sterile products, the FDA generally does not intend to take regulatory action related to sterility testing *“if sterility testing is initiated before batch release (also see Table D in Appendix B for [Beyond-Use Dates] for products released without a completed sterility test).”*

The FDA also generally does not intend to take regulatory action for sterile products if the conditions described in Table B of Appendix A [9] apply. If multiple conditions apply, the least stringent option for each individual batch release test among the applicable conditions should be chosen from Appendix A.

Additionally, Appendix B of the draft GFI – CGMPs for 503B Facilities [9] describes conditions under which the FDA generally does not intend to take regulatory action regarding stability testing and expiration date requirements. See Chapter 9 of this ISPE Guide for additional information on stability and expiration date requirements.

2.4 State Pharmacy Rules Overview

According to federal law, 503B compounding facilities do not need to be a registered/licensed pharmacy even if they are receiving prescriptions for individual patients. However, depending on the state(s) where the 503B compounding facility manufactures and where they distribute product, there are additional state pharmacy rules and requirements that the facility should adhere to regarding registrations and/or licensing, which may include the need to register as a licensed pharmacy.

The NABP maintains a website [34] with links to SBOP websites that gives the most up-to-date information on the state laws with which facilities need to comply. However, as some states are regulated by a group other than a SBOP, 503B facilities should be familiar with the requirements of the regulating body in the states in which they manufacture and distribute. These websites typically provide information on the following topics, at minimum:

- How to obtain a registration/license and registration/license renewal
- Regulations to be followed depending on the role/company type – pharmacist/pharmacy, operator/compounding pharmacy, compounding facility, etc.
- General requirements including policies and procedures that should be put in place, inspections, appropriate storage, handling and security, notifications, etc.

The following should be posted in a prominent place at the facility:

- License details for the supervising licensed pharmacist(s)
- SBOP license(s) or similar, if required/applicable

Many state rules and regulations are linked to the federal (CGMP) regulations, although some may be associated with the following USP chapters [19]:

- USP <795> Pharmaceutical Compounding – Nonsterile Preparations
- USP <797> Pharmaceutical Compounding – Sterile Preparations
- USP <800> Hazardous Drugs – Handling in Healthcare Settings

Beyond the many regulations, there are three state requirements that apply to the majority of states that regulate 503B facilities:

- Facilities must be licensed/registered in the state in which they are located.
- Facilities must be licensed/registered in the states to which they distribute compounded medications.
- Compounding of preparations must be performed by, or directly supervised by, a licensed pharmacist.

3 Quality System

3.1 Introduction

Registered 503B outsourcing facilities are subject to CGMP inspections by the FDA. [5] CGMP regulations define the minimum requirements needed during the processing, testing, packaging, and distribution of drug products. In order to comply with CGMPs, 503B outsourcing facilities need to establish a QS. An important aspect of the QS is the establishment of the Quality Unit (QU) that is independent from other units within the outsourcing facility's organization. The QU's responsibilities include approving procedures, investigating deviations, and reviewing records. The QU is responsible for approving and releasing or rejecting incoming components, packaging materials, and final compounded products.

Another critical aspect of the QS in 503B facilities is the control and use of written procedures that provide clear, concise instructions for performing tasks. 503B facilities are required to have Standard Operating Procedures (SOPs) for compounding drug products as well as for executing activities performed as part of the QS. Procedures standardize the facility's processes and are a key tool in training personnel.

The draft GFI – CGMPs for 503B Facilities [9] uses a risk-based approach to CGMP requirements that focuses on control of RMs, facility design and maintenance, production techniques and controls, and personnel practices as the most critical aspects in ensuring quality for all drug products. Other CGMP requirements described in the appendices of the draft GFI are based on the FDA's regulatory discretion, i.e., their likelihood of taking regulatory action depends on the level of risk (high or low) of the compounding activities conducted at the 503B outsourcing facility. FDA regulatory discretion is discussed in Section 2.3.3 of this ISPE Guide.

This chapter provides information on key aspects of the QS with greater emphasis on the sections mentioned above as the focus of the FDA. While the QS can be scaled based on the organization's preferences and the type of drugs being compounded (such as high-risk or lower risk drugs), all aspects of the QS discussed in this chapter should be present in a 503B outsourcing facility.

3.2 Quality Responsibilities

3.2.1 Quality Unit

In CGMP manufacturing, the QU is frequently split into two sections: Quality Assurance (QA) and QC. The draft GFI – CGMPs for 503B Facilities [9] applies the phrase "quality control unit" to describe the responsibilities associated with the overall QU. As either approach is acceptable, this ISPE Guide uses the QA and QC naming convention and division of responsibilities, recognizing that 503B manufacturers may combine QA and QC functions under one QU to carry out quality assurance activities.

A QU needs to be established to provide independent, unbiased oversight of 503B compounding activities. QA activities are required to ensure compliance with the applicable 21 CFR 211 sections (§§ 211.22, 211.180, 211.192, 211.198) [21]. Quality oversight of production activities is necessary to ensure that the final compounded drugs are of the appropriate Strength, Quality, Identity, Purity, and Potency (SQIPP). Additionally, the QU is responsible for ensuring proper controls are in place during manufacturing of both sterile and non-sterile products so that the finished drug products are free from microbiological contamination.

The FDA requires the QU to maintain an organizational reporting structure independent of any functions outside of Quality. [35] Quality should not have a reporting structure, directly or indirectly, to production and should not take on any aspect of production activities for compounded drugs. The QU should be adequately staffed to ensure that sufficient Quality oversight can be provided to the facility and compounding activities. This may be accomplished through outsourcing activities, such as auditing, testing, etc., to qualified vendors/consultants.

The role of the QU is defined in 21 CFR 211 Subparts B, E, F, G, and H [21]. The required procedures describing the Quality framework for the outsourcing facility and QU responsibilities include:

- Establishment of written procedures for production and process controls that must be followed and documented at the time of performance. The procedures must be *“designed to assure that the drug products have the identity, strength, quality, and purity they purport or are represented to possess.”* (§ 211.100)
- Establishment of controls to prevent cross-contamination in the event that powder drugs are used in production. *“FDA recommends the physical segregation of areas in which powder drugs are exposed to the production environment.”* [9]
- Establishment and approval of drug product (§ 211.110), component (§ 211.84), and labeling (§ 211.122) specifications. This includes drug products compounded at the outsourcing facility and any drug product that is under contract by another company (§ 211.22).

Note: In the event that a drug product is manufactured, processed, packed, or held under contract by another company, the QU must provide oversight for these activities.

- Ensuring that adequate laboratory facilities are available for the testing and disposition of drug product, drug product containers, closures, in-process materials, packaging materials, and labeling. (§ 211.22)
- Establishment of sterility assurance activities for sterile products and of microbial quality for non-sterile products at the outsourcing facility as well as contract companies used for manufacturing any stage of the drug product (§ 211.113 and 211.167)
- Establishment of sampling procedures, including sampling tools, sampling process, and number of samples needed per lot (§ 211.160)
- Ensuring that each drug product batch, including drug product, drug product containers, closures, in-process materials, packaging materials, and labeling, is sampled, tested/examined, and reviewed against approved release specifications and approved/released prior to use in drug compounding [9]
- Establishment of warehousing and distribution procedures as outlined in 21 CFR 211 Subpart H
- Establishment of procedures for adding sterility results to compounding records in the event that sterile product is released with pending sterility testing results [9]
- Review of compounding operations, on an annual basis at a minimum, to determine if specifications or control procedures require revision. This review should include identifying trends and evaluating quality indicators for the following [9]:
 - Finished product testing results
 - Results of Environmental Monitoring (EM) and personnel monitoring
 - When water is used as a compounding component, results of water system testing for water that is purified/processed on-site or results of testing from the supplier/outsourcing facility for water purchased as an incoming component
 - Media fills/process simulations performed since the last review
 - Operations to determine if procedures are being followed and proper aseptic techniques are employed during drug compounding

- Complaints, nonconformances, failures (such as for system or equipment), and variation in yield
- Discrepancy and failure investigations along with identification and oversight of corrective actions, including changes implemented to prevent recurrence, with regard to any of the following as based on the draft GFI – CGMPs for 503B Facilities [9]:
 - Compounded drug evaluation testing, such as “sterility testing, endotoxin levels, content assay, impurity assay, particulate matter, reconstitution time, content uniformity, preservative content testing, microbial enumeration, tests for specified microorganisms, and weight, volume, or counts.”
 - Unexpected results or trends, such as Out of Trend (OOT)
 - Validation/revalidation failures (e.g., process validation, sterilization, depyrogenation)
 - Stability failures including drug product degradation and failures determined to be caused by factors other than degradation
 - EM and personnel monitoring results that exceed action or alert limits
 - Failures associated with aseptic techniques and procedures
 - Failures of drug product, RMs, components, or packaging/labeling
 - Process deviations or equipment malfunctions that involve critical equipment (e.g., sterilizers, homogenizers, mixer, lyophilizers)
 - Complaints that indicate possible compounded drug contamination or any complaint that identifies potential patient risk, such as particulate matter, precipitating particles, changes in drug appearance or color, etc.

3.2.2 Release Testing

The draft GFI – CGMPs for 503B Facilities states [9]: “[p]rocedures for release must be established to ensure that each batch of a drug product is not released until the following have been completed (see §§ 211.22, 211.165, 211.167, 211.192):

- An appropriate laboratory determination has been conducted to ensure that each batch of a drug product conforms to specifications.
- A review of environmental and personnel monitoring data, if applicable, has been conducted to ensure that manufacturing conditions were acceptable during production of the batch.
- Associated laboratory data and documentation have been reviewed by the quality control unit, and they demonstrate that the drug product meets specifications.
- A designated qualified individual from the quality control unit has authorized final release.

Under certain conditions described in Appendix A [of the GFI – CGMPs for 503B Facilities], FDA generally does not intend to take action against an outsourcing facility regarding [these] release testing requirements.”

3.2.3 Laboratory Controls

Contract laboratories must be qualified under the QU supplier/vendor qualification program. The draft GFI – CGMPs for 503B Facilities [9] states: *“When an outsourcing facility seeks the services of a contract facility to perform all or part of the testing of a drug, the outsourcing facility’s quality control unit is responsible for approving and rejecting drugs tested by the contractor.”* Refer also to the FDA GFI Contract Manufacturing Arrangements for Drugs: Quality Agreements [36].

Note: FDA recommends that contract facilities performing testing of a drug have ISO 17025 [37] accreditation.

3.3 Training

Per 21 CFR 211.25 [21], each person engaged in or supervising the manufacture, processing, packing, or holding of a drug product should be qualified to perform the assigned functions through any combination of education, training, and experience. Training should be provided in the area in which the employee works, and each employee should be trained according to CGMPs as outlined in 21 CFR 211. CGMP training *“shall be conducted by qualified individuals on a continuing basis and with sufficient frequency to assure that employees remain familiar with CGMP requirements applicable to them.”*

The training record for a given individual should include a Curriculum Vitae (CV), when available, that provides information on their education (including training classes) and prior experience relevant to their role at the compounding facility. In addition, the training record should contain information on each training completed at the facility or externally, including the trainer name, course name/unique number, and the type of training, e.g., computer, in-person, job specific/On-the-Job Training (OJT).

Per 21 CFR 211.25, personnel training should be related to the operations that the employee will be conducting in their role. Likewise, each person supervising someone in operations must also be trained on any operations that they are responsible for supervising.

Per 21 CFR 211.34, consultants brought into a facility must also have the appropriate training or experience for the role they are performing. The outsourcing facility is required to maintain records of the name, address, and qualifications of the consultants, and what service they provided.

Training must be conducted by a qualified individual who possesses a combination of education, certification, or experience in the topic in which they provide training. The effectiveness of the training should be evaluated through assessment testing or via trainee demonstration of the skills, with documented evidence of the completed assessment. The skillset demonstration needs to be performed in the presence of a qualified trainer and should be properly documented. If performance issues indicate that the trainee did not appropriately learn the skillset, then retraining as well as evaluation of the individual’s appropriateness for the role should be conducted; the trainee shall not perform any function for which the training indicates that the trainee did not properly learn the skillset.

Training for personnel working in sterile processing should include an overview of the concepts of sterility, including the reasons a product is required to be sterile, aseptic gowning, and techniques required to ensure the sterility of the product. Job-specific aseptic training may employ demonstration of proper aseptic techniques followed by proficiency evaluations and written assessments. A gowning certification training program is recommended as well as additional OJT to ensure aseptic personnel are qualified to perform their job functions. An employee should not conduct aseptic processing prior to demonstrating proficiency in aseptic compounding activities per established aseptic training requirements, including satisfactory performance in a media fill. [9] Retraining on sterile aseptic techniques should be considered for inclusion in the annual training program.

3.4 Document Control

The policies, procedures, and instructions that define a manufacturer's QS requirements are subject to document control procedures per 21 CFR 211 Subpart J [21]. 503B compounded drugs manufacturers should have governing written procedures that describe the controls in place for issuing, approving, storing, and maintaining the CGMP documentation. Such controls include:

- Document lifecycle, including preparation, review, revision, approval, document obsolescence, and archival
 - Document development, review, revision, and approval to ensure that documents are accurate and meet the requirements of the QS
 - Periodic review of the documents to assess if changes/updates are needed to comply with current practices with respect to regulations, the firm, and the industry
 - Document obsolescence and archival to ensure that obsolete or superseded documents are retained per the record retention policy and managed to ensure that only the current and approved version is available for use
- Assignment of unique numbers as well as version numbers, effective dates, and change history
- Distribution and maintenance of documents to ensure that documents are made available for use to applicable personnel at the point of use

3.5 Change Management

The change management (change control) process for 503B compounded drug manufacturers should be consistent with the requirements presented in ICH Q10 [38] and the FDA guidance, Quality Systems Approach to Pharmaceutical CGMP Regulations [35]. Change management provides a system by which changes are evaluated prior to implementation to ensure that a change does not negatively affect the SQIPP, safety, and/or efficacy of the compounded drug product. The QU maintains primary responsibility for the direction and oversight of the change management process.

Within the governing procedure(s) for change management, the process should include documentation of the following:

- Reason for the change
- Type of change (permanent or temporary)
- Evaluation of the proposed change
- Impacted documents, products, processes, specifications, training needs, validation, etc. after change implementation
- Timing of the proposed change
- Personnel responsible for making, implementing, and approving the change, as well as other activities such as procedure, training, and specification updates and timelines for change activities and implementation

3.6 Deviations and Investigations

A deviation (sometimes referred to as a nonconformance) is an event in which the compounded product or any materials, components, labels, etc. depart from established procedures, processes, instructions (batch records), specifications, or established standards. Deviations can occur at any time during compounding including processing, filling, packaging, and testing/holding of a component, RM, intermediate, or drug product. Deviations may have an impact on the SQIPP or safety of a product and, therefore, each event is to be investigated and documented.

An investigation includes the documented review of the occurrence to determine if there is any impact to the safety and/or SQIPP of a product(s) due to the event. Once the need for an investigation is identified, it should be completed in a timely manner. It is recommended that an investigation team be assembled that, based on need, include appropriate members of technical, development/R&D, Quality, and operations/manufacturing (along with individuals involved in the event where possible) to ensure a complete investigation. Key parts of an investigation include:

- Clear, concise, and factual statements of the event, including date of occurrence, person that discovered the event, and product (RM/component/API/in-process/finished/equipment/batch record/SOP step) details; avoid opinion and conjecture
- Immediate actions taken to remediate the event/issue until the investigation is completed, when applicable
- Identification of all impacted products, lots, processes, equipment, etc.
- Review and analysis of the event in a logical manner, including a scientific evaluation of all laboratory data and testing related to the event, which must include a review in accordance with data integrity principles
- Evaluation of similar products/processes to determine if they need to be included in the investigation
- Historical review of related events to determine recurrence/trend
- Identification of a root cause and establishment of appropriate corrective action and preventive action to prevent recurrence
- Statement of regulatory review and notification, when required
- Documented Quality review and approval of the completed investigation report, with Quality as the last approver prior to closure

Written procedures must be in place to govern the deviation investigation and documentation processes. The deviation procedure designates responsibilities for the stages in the investigation with required reviewers and approvers. It is good practice for the procedure to include root cause analysis techniques such as generation of fishbone diagrams and application of 5 Whys. The deviation investigation should also document an assessment of impact to product currently on the market or that may have been recently dosed to a patient; this may involve application of risk assessment tools.

Laboratory investigations must be initiated when an Out of Specification (OOS) testing result is obtained. Each OOS must be investigated to determine its root cause; where some potential root causes may be ruled out, the investigation should document the sound scientific reasoning providing support for doing so. OOS results determined to not be associated with laboratory error must be further investigated to determine if anything in the manufacturing/compounding process contributed to or caused the event. Further information on investigating OOSs can be found in the FDA GFI Investigating Out-of-Specification (OOS) Test Results for Pharmaceutical Production [39].

3.7 Corrective Action and Preventive Action

The Corrective Action and Preventive Action (CAPA) program is one of the most important and essential parts of the QS. The procedures governing the CAPA program prescribe the requirements for addressing the identified cause of a nonconformance and determining the actions to be taken in order to prevent recurrence. Additionally, CAPAs may be identified as the result of continual improvement, an audit, walkthrough, or other means. The identified preventive actions may help prevent further deviations by implementing/revising procedures, adding additional steps, clarifying batch records/procedures, identifying trends, etc. The CAPA program extends to all areas of the company, including production and process control, design, purchasing, laboratory, and audit findings.

Written procedures must be in place for the CAPA process. Once the need for a CAPA is identified through a nonconformance investigation or other means, the CAPA process should start in a timely manner. Key parts of the CAPA report include:

- Clearly defined problem statement
- Assessment of all potential impacts, including product, process, validation, equipment, documentation
- Assessment of previous CAPAs implemented for similar/like issues and effectiveness
- Assessment of the risks
- Action plan that details the required tasks to be completed to appropriately address the issue identified and prevent recurrence
- Defined effectiveness checks after implementation of the action plan to ensure that the CAPA addressed the issue and prevented recurrence

3.8 Batch Production Record and Batch Record Disposition

Batch production records are known by many names in the pharmaceutical industry, such as Master Batch Records (MBRs) and Batch Records (BRs)⁷. In 503B outsourcing facilities, they may be called Master Formulation Records (MFR). MBRs are required for each process created for a compounded drug and need to document all stages of the compounding process. MBRs may be one large document inclusive of the entire process or broken down by major processes (for example, dispensing, compounding, filling, labeling). Any of these terms and approaches are acceptable, as long as the site's written procedures are followed.

From a master production record, BRs are issued with a unique identifying number for each batch/lot⁸ of the compounded drug product to be manufactured. BRs are controlled/issued by the QU as needed for batch production and are required for each lot of compounded drug to document all stages of the compounding process. The BR should minimally include the following, in accordance with 21 CFR 211.188 [21]:

- An accurate reproduction of the approved master production or control record, issued and signed by the QU and assigned a BR number.
- Documentation of each significant step in the manufacture, processing, packing/packaging, or holding of the batch completed, including:
 - Date of manufacture

⁷ Within this Guide, MBR and BR are considered equivalent and used interchangeably.

⁸ Within this Guide, batch and lot are considered equivalent and used interchangeably.

- Identification of person performing or checking each significant step
- Defined parameters and specifications for each significant step, as established during validation/qualification activities
- Identification of major equipment and lines used
- Identification of each lot of component, containers/closures, or in-process materials used
- Weights/amounts of components used
- Defined limits for actual yield and theoretical yield at appropriate phases of compounding
- Inspection of the packaging and labeling area before use
- Complete labeling control records, including copies of all labeling used
- Sampling performed
- In-process and laboratory control results
- Any investigation that may have been initiated
- Results of inspections performed during compounding

All completed BRs are required to be reviewed and approved by the QU to ensure that the batch has been manufactured to all established specifications and procedures prior to release and distribution (21 CFR 211 Subpart J [21]). This review should include review of QC raw data, QC audit trails, and audit trails for any automated equipment used for compounding, where applicable. Any discrepancies or failures of the batch or component to meet specifications must be thoroughly investigated. Based on review of all this information, the QU determines the final disposition of the batch/lot.

3.9 Logbooks

Many outsourcing facilities record data/information in logbooks, which are maintained at the source of the work. Issuance, usage, and archival of logbooks should be governed by an SOP and referenced in the BR as appropriate. Completed logbook pages should be reviewed by the QU, preferably prior to batch release, to ensure that all entries have been properly documented and that the entries in the logbook, audit trail, and executed BP align with each other. Facilities often use logbooks for the following areas:

- Equipment logs: Used to record calibration activities (when required), usage dates and batch numbers, API, cleaning method and date, and who performed compounding and cleaning, at a minimum. For example, scales and balances may require daily verification, and these activities may be noted in a logbook or per other proceduralized method. Equipment logs are maintained in the designated area with the piece of equipment.
- Room logs: Used to record what batches numbers were processed in the room, including details such as processing dates, API usage, equipment used, cleaning method, date and who performed cleaning, who performed compounding, who entered/exited the room during compounding, at minimum. Room logs or a separate logbook should be used as a room sign-in during operations in cases where badge access is not required/traceable. Room logs are maintained in the designated area for the room.
- Laboratory notebooks: Issued and used per the procedure governing laboratory notebook control and usage.
- Other logs: Used to account for RM components, container/closures, etc.

3.10 Complaint Handling

A complaint is a notification by internal or external customers regarding the quality or purity of a product. It may be provided as either written or verbal communication, expressing dissatisfaction with the product. Written procedures for handling written or verbal customer complaints should be in place per 21 CFR 211.198 [21]. Key points for customer complaint evaluation include:

- Clear, concise written procedures and extensive training of personnel performing the complaint review/investigation to ensure proper, consistent complaint handling.
- 100% of all complaints received must be documented, including receipt of any returned sample or photographs depicting impacted sample/nature of the complaint. As much information as possible should be gathered at the intake call and if possible, the complainant's name and contact information should be obtained for follow-up if needed.
- All complaints need to be investigated to confirm if the complaint is valid. If the complaint is not verified due to lack of information, such as no sample provided, missing lot number, etc., then documentation should include the reason for not completing an investigation as well as the responsible person making the determination that the complaint does not require investigation.
- All complaint records are to be maintained for 1 year after the expiration date of the drug product or 1 year after the complaint was received, whichever timeframe is longer. The file should be maintained at the drug product manufacturing site or be readily available during an inspection.
- The QU should perform the final review and approval of all completed complaint write-ups, including those complaints that could not be investigated due to lack of information/samples. Per § 211.198, *"Such procedures shall include provisions for review to determine whether the complaint represents a serious and unexpected adverse drug experience which is required to be reported to the Food and Drug Administration in accordance with §§ 310.305 and 514.80..."*
- There should be a process to determine the cause of the defect and evaluate the impact of the defect on other lots of products. In cases where a serious and unexpected ADE has occurred, it must be reported to FDA (see Section 3.10.2 of this ISPE Guide).
- A designated file/repository for compounded drug complaints should be maintained at the compounding, processing, or packaging facility.

Manufacturers typically focus first on assessing risk to the customer and on verifying whether a quality problem exists, the source of the problem, and the appropriate correction and/or corrective action. However, it is important to ensure that all information related to any specific product line is aggregated so that adequate investigation and data analysis may be performed to determine if any trends exist. This is the challenge that most manufacturers face: ensuring that all information representing a complaint is properly documented and handled, regardless of whether the information comes from customers external or internal to the manufacturer.

3.10.1 Complaint Investigation

During complaint investigation, it is important that an adequate complaint investigation is performed to determine if the complaint can be confirmed and verified, to discover the root cause of the complaint, and to identify the corrective action. If the complaint is not verifiable, the rationale for not performing an investigation needs to be documented. The extent of the investigation should be directly proportional to the magnitude and risk of the reported issue. These investigations may involve a review of records (i.e., BRs, risk management files, complaint files, or other QS records) to assess risk and determine if deviations occurred in the compounding process or if similar failures have occurred with the product in question or other related product lines. Failure investigations may also involve testing returned or retained samples, auditing suppliers, or other actions appropriate to identifying the root cause(s) for the complaint.

For customer complaint closure, the investigation package should include:

- Root cause summary
- Indication of whether the product was replaced and, if applicable, a reference to the order number
- Final complaint category
- Indication of whether a CAPA is required and, if applicable, a reference to the CAPA number
- Date that the investigation was approved or date of complaint closure
- Adverse event report if the complaint was determined to be a serious and unexpected adverse event (see Section 3.10.2 of this ISPE Guide)
- Summary of communication with the complainant including contacting the complainant to provide a follow-up summary of the investigation

Complaints should be trended annually. If a negative trend is observed (i.e., if the rate of complaints is increasing), an investigation must be opened to determine the root cause of the negative trend and a CAPA may be required. When the complaint is identified as a potential adverse event (see Section 3.10.2 of this ISPE Guide), the complaint should be investigated promptly to meet the 15-day FDA reporting requirement. (21 CFR 310.305 [24])

3.10.2 Adverse Events and Reporting

An adverse drug experience (ADE, also called an adverse event) is *“Any adverse event associated with the use of a drug in humans, whether or not considered drug related, including the following: An adverse event occurring in the course of the use of a drug product in professional practice; an adverse event occurring from drug overdose whether accidental or intentional; an adverse event occurring from drug abuse; an adverse event occurring from drug withdrawal; and any failure of expected pharmacological action.”* [24] (21 CFR 310.305) ADE complaints should be documented and promptly addressed as they can result in serious impact to patient health and legal liability. The FDA requires manufacturers, packers, and distributors of compounded marketed prescription drug products to establish and maintain records and make reports to FDA of all serious, unexpected adverse drug experiences associated with the use of their drug per Section 503B(b)(5) of the FD&C Act [5] and 21 CFR 310.305. [24]

21 CFR 310.305 further defines the following ADE terminology:

“Life-threatening adverse drug experience. Any adverse drug experience that places the patient, in the view of the initial reporter, at immediate risk of death from the adverse drug experience as it occurred, i.e., it does not include an adverse drug experience that, had it occurred in a more severe form, might have caused death.”

“Serious adverse drug experience. Any adverse drug experience occurring at any dose that results in any of the following outcomes: Death, a life-threatening adverse drug experience, inpatient hospitalization or prolongation of existing hospitalization, a persistent or significant disability/incapacity, or a congenital anomaly/birth defect.”

“Unexpected adverse drug experience. Any adverse drug experience that is not listed in the current labeling for the drug product. This includes events that may be symptomatically and pathophysiologically related to an event listed in the labeling, but differ from the event because of greater severity or specificity... ‘Unexpected,’ as used in this definition, refers to an adverse drug experience that has not been previously observed (i.e., included in the labeling) rather than from the perspective of such experience not being anticipated from the pharmacological properties of the pharmaceutical product.”

Reporting ADEs alerts the FDA of potential quality problems that may be linked with a particular product or an issue with standard conditions or processes at the 503B facility. ADE reporting helps the FDA distinguish between quality issues limited to a particular facility or broader issues related to components or APIs used in compounding. Additionally, ADEs may be reported for issues unrelated to compounded drug quality, such as a patient's underlying medical condition.

Many companies employ a call center for intake of ADEs; however, anyone told about a serious or unexpected ADE is responsible for ensuring it is reported to the FDA. For example, if an employee is told by someone that they had an ADE that qualifies as a serious and/or unexpected ADE to one of the company's products, the employee should take down the information and report it to the group responsible for collecting this information for the facility. As much information as possible related to the ADE should be obtained by the drug compounder/intake call employee. ADE reporting requirements state that serious and/or unexpected ADEs must be reported as soon as possible, but no later than 15 calendar days from initial receipt of the information by the person whose name appears on the label. The information to be provided in the report are detailed in 21 CFR 310.305 [24], with key points as follows:

- Information related to the identifiable patient
 - Whenever possible, this should include patient code, contact information, age or age range, gender, and weight. Sometimes it is not possible to obtain all of this information, but significant effort should be made to obtain as much information as possible.
 - Patient identity must not be revealed when submitting reports to the FDA of an adverse event. In order to keep this information anonymous, the patient is assigned a unique identifying code in lieu of including information such as the patient's name or address. A cross-listing with the patient information should be maintained at the intake.
- Identifiable person that reported the serious and/or unexpected ADE, even if this person is the patient.
- Suspect drug information (all available information)
- Specifics of the serious and/or unexpected ADE
- A copy of the Drug Product Labeling in electronic format

In some cases, a medical event that does not result in the above outcomes could also be determined to be a serious ADE upon medical review per the GFI, these are called "important medical events."

"Important medical events that may not result in death, be life-threatening, or require hospitalization may be considered a serious adverse drug experience when, based upon appropriate medical judgment, they may jeopardize the patient or subject and may require medical or surgical intervention to prevent one of the outcomes listed in this definition. Examples of such medical events include allergic bronchospasm requiring intensive treatment in an emergency room or at home, blood dyscrasias or convulsions that do not result in inpatient hospitalization, or the development of drug dependency or drug abuse" (21 CFR 310.305(b) [24]).

It is considered industry best practice that an appropriate medical professional evaluates ADEs to determine if the ADE reported is a serious and/or unexpected ADE.

All serious and/or unexpected ADEs associated with use of the compounded drug product must be reported by the outsourcing facility. When reporting the event, the facility must submit the report as a 15-day "Alert Report." A copy of the compounded drug product labeling (all labeling components) must accompany the ADE report. Information regarding the four data elements should be actively investigated by the outsourcing facility after receiving the information about the ADE:

1. An identifiable patient

2. An identifiable reporter
3. A suspect drug
4. A serious adverse event

In the case of a death, outsourcing facilities should also provide any available information on what led to the death, consult the FDA Guidance for Industry: Adverse Event Reporting for Outsourcing Facilities Under Section 503B of the Federal Food, Drug, and Cosmetic Act [40] for additional requirements.

The outsourcing facility should promptly investigate adverse events that are the subject of a 15-day “Alert Report.” It should “*exercise due diligence to obtain information about any of the remaining elements and should keep records of its efforts to obtain this and other relevant information (e.g., dates of discussions with the reporter to determine how many patients experienced a particular adverse event or dates of discussions with a health care facility to obtain contact information for an identifiable person who purports to have knowledge about the patient, adverse event, or drug involved).*” [40]

After the initial adverse event report is submitted, these reports should be promptly investigated (see Section 3.10.1). A follow-up report must be submitted to the FDA within 15 calendar days of receipt of any new information or as requested by FDA. Additionally, the outsourcing facility should report any new information concerning the initial report, as it is obtained, as the information could assist the FDA in investigating an adverse event. If additional information is not obtainable, the outsourcing facility should maintain records of the steps taken to seek the additional information. Reporting on adverse events should continue as long as the outsourcing facility continues to obtain new information regarding the suspect compounded drug product or the adverse event. [40]

Per 21 CFR 310.305 [24], records of adverse events reported to regulatory authorities, including raw data and any correspondence related to the adverse event, must be maintained for a minimum of 10 years and access to the records must be provided to the FDA for review, copying, and verification. The regulations acknowledge that the report does not mean an admission that the drug caused or contributed to any adverse event. [40]

3.10.3 Field Alert Reporting

According to the draft GFI – CGMPs for 503B Facilities [9], 503B outsourcing facility must have written procedures specifying the actions required if the drug product fails to meet a criterion for sterility:

- Immediate notification of the sterility failure to facilities that received the drug product, along with any appropriate information and recommendations to aid in the treatment of patients to which the compounded drug product was administered.
- The notification is documented.
- Written notification to the FDA within 5 working days of the discovery of the sterility failure via email to: OFAlertReport@fda.hhs.gov.

In accordance with 21 CFR Part 211 [21], investigations initiated for deviations, complaints, or adverse events concluding that the product quality of a batch or batches was compromised should be evaluated as to whether a field alert/recall is warranted. See Section 3.11 for information on recalls.

3.11 Recalls

A product recall is a voluntary action that may be taken to remove potentially violative product distributed to the market that may present a risk or injury to public health based upon internal data (i.e., stability data, quality investigation) or external data (complaint). A product recall may be initiated on the company's own initiative, by FDA request, or, in the event that a firm's actions are not effective in recalling the product, FDA may consider taking other appropriate actions. In March 2022, FDA issued the Guidance for Industry and FDA Staff – Initiation of Voluntary Recalls Under 21 CFR Part 7, Subpart C [41] to establish recall initiation procedures, ensure timely identification of and response to product problems that might lead to a recall, and to promptly issue recall communications and press releases or other public notices.

Companies should have recall policies and procedures for making a recall decision, documentation required for the recall, and a process to initiate a product recall or correction. The FDA provides a website of recall resources: www.fda.gov/safety/recalls-market-withdrawals-safety-alerts/recall-resources.

The FDA reviews a firm's overall recall strategy including associated recall notifications and assigns a recall classification level. The recall classification is the numerical designation assigned based on the relative degree of health hazard presented by the product being recalled. The levels as stated in 21 CFR 7.3 are as follows [24]:

- *“Class I is a situation in which there is a reasonable probability that the use of, or exposure to, a violative product will cause serious adverse health consequences or death.*
- *Class II is a situation in which use of, or exposure to, a violative product may cause temporary or medically reversible adverse health consequences or where the probability of serious adverse health consequences is remote.*
- *Class III is a situation in which use of, or exposure to, a violative product is not likely to cause adverse health consequences.”*

The company should initiate an investigation to determine the root cause of the product issue and determine the corrective actions to take to eliminate the issue. The firm should not delay the initiation of the voluntary recall pending investigation. The FDA should be updated on the progress and resolution of the investigation periodically (between 2 and 4 weeks, depending on the relative urgency of the recall) until the recall has been terminated. Recall status reports should include (21 CFR 7.53 [24]):

- Dates and method of customer notification
- Number of customers notified
- Number of customers that responded
- Quantity of recalled product returned or otherwise accounted for
- Number of customers that did not respond
- Estimated time frame for completion of the recall
- Details of the company's recall effectiveness checks

It is good practice to also include:

- Root cause of the problem that resulted in the recall
- A summary of the CAPA to prevent future occurrences of the problem, specifically focusing on:
 - Actions taken pertaining to the recall strategy
 - Investigation results
 - Underlying cause(s)
 - Actions taken to prevent recurrence

Termination of the recall can be requested when it is reasonable to assume that the recalled product has been removed and proper disposition or correction has been made. (21 CFR 7.55 [24]) A final status report and document of recalled product disposition should be provided to the FDA Division Recall Coordinator, after which the FDA will consider formal termination of the recall.

4 Supply Chain and Materials

4.1 Introduction

Section 503B [5] states that a compounder must use bulk APIs and components that comply with the standards of an applicable USP-NF monograph, if a monograph exists. In addition, as stated in Chapter 2, only bulk APIs listed on the current 503B Bulks List [16], which is updated periodically, can be used for compounding drug products, unless the drug product appears on the Drug Shortages List [28] in effect under Section 506E at the time of compounding, distribution, and dispensing. Per 21 CFR 210.3 Definitions [20], a component is *“any ingredient intended for use in the manufacture of a drug product, including those that may not appear in such drug product.”*

21 CFR 211, Subpart E [21] sets requirements for the selection, purchase, receipt, handling, and use of APIs, drug products, materials (excipient and processing aides), and components (e.g., Container-Closure System (CCS)). These requirements apply to all compounded pharmaceutical products prepared by a 503B compounding facility.

4.2 Supplier/Vendor Qualification Program

Supplier/vendor selection should be made with preference given to drug products, components, and packaging and labeling materials manufactured at FDA-registered facilities that comply with the USP-NF monograph for the items to be purchased. API and drug products must be obtained from an FDA-registered facility or CGMP facility. It is preferable that the selected components have a USP-NF monograph.

Once a supplier/vendor is selected, the outsourcing facility should enter into a quality agreement with the supplier/vendor of each drug product, component, and packaging and labeling material that will be used in the final compounded product, as discussed in the FDA GFI Contract Manufacturing Arrangements for Drugs: Quality Agreements [36]. The quality agreement should minimally contain:

- Description of the testing performed prior to release for each lot and shipment
- Specific tests required on the COA to meet the compounding specifications for the material
- Defined responsibilities for incoming drug product/component sterility, when applicable
- Requirements for supplier/vendor notification to the compounding facility of changes to material testing specifications, test methods, manufacturing location, or manufacturing process
- Immediate notification of a quality issue of any item shipped or located at the outsourcing facility

The process for vendor qualification should be established by the outsourcing facility and documented to ensure that materials are purchased only from qualified suppliers/vendors. Approval by the QU of vendors and the signed quality agreements are the minimum requirements for supplier/vendor qualification. Additionally, a combination of the following is recommended:

- Supplier/vendor audits, where the frequency of audits is determined by the criticality of the vendor
 - For example, suppliers of sterile components should be audited more frequently than suppliers of packaging components
- Supplier/vendor material testing prior to supplier/vendor qualification

- Annual testing by the compounding facility of materials received from supplier/vendor
 - For service providers, an annual evaluation of services provided is used, where applicable (see Section 6.4 for more information on reduced testing for materials from qualified vendors).
- Periodic evaluation of deviations associated with the supplier/vendor or service provider

4.3 Critical Suppliers of Raw Materials, Components, or Services

4.3.1 Drug Products, Components, Packaging, and Labeling

RMs, processing aides (e.g., compressed gases), and components used in compounding should be obtained from an FDA-registered facility whenever possible. At a minimum, the bulk API must be manufactured in an FDA-registered facility. If components are not available from an FDA-registered facility, the 503B outsourcing facility should perform an evaluation (audit, etc.) to ensure that the components obtained are of high quality with appropriate identity and purity. Drug products and components cannot be used for compounding if they appear on the FDA-published list of drug products that have been withdrawn or removed from the market for safety or effectiveness reasons (21 CFR 216.24 [24]).

When selecting materials for use in compounding, the type of compounding (sterile versus non-sterile) should be considered. Appropriate controls should be exercised and documented for the purchasing, receipt, storage, material movement (quarantine, approval, rejection, release, return to manufacturer, destruction), and usage of drug products, components, and packaging and labeling in drug compounding.

Sterile drug products and components require additional written controls to ensure that sterility is maintained, where required, from receipt through compounding. Drug products and components should be labeled by the manufacturer with the drug product/component name, lot/batch number, and expiration date. When selecting drug products and components for the final compounded drug product, the impact of manipulation of the drug product/component relative to the intended use in compounding should be evaluated.

All RMs and components should have specifications established by the manufacturer that are scientifically based and comply with respective USP-NF monographs, if they exist. These specifications should address the critical quality attributes that ensure the quality of the finished drug product and that are appropriate for the route of administration of the compounded drug product. Specifications should minimally include those tests described in ICH Q6A Section 3.2, Universal Tests/Criteria [42].

APIs, drug products, RMs, and components received from all suppliers must be accompanied by a valid COA that must be reviewed for compliance to established internal specifications. Bulk containers must be labeled in accordance with Occupational Safety and Health Administration (OSHA) [43] procedures and associated Safety Data Sheets (SDS) must be available to personnel handling the material.

The selection of packaging materials and storage of materials are discussed in more detail in Chapter 7.

Upon receipt at the outsourcing facility, the following controls should be in place:

- Storage of all materials, APIs, and drug products in a manner that maintains the material according to the manufacturer-recommended storage conditions
- Written procedures detailing warehousing and specific storage areas/requirements
- Special storage conditions, such as temperature and humidity, specified by the supplier/vendor

- Segregation of materials as appropriate (see additional information in Chapter 7) while storing materials in a manner that prevents contamination, chemical reaction, or compromises material quality
- Storage of materials off the floor to allow for ease of cleaning storage areas
- Written warehousing procedures for management of inventory, including movement of final drug product to end user in order to trace compounded drugs in the event of a recall.
- Management of warehouse inventory should consider a First In, First Out (FIFO) method to ensure proper rotation of warehouse materials.

QC is responsible for approval/rejection of drug products, in-process materials, components, and labeling and packing materials. These QC responsibilities are discussed in more detail in Chapter 3 Quality System and reduced testing is discussed in Section 6.4.

4.3.2 Reserve Samples

An appropriately identified reserve sample representative of each lot or batch of drug product should be retained and stored in accordance with the compounded drug labeling. See Section M in the draft GFI – CGMPs for 503B Facilities [9] for additional information.

4.3.3 Services

Written procedures must be in place for managing the selection and qualification of providers of outsourced services, such as contract laboratories, consultants, uniform services, software vendors, etc. Each service vendor must meet the vendor qualification program requirements as well as enter into a quality agreement with the 503B outsourcing facility.

4.4 Shipping

Compounded drugs must be shipped from the 503B outsourcing facility in a manner that preserves the SQIPP and safety of the product. Packaging storage containers must be closed tightly, and the compounded drug must not interact physically or chemically with the packaging, as discussed in Section F of the draft GFI – CGMPs for 503B Facilities [9]. Procedures for shipping should be in place to ensure that temperature, humidity, and light requirements are maintained during shipment/delivery to the patient or healthcare provider. Shipping studies may be needed depending on the conditions for shipment/delivery of the compounded drug product.

Shipping requirements should also consider the Beyond-Use Date (BUD) of the compounded drug. (See Chapter 9.)

Where a manufacturer of 503B products cannot ensure that product will be shipped under the specified storage conditions, temperature studies and/or studies under accelerated conditions (e.g., stability studies) are recommended to determine the impact of temperature excursions on product quality. Shipping studies, representative of yearly temperatures during different seasons, may be considered to ensure that product storage requirements are maintained. Additionally, to determine the actual shipping conditions, using a temperature/humidity data logger or other monitoring device is recommended.

4.5 Drug Enforcement Administration Controlled Substances

The DEA assigns products into categories called Schedules (I-V) that are defined by the drug's acceptable medical use and the drug's abuse or dependency potential. Materials (primarily APIs) classified as controlled substances can only be purchased from qualified vendors who have the appropriate licenses to handle the controlled substance, and can only be purchased by someone with the appropriate license to handle the controlled substance. [9] For example, Schedule I drugs are defined as drugs with no currently accepted medical use and a high potential for abuse.

Controlled substances require special storage, depending on the DEA schedule. They also may require special documentation for transport. There are many regulations around DEA-controlled substances that are not specifically covered in this ISPE Guide, but can be found in 21 CFR Parts 1301, 1305, 1306, 1307, 1308, and 1309 [24].

5 Facilities and Equipment

5.1 Introduction

This chapter provides an overview of the requirements that should be in place for the facilities and equipment used in the manufacturing, processing, packing, or holding of drugs compounded in 503B outsourcing facilities. This chapter also covers insanitary conditions, as listed in Section 501(a)(2)(A) of the FD&C Act [4], and as further defined in the GFI – Insanitary Conditions [33]. (See Section 5.4.)

It should be noted that clearly designated separation is required in a facility that performs 503A compounding, 503B compounding, and/or pharmaceutical manufacturing. If the 503A and 503B have the same physical address, common areas/system air handling systems, or rooms, then the FDA considers the 503A portion of the facility as 503B and subject to regulation as such (per GFI – 503B Facility Definition, Section III [25]).

5.2 Facilities

5.2.1 Facility Design

Proper design of a facility is required under 21 CFR 211, Subpart C, of the CGMP regulations on Buildings and Facilities [21]. The facility design at the site should be appropriate for its intended use, which in this case, is compounding the drug products types determined by the firm.

21 CFR 211.42 Design and construction features [21] outlines regulations applicable to 503B firms for facilities oversight. It states: *“Any building or buildings used in the manufacture, processing, packing, or holding of a drug product shall be of suitable size, construction, and location to facilitate cleaning, maintenance, and proper operations.”* Examples of expectations regarding this requirement are:

- Clean rooms should be designed with easy-to-clean surfaces
- Facilities should be designed considering the flow of personnel, equipment, and materials. For example, the gowning room should be well positioned to support flow of personnel into clean rooms.

Good practice is to add redundancy on Primary Engineering Controls (PEC) and monitoring systems where the risk is high if these systems were to go down, e.g., uninterruptable power supply systems, backup generators, and duplicate sensor hardware. This may be considered as a business risk rather than a product quality issue. However, if any of these systems are used to monitor quality systems essential to CGMPs, the risk could be considered as both; therefore, a risk assessment should be performed to determine the PEC impact on product quality.

- The following apply to sterile operations [9]: *“Sterile drugs should be produced only in ISO 5 or better air quality as determined under dynamic conditions.”* (see Table 5.1 for ISO cleanroom classification standards)
- *“The facility should be designed and operated with cascading air quality (e.g., by proper air classification and air pressurization) to protect the ISO 5 zone...The facility layout, room separation, and process flow must be designed to prevent the influx of contamination from adjacent areas and rooms of lower air quality and to avoid any disruption of HEPA unidirectional flow.”*
- *“The air cleanliness classification of the area surrounding the ISO 5 zone immediately adjacent to the aseptic processing line should, at a minimum, meet ISO 7 standards under dynamic conditions.”*
- *“If an isolator is used, the surrounding area should, at a minimum, meet ISO 8 standards under dynamic conditions.”*

- *“If a restricted access barrier is used (e.g., a glove box), the surrounding area should, at a minimum, meet ISO 7 standards under dynamic conditions.”*
- *“ terminally sterilized drugs should be produced in ISO 8 or better air quality as determined under dynamic conditions.”*
- A room should be designed to allow suitable space for a number of operators to be present in the ISO 5 area at a given time.

Table 5.1: ISO Classification of Particulate Matter in Room Air

Source: FDA draft GFI – CGMPs for 503B Facilities [9].

ISO Class Name	Particles/m ³
3	35.2
4	352
5	3,520
6	35,200
7	352,000
8	3,520,000
Limits are in particles of 0.5 µm and larger per cubic meter (current ISO) measured under dynamic conditions. Adapted from ISO 14644-1:2015, Cleanrooms and associated controlled environments – Part 1: Classification of air cleanliness by particle concentration.	

The FDA recommends that rooms used for cleanroom/sterile operations contain only equipment needed for those operations and to not store other equipment in them, particularly during operations. Equipment should be placed in the room so that it does not obstruct air vents and airflow that could potentially compromise aseptic operations. [9]

Building design needs to take into account the flow of personnel, equipment, and materials through the facility. Regarding materials specifically, the flow of components, drug product containers, closures, labeling, in-process materials, and final compounded drug products through the building needs to be designed to prevent contamination. Additionally, there should be appropriately sized storage areas with adequate separation where needed/required, designated for equipment and materials to prevent mix-ups between different components, drug product containers, closures, labeling, in-process materials, or drug products to prevent contamination.

21 CFR 211.42 [21] states: *“separate or defined areas or such other control systems for the firm’s operations... are necessary to prevent contamination or mixups during the course of the following procedures...”* These requirements apply to outsourcing facilities and include:

- *“Receipt, identification, storage, and withholding from use of components, drug product containers, closures, and labeling, pending the appropriate sampling, testing, or examination by the quality control unit before release for manufacturing or packaging”*
 - Area for placement of *“rejected components, drug product containers, closures, and labeling before disposition”*
 - Area for *“storage of released components, drug product containers, closures, and labeling”*
 - *“Storage of in-process materials”*

- *“Quarantine storage before release of drug products”*
- *“Storage of drug products after release”*
- Areas designated for operations:
 - *“Manufacturing and processing operations”*
 - *“Packaging and labeling operations”*
 - *“Control and laboratory operations”*

Areas designated for aseptic processing need to include the following, as appropriate [21]:

- *“Floors, walls, and ceilings of smooth, hard surfaces that are easily cleanable”*
- *“Temperature and humidity controls”*
- *“An air supply filtered through high-efficiency particulate air filters under positive pressure, regardless of whether flow is laminar or nonlaminar”*
- *“A system for monitoring environmental conditions”* that is well-defined and documented and has the following:
 - The ability to evaluate routes of microbial contamination that could come from air, surfaces, process, operations, or personnel and ability to detect excursions.
 - The draft GFI – CGMPs for 503B Facilities [9] also states: *“[A] system for environmental monitoring must include the establishment of pressure differential limits (see § 211.42), and control systems should include built-in alarms to detect excursions...[of] pressure differentials, humidity, and temperature...”*
- A system for cleaning and disinfecting the room and equipment to produce aseptic conditions
- A system for maintaining equipment used to control the aseptic conditions

21 CFR 211.42(d) [21] states: *“Operations relating to the manufacture, processing, and packing of penicillin shall be performed in facilities separate from those used for other drug products for human use.”* On this topic, the draft GFI – CGMPs for 503B Facilities [9] states: *“However, FDA has clarified that separate buildings may not be necessary, provided that the manufacturing operation involving penicillin is isolated (i.e., completely and comprehensively separated) from the areas in which non-penicillin products are manufactured...Additionally, appropriate controls related to movement of equipment, product, and personnel should be established to prevent cross contamination of non-beta-lactam products.”*

5.2.2 Facility Design Qualification

As stated in the FDA Guidance for Industry: Process Validation: General Principles and Practices⁹ [44]:

“Process validation involves a series of activities taking place over the lifecycle of the product and process. This guidance describes process validation activities in three stages.

- *Stage 1 – Process Design: The commercial manufacturing process is defined during this stage based on knowledge gained through development and scale-up activities.*

⁹ Referred to as GFI – Process Validation throughout this ISPE Guide.

- *Stage 2 – Process Qualification: During this stage, the process design is evaluated to determine if the process is capable of reproducible commercial manufacturing.*
- *Stage 3 – Continued Process Verification: Ongoing assurance is gained during routine production that the process remains in a state of control.”*

Per the GFI – Process Validation [44], Stage 2 Process Qualification – Element 1 applies to the facility, utilities, and equipment used for the manufacture of (compounded) drug products, and “*refers to activities undertaken to demonstrate that utilities and equipment are suitable for their intended use and perform properly.*” Therefore, the facility design determined in Section 5.2.1 must be qualified to show that it is operating appropriately and as designed based on its intended use. The GFI – Process Validation gives an outline of activities generally included in the qualification of utilities and equipment.

Qualifications of a facility design are often detailed under an overall site Validation Master Plan (VMP). The site VMP may include the utilities and equipment plans or they can be done as individual validation plans. However, they should be referenced in the master plan and the QU should review and approve the qualification plan and report (21 CFR 211.22 [21]). When developing the qualification plan for any of these (facility, utility, or equipment), it is recommended to consider the following [44]:

- Requirements for use and incorporate risk management where appropriate
- Identify tests or studies to use with appropriate criteria to assess the outcome
- Timing of qualification activities, including if one should be done prior to another
- Responsibilities for writing, reviewing, approving, and executing VMPs
- Procedures for documentation and approval of the qualification activities
- How changes or deviations will be managed during qualification activities

The following are examples of qualifications that may be required for different types of facility designs.

For Non-Sterile and Sterile Compounded Drug Products:

- The room can be cleaned to appropriate standards
 - Established procedures for cleaning and disinfecting should include which cleaning agents can be used, how often they can be used, how they are to be used, and if needed, a cleaning agent rotation schedule.
 - *“For multiuse facilities and nondedicated equipment, changeover and cleaning procedures for equipment and utensils must be established and followed to prevent contamination, including cross-contamination between products (see §§ 211.42, 211.67).” [9]*
 - Demonstrate the ability to remove difficult-to-remove API/product.

For Sterile Compounded Drug Products:

- A system for cleaning and disinfecting the room and equipment to produce aseptic conditions:
 - *“A system for cleaning and disinfecting all critical areas to produce aseptic conditions includes sporicidal and other sterile disinfectants and lint-free sterile wipes (see § 211.42).” [9]*

- The results of EM testing help to determine if the cleaning/disinfecting procedures are adequate and effective. *“Conducting disinfectant effectiveness testing may be useful in guiding revision of the disinfection program in such cases.”* [9]
- ISO zones should be qualified appropriately based on the room requirements, which minimally include temperature, humidity, and differential pressure. The draft GFI – CGMPs for 503B Facilities states: *“[A] system for environmental monitoring must include the establishment of pressure differential limits (see § 211.42), and control systems should include built-in alarms to detect excursions...of pressure differentials, humidity, and temperature...”* If excursions occur and cannot be immediately corrected, production should stop until it is back in range. [9]. Depending on the length of downtime, consideration should be given to removal of the product from the compounding/production area back to the designated controlled storage area until the excursion is resolved. Additional information can be found in Section 5.2.3.
- Additionally, the draft GFI – CGMPs for 503B Facilities states, for example, that the ISO 5 zone or critical area must be qualified. Qualification should minimally include conducting and documenting the following studies and tests [9]:
 - *“Airflow studies (e.g., an in-situ smoke study) should be conducted under simulated operational conditions to evaluate airflow patterns because of the risk for contamination of exposed product in the critical area. These studies should be conducted at the critical area to demonstrate unidirectional flow...under dynamic conditions”*
 - > If there is an indication of poor airflow (e.g., non-unidirectional or turbulent), it must be corrected before the room can be qualified. These studies should be repeated when any changes are made to the critical area that might affect airflow.
 - > Smokes studies should be conducted for different setups within an ISO 5 environment as well. For example, *in situ* smoke studies done for filling of syringes in a hood would not necessarily qualify IV bag filling. Consideration and justification needs to be made based on the amount and placement of equipment and materials for each operation.
 - High-Efficiency Particulate Air (HEPA) filters periodic testing/recertification should be performed at least every 6 months to ensure that appropriate airflow and quality are maintained. These tests should include integrity testing of the HEPA filters, particle counts, and air velocity checks.
 - *“Velocities of unidirectional air should be measured 6 inches from the HEPA filter face and at a defined [(and documented)] distance close to the work surface in the ISO 5 area.”*
 - Portable ISO 5 units moved from one location to another should be requalified before resuming use in sterile compounding.
- The qualification of the system design to monitor environmental conditions should include [9]:
 - An EM program to evaluate the potential routes of microbial contamination that could arise from the air, surfaces, process, operation, and personnel practices with a manner for detecting excursions. *“However, environmental monitoring equipment should not interfere with aseptic operations (e.g., instruments should not interfere with validated and appropriate airflow patterns). In particular, the program should:*
 - > *Cover all production shifts and include monitoring during normal production conditions.*
 - > *Include at least daily monitoring of the ISO 5 zone during operations.*
 - > *Establish alert and action limits and define appropriate responses for when excursions occur.*

- > *Describe the use of sampling (e.g., contact plates, swabs, active air samplers), alert and action limits and responses, and testing methods (e.g., media, plate exposure times, incubation times and temperatures) that are designed to detect environmental contaminants, including changes in microflora type and amount, and the scientific justification for the testing methods selected.*
- > *Be supported by a scientific justification for sampling locations, based on risk [evaluation], and sampling methods, which may be based on risk and peer-reviewed literature.*
- > *Investigate results that exceed established limits or demonstrate adverse trends; determine product impact; and execute appropriate corrective actions.” [9]*
- Personnel monitoring, which is a part of EM [9]:
 - > *“Include a routine program for daily/shift monitoring of operators’ gloves and an appropriate schedule for monitoring other critical sites of the gown (e.g., gown sleeves for hood work) during or immediately after completion of aseptic operations. Monitoring should take place before planned disinfection so that actual operating conditions are being assessed.*
 - > *Establish and justify limits that are based on the criticality of the operation relative to the contamination risk to the product.*
 - > *Call for an investigation of results that exceed the established levels or demonstrate an adverse trend, a determination of the impact on the sterility assurance of finished products intended to be sterile, and the development and execution of appropriate corrective actions.”*

5.2.3 Ventilation, Air Filtration, Air Heating and Cooling

21 CFR 211.46 Ventilation, air filtration, air heating and cooling [21] discusses the regulations applicable to 503B outsourcing facilities. Examples are outlined in the draft GFI – CGMPs for 503B facilities [9]:

- *“Adequate ventilation shall be provided.” (§ 211.46 [21])*
 - *“HEPA filters should be qualified to provide appropriate air quality and be periodically maintained and tested to ensure intended air quality. Discolored, dirty, or damaged HEPA filters should be repaired or replaced.” [9]*
- *“Equipment for adequate control over air pressure, micro-organisms, dust, humidity, and temperature shall be provided when appropriate for the manufacture, processing, packing, or holding of a drug product.” (§ 211.46 [21])*
 - *“Temperature and humidity must be maintained in cleanroom areas; such controls are critical to reduce micro-biological growth (see § 211.46). A specification for humidity should be established considering that higher humidity supports microbial growth, while too little humidity can cause problems with static electricity (which may be particularly problematic when working with powders) and may lead to increased particulates.” [9]*
 - *“Cleanroom temperature and humidity specifications should be maintained solely through the facility’s central heating, ventilation, and air conditioning (HVAC); peripheral devices such as stand-alone (de-) humidifiers and air conditioners should not be used because they generate airborne particles, are water sources, and may harbor microorganisms.” [9] Additionally, there are no monitoring controls on them.*
- *“Air filtration systems, including prefilters and particulate matter air filters, shall be used when appropriate on air supplies to production areas. If air is recirculated to production areas, measures shall be taken to control recirculation of dust from production. In areas where air contamination occurs during production, there shall be adequate exhaust systems or other systems adequate to control contaminants.” (§ 211.46 [21])*

- “Air-handling systems for the manufacture, processing, and packing of penicillin shall be completely separate from those for other drug products for human use.” (§ 211.46 [21])

5.2.4 Compressed Gases

Compressed gases are used in many areas of the production of drug products, for example:

- Compressed air or other gas used in an analytical instrument to deliver the sample through a column for testing
- Compressed air for a vacuum to work to transfer materials
- Nitrogen or other gases used to blanket a drug product sensitive to oxygen

Compressed gases may be purchased from a qualified vendor, per the supplier qualification program (see Section 4.2). Compressed gases come in different grades, depending on its use, with specifications based on the grade. They should be part of the QA testing program.

Sometimes a facility may choose to pipe compressed gas into a facility in order to supply areas more efficiently or cost effectively. In this case, the facility should qualify the overall system design and monitor it on a periodic basis to ensure that it is not contaminated, does not leak, and delivers the appropriate quantity/pressure of gas needed. It is recommended to use flow regulators at the point of use to control incoming gas pressure.

5.2.5 Sanitation

Regulations regarding sanitation are discussed in 21 CFR 211.56 [21]:

- Buildings used to manufacture, process, pack, or hold a drug product should be maintained in a clean and sanitary condition, be free of infestation by rodents, birds, insects, and other vermin (other than laboratory animals). Trash should be collected and disposed of in a timely and sanitary manner.
- Procedures should be in place and followed for the cleaning schedules, methods, equipment, and materials allowed to be used in cleaning the buildings and facilities, who is responsible for these activities, and how the cleaning is to be documented.
- Procedures should be in place and followed for allowable rodenticides, insecticides, fungicides, fumigating agents, and cleaning and sanitizing agents. The chosen agents should take into account preventing contamination of equipment, components, drug product containers, closures, packaging, labeling materials, or drug products. “Rodenticides, insecticides, and fungicides shall not be used unless registered and used in accordance with the Federal Insecticide, Fungicide, and Rodenticide Act (7 U.S.C. 135).”

5.2.6 Maintenance

In accordance with 21 CFR 211.58 Maintenance [21], “Any building used in the manufacture, processing, packing, or holding of a drug product shall be maintained in a good state of repair.” The FDA does not go into detail on the expectations for a maintenance program in an outsourcing facility, as they are the same as for traditional manufacturing facilities.

The *ISPE Good Practice Guide: Maintenance (Second Edition)* [45] is a useful reference about comprehensive maintenance programs and recommends using a risk assessment to determine the level of maintenance needed. A maintenance program should include the following aspects:

- System inventory and risk assessments
- Maintenance plans and schedules

- Change management
- Roles and responsibilities
- Documentation requirements, including check sheet, use logs, etc.
- Spare parts list
- Training requirements

The maintenance plan for a given piece of equipment should be determined during the commissioning and qualification phase, prior to putting the asset into service. Regulations are flexible regarding how the organization is to be structured; however, regulations specifically state that assignment of responsibility for maintenance and maintenance scheduling be included in the written procedures for maintenance (§ 211.67 [21]). It is an expectation that roles and responsibilities are established for the overview of the maintenance activities including:

- Establishing maintenance procedures (such as for preventive maintenance schedules, lock-out/tag-out, passivation, etc.)
- Authorizing changes
- Review and approval of work processes/orders
- Identification and investigation of deficiencies
- Maintenance-specific training
- Documentation requirements

Computerized Maintenance Management Systems (CMMSs) may provide capabilities for scheduling and maintaining documentation for maintenance management. However, smaller sites tend to use manual systems, which are also effective from a regulatory perspective. If a CMMS is used, it should be in compliance with 21 CFR Part 11 [46]. For further information, see the *ISPE GAMP® Guide: Records and Data Integrity* [47].

For both facilities and equipment, change management (refer to Section 3.5) is used to address changes that occur throughout the lifecycle, including non-routine repairs, modifications, and repairs. The impact of the change should be assessed and the need for requalification determined.

CGMP regulations state that “*records shall be kept of maintenance, cleaning, sanitizing, and inspection...*” (§ 211.67 [21]) CGMP systems and equipment should have a documented status, and when maintenance is being performed, a tag identifying the status of the equipment should be displayed visually. Appropriate cleaning should be performed before and after maintenance activities to minimize drug product contamination and exposure concerns. All maintenance and cleaning activities should be logged in the logbook or CMMS.

Any spare parts and materials (lubricants, etc.) used on the equipment need to be approved. It is recommended that any non-approved materials and parts are not allowed in the area to prevent accidental usage. Spare parts lists are documented and approved by the QU. Spare parts may be identified on the list as a like-for-like change, which enables installation on equipment without additional change management documentation requirements as they are functionally equivalent; however, the change is required to be documented on a work order or similar system.

Training is required in the maintenance group for CGMPs and job function, including the impact of their duties on drug product quality. See Section 3.3 for details on training requirements. Trade skills that personnel bring to a job are normally confirmed through a review of education and certification documents (e.g., certified electrician) during hiring, and training is not required on those specific skills.

5.2.7 Calibration

A calibration program is an essential element for CGMP compliance in the pharmaceutical industry, including 503B compounding. Calibration is needed in both process areas and laboratory instrumentation.

As with maintenance, the first resource for calibration information should be the manufacturer of the system or equipment. They are the best source to provide the recommended schedule for maintenance and calibration, as well as what should be done at each time period.

However, as noted in the *ISPE GAMP® Good Practice Guide: A Risk-Based Approach to Calibration Management (Second Edition)* [48], the criticality of the instrument and the likelihood of drift based on the environmental conditions in which it operates could warrant more frequent recalibration than required by the manufacturer. This allows the facility to focus their resources in the most important areas, which will maximize their effectiveness. The risk assessment process can help direct these decisions and reduce costs.

5.2.8 Other Aspects

There are other regulations listed under Subpart C of 21 CFR 211 [21] that are currently not mentioned in the draft GFI – CGMPs for 503B Facilities [9]; however, they are requirements for drug manufacturers and have important functions in ensuring that facilities maintain CGMPs. It is recommended that 503B firms consider applying the following 21 CFR 211 buildings and facilities regulations to their facilities:

- § 211.44 Lighting
- § 211.48 Plumbing
- § 211.50 Sewage and refuse
- § 211.52 Washing and toilet facilities

Reviews of all systems and utilities should be done to determine the level of qualification or validation required at initial design/build, and for ongoing maintenance. This should be documented by the site and reviewed periodically to determine if updates are needed.

Other regulations related to facilities mentioned in the draft GFI – CGMPs for 503B Facilities [9] include:

- 21 CFR 200.10 Contract facilities (including consulting laboratories) utilized as extramural facilities by pharmaceutical manufacturers [24]:
 - Contract facilities are often utilized by pharmaceutical manufacturers. The FDA, in accordance with Section 704(a) of the FD&C Act [4], is authorized to inspect contract facilities, including any factory, warehouse, or establishment in which compounded drugs are manufactured, processed, packed, or held.
 - Many outsourcing facilities regard contract facilities as an extension of their own facility. As such, the QU at the outsourcing facility is responsible for ensuring that the existence, appropriateness, and implementation of the requirements outlined in the draft GFI – CGMPs for 503B Facilities [9] are followed at the contract facility. The outsourcing facility should be aware of all work that is done at contract facilities on their behalf.
 - It is of note that “the Food and Drug Administration does not consider results of validation studies of analytical and assay methods and control procedures to be trade secrets that may be withheld from the drug manufacturer by a contracted extramural facility.” [24]

- 21 CFR 211.28 Personnel responsibilities [21]:
 - *“Personnel engaged in the manufacture, processing, packing, or holding of a drug product shall wear clean clothing appropriate for the duties they perform. Protective apparel, such as head, face, hand, and arm coverings, shall be worn as necessary to protect drug products from contamination.”*
 - It is recommended that facilities provide uniforms and dedicated shoes for their personnel to change into. Street clothes should not be worn into the manufacturing areas, even though they will be covered by further gowning.
 - Additionally for sterile drug products, the draft GFI – CGMPs for 503B facilities [9] states: *“Procedures for monitoring the environment and personnel for the presence of viable particles and nonviable particles should be established and followed.”* Additional clothing/gowning is necessary for working in specific areas. Gowning is discussed in more detail in Section 6.6. Training for personnel gowning should include both written procedures and OJT. It is important that gowning procedures for manufacturing areas *“ensure that there is no exposed skin on personnel involved in any production activities in, or that can directly affect, the ISO 5 area.”*

- 21 CFR 211.113 Control of microbiological contamination [21]:

- *“Appropriate written procedures, designed to prevent objectionable microorganisms in drug products not required to be sterile, shall be established and followed.”*

Production areas for non-sterile drug products should be designed to minimize the level of contamination where components, equipment, CCS, in-process materials, and drug products are prepared, held, stored, or transferred.

- *“Appropriate written procedures, designed to prevent microbiological contamination of drug products purporting to be sterile, shall be established and followed. Such procedures shall include validation of all aseptic and sterilization processes.”*

Note that the production areas for sterile drug products should be designed to minimize contamination where components, equipment, CCS, in-process materials, and drug products are prepared, held, stored, or transferred.

5.3 Equipment

5.3.1 Equipment Qualification

Equipment qualification is required in 503B outsourcing facilities as in all CGMP facilities. Initial equipment qualification should be conducted according to an approved protocol or plan and include multiple steps, some of which may be performed concurrently, and appropriately documented. A plan should be developed, based on risk assessments, for when equipment needs to be requalified. See Table 5.2 for equipment qualification details.

Table 5.2: Equipment Qualification Steps [49]

Qualification Step	Goal	Details
Installation Qualification (IQ)	Identify and document equipment	- Verify proper installation – critical components are installed correctly and in accordance with design documentation requirements - Integration with other equipment and utilities is correct - Document via operating manuals, equipment drawings, etc., and record any deviations and evaluate for impact
Operation Qualification (OQ)	Verification of alignment with the functional specifications	- Verify operation within specified parameters such as temperature, pressure, flow, etc. - Involves the verification of the proper operation of controllers, alarms, recorders, interlocks, etc. - Confirmation that the equipment or system can sequence through its operating steps - Verify that key functions and process parameters are checked and meet the operating specifications - Confirm no undesirable operations - Ensure the system appropriately responds to alerts/fault/failures/alarms and conditions - May include scale-up or optimization by testing different parameters - Ensure that critical components and systems can operate within established limits and tolerances “Worst-case” demonstration that equipment performs as expected at the extremes of the proposed ranges of operation (Proven Acceptable Range)
Performance Qualification (PQ)	Verification of the ability of a process to meet user requirements specifications	- To conduct PQ – the following documents are needed: Piping and instrumentation diagrams, approved SOPs, load configurations, test methods, calibration certificates - Integrate procedures, personnel, systems, and materials to verify that the pharmaceutical grade utilities, environment, equipment, and/or support systems produces the required output - Conduct PQ production under conditions that simulate those that used during actual manufacturing - Issue a final report that includes a summary of results, any OOS results, outlying data, deviations, investigations, non-conformances, etc., encountered during validation

A requalification plan should be available at the facility that shows the tests needed to requalify each piece of equipment and at what frequency. This may be part of the VMP, an SOP, or kept as part of the calibration and maintenance programs, as they are often scheduled and/or executed by the same personnel.

5.3.2 Equipment CGMP Requirements

503B outsourcing facilities are required to follow Subpart D of 21 CFR 211 [21] for equipment oversight, which includes design, size and location, construction, cleaning and maintenance, automatic, mechanical, and electronic equipment, and filters. Additionally, requirements for written records for cleaning and use logs are detailed in 21 CFR 211.182.

21 CFR 211.63 [21] states that *“equipment used in the manufacture, processing, packing, or holding of a drug product shall be of appropriate design, adequate size, and suitably located to facilitate operations for its intended use and allow for its cleaning and maintenance.”* Equipment design needs to take into consideration sanitization or sterilization needs to prevent contamination.

21 CFR 211.65 [21] Equipment construction states: *“Equipment shall be constructed so that surfaces that contact components, in-process materials, or drug products shall not be reactive, additive, or absorptive so as to alter the safety, identity, strength, quality, or purity of the drug product beyond the official or other established requirements.”* Additionally, *“[a]ny substances required for operation, such as lubricants or coolants, shall not come into contact with components, drug product containers, closures, in-process materials, or drug products.”*

CGMP equipment should be identified by a unique number attached to the equipment to track use and maintenance activities. Equipment should have a documented status indicating if the equipment is dirty, out of service (for maintenance), clean (and cleaning expiration based on Clean Hold Times (CHTs)), or decommissioned/to be removed. Additionally, the equipment logbook should be kept up-to-date with the status so it is clear what was the last product/batch run on the equipment.

21 CFR 211.182 [21] discusses the use of equipment cleaning and use logs:

- *“A written record of major equipment cleaning, maintenance (except routine maintenance such as lubrication and adjustments), and use shall be included in individual equipment logs that show the date, time, product, and lot number of each batch processed. If equipment is dedicated to manufacture of one product, then individual equipment logs are not required, provided that lots or batches of such product follow in numerical order and are manufactured in numerical sequence. In cases where dedicated equipment is employed, the records of cleaning, maintenance, and use shall be part of the batch record. The persons performing and double-checking the cleaning and maintenance (or, if the cleaning and maintenance is performed using automated equipment under § 211.68, just the person verifying the cleaning and maintenance done by the automated equipment) shall date and sign or initial the log indicating that the work was performed. Entries in the log shall be in chronological order.”*

21 CFR 211.67 Equipment cleaning and maintenance [21] includes not only equipment, but accessory items such as utensils, stating: *“Equipment and utensils shall be cleaned, maintained, and, as appropriate for the nature of the drug, sanitized and/or sterilized at appropriate intervals to prevent malfunctions or contamination that would alter the safety, identity, strength, quality, or purity of the drug product beyond the official or other established requirements.”* The draft GFI – CGMPs for 503B Facilities [9] gives additional recommendations on how to apply these requirements in an outsourcing facility. For example:

- For multiuse facilities and/or nondedicated equipment, procedures should be in place for changeover and cleaning of equipment and utensils to prevent contamination, including cross-contamination between products.
- The cleaning procedures for equipment that comes into contact with the drug product should be evaluated to ensure the adequacy of cleaning, prevent cross-contamination, and ensure sterility and cleanliness at time of use.
- *“Critical equipment surfaces that come in contact with sterile drug products, containers, and closures must be sterilized at appropriate intervals (see § 211.67); disinfection alone is not sufficient... Single-use disposable equipment and supplies that are purchased presterilized and depyrogenated and are discarded after one use need not be resterilized.”*

- *“For sterility and endotoxin limits, a valid COA may be used in lieu of testing by the outsourcing facility for single-use disposable equipment (see §§ 211.65, 211.113). If the outsourcing facility does not use presterilized and depyrogenated single-use disposable equipment (e.g., filters, transfer tubing, temporary holding vessels), the equipment must be sterilized and depyrogenated before use through processes that have been validated (see §§ 211.65, 211.67(a) and (b), 211.100, 211.113).”*

An option for outsourcing facilities is to use single-use disposable equipment, which reduces some of the cleaning requirements and time between batches as well as the potential for contamination. *“The suitability of single-use disposable equipment for its use in processing may be determined by the use of a valid COA from the supplier in lieu of testing or examination by the outsourcing facility (see §§ 211.65, 211.113).”* The integrity of the packaging of the single-use disposable equipment should be verified upon receipt before use. The single-use disposable equipment should also be inspected for damage or contamination following use. (Draft GFI – CGMPs for 503B Facilities [9])

Procedures expected to be in place for the cleaning and maintenance of equipment and accessory items include:

- Responsibility for cleaning and maintaining equipment
- Maintenance and cleaning schedules, including, where appropriate, sanitizing schedules
- Detailed descriptions of the methods, equipment, and materials used in cleaning and maintenance operations
- Methods of disassembling and reassembling equipment, as necessary, to ensure proper cleaning and maintenance
- Removal or obliteration of previous batch identification
- Protection of clean equipment from contamination prior to use
- Inspection of equipment for cleanliness immediately before use

Chemical passivation of stainless steel tanks/equipment should be performed prior to initial use as well as periodically throughout the equipment lifecycle to ensure the tank surface remains nonreactive. Tanks should be routinely examined for rouging and other defects.

Records of maintenance, cleaning, sanitizing, and inspection must be kept as specified in 21 CFR 211.180 and 211.182 [21]. These records can be maintained electronically or via written measures, such as equipment logbooks.

21 CFR 211.68 [21] states that automatic, mechanical, or electronic equipment, including computers, or related systems that are appropriate for the function, may be used in the manufacture, processing, packing, and holding of a drug product. Key requirements related to automatic, mechanical, and electronic equipment are as follows:

- *“If such equipment is so used, it shall be routinely calibrated, inspected, or checked according to a written program designed to assure proper performance. Written records of those calibration checks and inspections shall be maintained.”* This equipment must be qualified as capable of performing its intended functions or operations before first use. Where available, equipment with audit trail capability should be used.
- *“Appropriate controls shall be exercised over computer or related systems to assure that changes in master production and control records or other records are instituted only by authorized personnel. Input to and output from the computer or related system of formulas or other records or data shall be checked for accuracy. The degree and frequency of input/output verification shall be based on the complexity and reliability of the computer or related system. A backup file of data entered into the computer or related system shall be maintained except where certain data...are eliminated by computerization or other automated processes. In such instances a written record of the program shall be maintained along with appropriate validation data. Hard copy or alternative systems, such as duplicates, tapes, or microfilm, designed to assure that backup data are exact and complete and that it is secure from alteration, inadvertent erasures, or loss shall be maintained.”*

- *“Such automated equipment used for performance of operations addressed by §§ 211.101(c) or (d), 211.103, 211.182, or 211.188(b)(11) can satisfy the requirements included in those sections relating to the performance of an operation by one person and checking by another person if such equipment is used in conformity with this section [§ 211.68], and one person checks that the equipment properly performed the operation.”*

The use of filters in outsourcing facilities is governed by 21 CFR 211.72 Filters [21]. Different types of filters are used for different purposes within manufacturing facilities:

- *“Filters for liquid filtration used in the manufacture, processing, or packing of injectable drug products intended for human use shall not release fibers into such products.”*
- When fiber-releasing filters are necessary to be used in the manufacture, they must be followed by *“an additional nonfiber-releasing filter having a maximum nominal pore size rating of 0.2 micron (0.45 micron if the manufacturing conditions so dictate) shall subsequently be used to reduce the content of particles in the injectable drug product.”*
- *“The use of an asbestos-containing filter is prohibited.”*

Additionally, the draft GFI – CGMPs for 503B Facilities [9] states:

- *“For sterile drug products that are filter-sterilized, prefiltration bioburden limits should be established and measured before sterile filtration, unless all components consist of FDA- approved sterile drug products and/or components purchased and certified to be sterile and nonpyrogenic. A sterile pharmaceutical sterilizing-grade filter appropriate for the drug product (e.g., chemically compatible) should be used. The filter must be compliant with § 211.72 and filter integrity testing should be conducted after each filtration or production run.”*

5.4 Insanitary Conditions

Section 501(a)(2)(A) of the FD&C Act [4] discusses insanitary conditions, a key area the FDA focuses on for compounding facilities. If a facility is fully compliant with CGMPs, they should not have findings for insanitary conditions. In the GFI – Insanitary Conditions [33], the FDA gives specific examples of the insanitary conditions of concern for different areas of compounding facilities and laboratories. The listing in the GFI is broken down by product type and area; an abbreviated list is provided here:

Insanitary conditions for sterile and non-sterile drugs:

- Vermin (e.g., insects, rodents) observed in the production area or areas immediately adjacent to production
- Visible microbial contamination, such as mold or bacteria, in the production area
- Foreign matter in the production area, such as rust, hairs, paint chips, etc.
- Handling of highly potent drugs without adequate containment and segregation from other drug products
- Producing drugs while construction is being done in an adjacent area without adequate controls to prevent contamination to the production area or drug product
- Standing water or evidence of water leakage in the production area or adjacent area
- Processing of beta lactams without appropriate controls in place to prevent cross-contamination to other products
- Using active ingredients, inactive ingredients, or processing aides that have or may have higher levels of impurities compared to compendial or pharmaceutical grade equivalents or labeled with “not for pharmaceutical use” or other equivalents, per the GFI – Insanitary Conditions [35]

Insanitary conditions for sterile drugs only:

- Gowning and Aseptic Practices:
 - Engaging in aseptic processing while wearing non-sterile gowning
 - Not changing gowning apparel when reentering from a non-classified area
 - Not changing or disinfecting gloves frequently enough
- Equipment/Facilities:
 - Rooms not properly classified for activities conducted in them
 - Exposing sterile product to non-sterilized or non-depyrogenated materials, supplies, and container-closures
 - Lack of adequate EM including room monitoring
 - Lack of adequate personnel monitoring
 - ISO 5 rooms not certified, or not certified appropriately
 - Facility design or operation with poor flow of personnel or materials that permits the influx of lesser quality air into a higher quality air area. For example, ISO 5 areas that are open to non-classified areas.
 - Lack of monitoring or detection for changes in air quality or room pressure differentials
 - Production areas with unsealed or loose ceiling tiles
 - Production areas or equipment with hard-to-clean surfaces
 - Lack of HEPA-filtered air, or inadequate HEPA-filter coverage or airflow, over the critical area
 - Poorly designed ISO 5 areas that contain overhangs or ledges that could collect dust, etc.
 - Excess equipment stored in ISO 5 areas, particularly particle-generating equipment
 - Sinks, drains, or water sources in the buffer room where the ISO 5 area is located; floor drains in the anteroom (sinks and drains are sources of microbial contamination)
 - Using lyophilizers not sterilized by routine sterilization cycles and not protected from contamination by sterilizing filters on vacuum break air lines/vents
- Sterilization:
 - Using a filter for product sterilization that is not adequate for the job, not pharmaceutical grade, or used in excess of its volume or pressure capability
 - Using a particle-shedding filter in any stage of a sterile drug production
 - Using a filter in drug production whose integrity is compromised; failing to conduct post-use filter integrity testing on filters used to sterilize products
 - Using parameters for sterilization (e.g., temperature, pressure, and time) that are not lethal to the resistant microorganisms

- Cleaning and disinfecting:
 - Use of non-sterile disinfecting agents and cleaning pads/wipes in ISO-classified areas
 - Failure to disinfect equipment and/or supplies at transitions from lower air quality areas to higher air quality areas
 - Improper use of cleaning agents, such as sporicidal agents, including using past expiry and not using disinfectant agents appropriately (e.g., not following the manufacturer's recommended contact time to achieve adequate levels of disinfection) – especially critical in **ISO 5** areas
- Other Insanitary Conditions:
 - Allowing operators with topical or respiratory infections or with open wounds to work in the compounding areas
 - Compounding with components, containers, or materials that have not been verified to assure they do not contribute endotoxins, depending on the product's intended use
 - Compounding under processing conditions that give insufficient assurance that the compounded drug product will meet the endotoxin specification appropriate for its route of administration
 - Failure to conduct media fills that simulate aseptic production at the facility under worst-case, most-challenging, and stressful conditions.

The GFI – Insanitary Conditions [33] discusses preventing insanitary conditions, which should start with identifying risks associated with the facility processes and procedures. Using risk management tools, controls should be implemented to monitor and prevent the identified insanitary conditions from occurring. If the insanitary conditions do occur, monitoring should enable detection before contamination to the compounded product or patient harm occurs. The FDA recommends that facilities consider engaging with third parties with expertise in risk management to help develop appropriate controls.

6 Manufacturing Controls in Compounding

6.1 Introduction

This chapter provides an overview of manufacturing controls that should be evaluated for compounding both bulk and drug products. Controls of RMs (APIs and excipients), CCSs, equipment, documentation, personnel gowning, and aseptic techniques (for sterile bulk and drug products) are necessary for compounding. These requirements include, at a minimum, control systems and procedures for maintaining suitable facilities, equipment, excipients and APIs, documentation of activities, and personnel gowning.

6.2 Documentation

Maintaining records of all activities related to compounding is essential to document that each step is properly conducted to ensure the SQIPP of the drug product produced. Initial receipt and release of materials are done in a different system than the BR, but should be outlined in written procedures at the outsourcing facility and followed. Some sites assign a new lot number to incoming materials, while others use the lot number assigned by the original manufacturer. Regardless of which method is followed, it needs to be stated in written procedures and the lot numbers easily traceable to the manufacturer lot number.

BRs are used to document all activities related to the manufacturing, including weighing/dispensing, mixing, compounding, filtering, filling, capping, sealing, inspection of vials, labeling, and packaging, as discussed in Section 3.8. Because outsourcing facilities compound a variety of different products, BRs can range significantly in size, from a few pages to many, but should be the length necessary to adequately describe and document the steps to be performed. Every BR requires one of the following:

- When BRs require manual recording, the technician/operator doing the step should document the action and a second person witnesses and verifies that it was performed and what was recorded (e.g., a weight from a balance, or a time from a timer)
- In facilities where the BR records values electronically, second person verification is not required (as the system serves as the secondary verification) as long as the electronic system is validated in accordance with 21 CFR 211.68 (c) [21]. For example, if a balance has an electronic printout that is generated when weighing, it can be attached in the BR and serve as the “second person” verification. (Written procedures should include information on how printouts should be attached in the BR, for example, tape or glue, and signing across the edge so it shows if the attachment falls off.)

BRs must be approved and issued by the QU/QA. Each page of the record must be prenumbered, and the record must be time and date-stamped by QU/QA prior to issuance.

Given the variability and complexity of BRs, it is not possible to present all variations. Table 6.1 provides a content overview for a generic BR.

Table 6.1: Batch Record Content Overview

Batch Record Section	Details
Maintenance and cleaning, and sanitizing information and instructions for equipment and processing lines	• Verify and record that the equipment and processing lines are clean and that line clearance from the previous batch was performed • Include date and time completed (and verification of being within CHT) • Cleaning information may be recorded in a logbook • Include instructions for when and how to perform these activities

Table 6.1: Batch Record Content Overview (continued)

Batch Record Section	Details
Bill of materials	- Complete formulation and component list - Include grade of material and part number that is traceable in the facility tracking system - Include weight or amount needed (and units)
Health and safety information	- Include information on the proper handling of materials (e.g., heating, cooling, and storage requirements) - Personal Protective Equipment (PPE) needed during manufacturing process by step (if needed) - Information for each step of the manufacturing process including potential hazards/reactivity
Overview of the manufacturing process	- High-level instructions explaining the entire process - May include information on PPE and cautions
Batch/lot or control numbers for each lot being packaged, labeled, and/or distributed	- If the BR is broken into sections for compounding, filling, etc., sometimes a different lot number is assigned per section. There should be a place to record the starting material (lot) from the previous section (e.g., the filling BR includes the lot number from the compounding BR). - If a finished product lot is used for compounding, that lot number is recorded
Start and end dates of the manufacturing process	- Record the date and time of the start and end of each major step (compounding, filling, etc.), as well as the overall (entire) batch - This shows that the process remained within validated hold times
List of personnel involved in the manufacturing process	- Records must be available at the site showing that each person is properly trained for the activities they are doing - When using a manual BR system, it is good practice to have a page at the beginning of each BR with each person's printed name and signed initials and signature so they can be identified throughout the document.
Lists of equipment and processing lines used	- Equipment should be listed with multiple numbers if more than one is qualified/validated (with specific one used in the batch/lot identified/marked) - Room numbers should be listed; if optional rooms are available/qualified, the room used should be identified/marked
Step-by-step documentation of the entire manufacturing process	- At the start of major process, date and time should be recorded - The finished/executed BR should include dates of completion for each step and initials of personnel who completed step, and secondary checker on critical steps. Where critical/indicated, times should be recorded
BR for Terminal Sterilization (TS) processes should include the approved validated loading diagrams for the autoclave chamber	The load of the chamber is an approved part of validated cycle, so operators need to verify that the correct load was used and sign off as such in the BR, as well as verify all validated settings used for the TS cycle
BR for aseptic processes and TS should include the EM requirements	The EM documentation must be included with the executed BR showing that the environmental conditions remained in compliance throughout the process

Table 6.1: Batch Record Content Overview (continued)

Batch Record Section	Details
BR for aseptic processes should include additional documentation on personnel and other observations	- During aseptic processes, there should be detailed logs of who is in the room at what times, including, at a minimum, which operators/technicians are at what stations, when anyone enters/exits the room, when someone enters/exits for EM, any maintenance (who and what), any excursions and what time they occurred - It is recommended that a “standard aseptic log” form is used documenting each aseptic process on which is recorded the lot number of the batch/lot and all operators involved in the process.
Results of in-process QC tests and status checks, including deviations from the process and any by-products or co-products	- These are often additional documents not part of the issued pages of the original BR and may come from the laboratory - These documents are considered original documentation to be added to the executed BR as part of the BR package
Yield calculations	Actual and expected/theoretical batch yields must be calculated at each major step
Packaging and labeling requirements and documentation	- Management of labels is required to ensure they are handled appropriately and kept under appropriate control (see Section 7.3.2 for additional information) - Labels are required to be held and issued by a group, usually QA. Remaining labels must be returned and all labels accounted for (including torn labels) - A copy of the approved label should be attached to the BR
Definitions of any abbreviations used	- Abbreviations in the document must be defined - Common abbreviations are often listed in an SOP
Complete change log and audit trail	- Anything that happens that does not exactly follow the BR should be detailed so that if aberrant results are found, it can help the investigation - Deviations from the BR should be brought to the attention of the supervisor/pharmacist in charge and Quality as soon as possible
Documentation showing the drug product meets established specifications	This is done by QA with review of the finished BR and testing results

Any deviation must be documented in the record, investigated, and approved by the QU prior to release of the batch.

Before a batch can be released, all testing must be successfully completed (e.g., sterility testing) and the testing results confirmed as passing by the QU/QA. In addition, the QU/QA needs to have reviewed the complete BR package, including at a minimum:

- The original BR issued by the QU/QA executed by production
- All data generated during the batch, such as EM data, aseptic processing logs, laboratory data (in-process and release), etc.
- Any deviations or investigations, including OOS

From this BR package, the QU/QA should create a complete batch/lot release package containing the information used to make the determination to release or not release the batch.

6.3 Compounding Considerations

6.3.1 Cleaning and Cleaning Validation

Cleaning is a critical component in compounding. The standard of clean for a pharmaceutical process room or equipment is more than visual. The FDA requires validation of cleaning processes, methods used to demonstrate that the equipment is clean, cleaning agents, and the length of time a room and equipment can be held as clean before it requires recleaning (cleaning hold time). The FDA expects firms to have written procedures on how cleaning processes will be validated as well as written procedures on the cleaning processes and cleaning agents used for various pieces of equipment. Validation should be done under a protocol with a report documenting the results and effectiveness of the cleaning validation. After cleaning validation is completed, generation of a cleaning schedule, including cleaning agent rotation, is recommended. (FDA Guide to Inspections of Validation of Cleaning Processes [50])

Cleaning validation is achieved by demonstrating that the API and hard-to-clean excipients are removed to an “acceptable level.” The “acceptable level” is usually calculated as the Maximum Allowable Carryover (MACO) and is based on Health Based Exposure Limits (HBEL), which are determined by either the Acceptable Daily Exposure or Permitted Daily Exposure when this data is available.

Cleaning processes can be bucketed, e.g., developing a process for each different API, a process for water-soluble residues, another process for non-water-soluble residues, and a process for cleaning between batches of the same product versus product changeover. During development and validation of cleaning methods, it is important to identify the hardest-to-clean product for use in validation. Residues from the cleaning process (detergents, solvents, etc.) also must be removed by the cleaning process. As additional products are introduced into the compounding facility, cleaning validation should be re-evaluated to ensure that the worst-case cleaning scenarios validated are still pertinent.

Sampling for cleaning can have its own challenges, although there are two types of sampling generally found to be acceptable: (1) direct sampling from the surface of the equipment via swabs or other means or (2) collection of rinse solutions. Additionally, the determination of where to sample is important, as samples need to be representative of the entire cleaning process. Sampling should include the hardest-to-clean locations, and statistics should be used to determine sample size. These cleaning validation and sampling processes should be appropriately documented in written procedures.

Analytical methods are required to be developed and validated by the QC laboratory to measure the results of routine cleaning. This process is detailed in Chapter 8. Developing tests for cleaning validation samples is more challenging than other analytical test methods because the cleaning samples are more diluted and measure at much lower levels of detection.

Results of cleaning processes should be trended similarly to other analytical results, and if deviations, nonconformances, or negative trends occur, the cleaning process should be evaluated to determine if changes need to be made.

Hold times are determined for equipment and form part of the validation of the cleaning procedures. Dirty Hold Time (DHT) is the period the equipment is held (and when appropriate, the equipment storage conditions) before it is cleaned after use. CHT is the time period between when the equipment is cleaned and held under appropriate conditions (to maintain it in the clean state) prior to use for processing. These controls provide confidence that the potential for microbial contamination is limited and that there is insufficient time for potential buildup of contamination. Among other risks, this buildup could cause degradation products that cannot be removed by standard cleaning procedures.

Processing rooms are often given DHTs and CHTs for similar reasons, particularly in sterile manufacturing. This should be evaluated in the facility risk assessment and decisions made based on lowering the overall risk to the facility when determining room DHTs and CHTs.

Information on cleaning processes and validation can be found in FDA Guide to Inspections of Validation of Cleaning Processes [50] and the Active Pharmaceutical Ingredients Committee (APIC) Guidance on Aspects of Cleaning Validation in Active Pharmaceutical Ingredient Plants [51].

6.3.2 Critical Aspects of Compounding

As stated at the beginning of this ISPE Guide, the FDA describes drug compounding as “the process of combining, mixing, or altering ingredients to create a medication tailored to the needs of an individual patient.” [1] Compounding may include combining two or more commercially available drugs already approved by the FDA, or may be more complex wherein APIs and excipients—including water—are weighed, mixed, heated (if necessary), etc., to create the bulk product. The bulk product may be stored in a bulk container prior to manufacture of the final dosage form or filling into a final product container for patient delivery. During compounding, certain adjustments to the bulk product may be required to ensure that it meets specifications with respect to, for example, pH, viscosity, or density. Normal compounding includes mixing and adjusting the components—API, excipients, and water—to the proper specifications and adding a sufficient amount of water or other solvent (such as alcohol) at the end of the process to derive the correct volume of the final bulk product at the intended concentration. In-process samples should be obtained and tested throughout the process to ensure that the final bulk product has the proper quality attributes (§ 211.110 [21]).

6.3.2.1 Water Quality

Water quality is critical depending on where it is used in the process. There is an acceptable grade of water for cleaning, an acceptable grade for non-sterile preparations, and an acceptable grade for sterile/parenteral preparations, as discussed in the General Chapter USP <1231> Water for Pharmaceutical Purposes [19]. Figure 6.1 shows the types of water, their preparation, and uses from USP <1231>. See Table 6.2 for the primary water types used in pharmaceutical manufacturing and testing requirements as outlined in the USP.

Figure 6.1: Water for Pharmaceutical Purposes [19]

Source: USP <1231>. Used with permission.

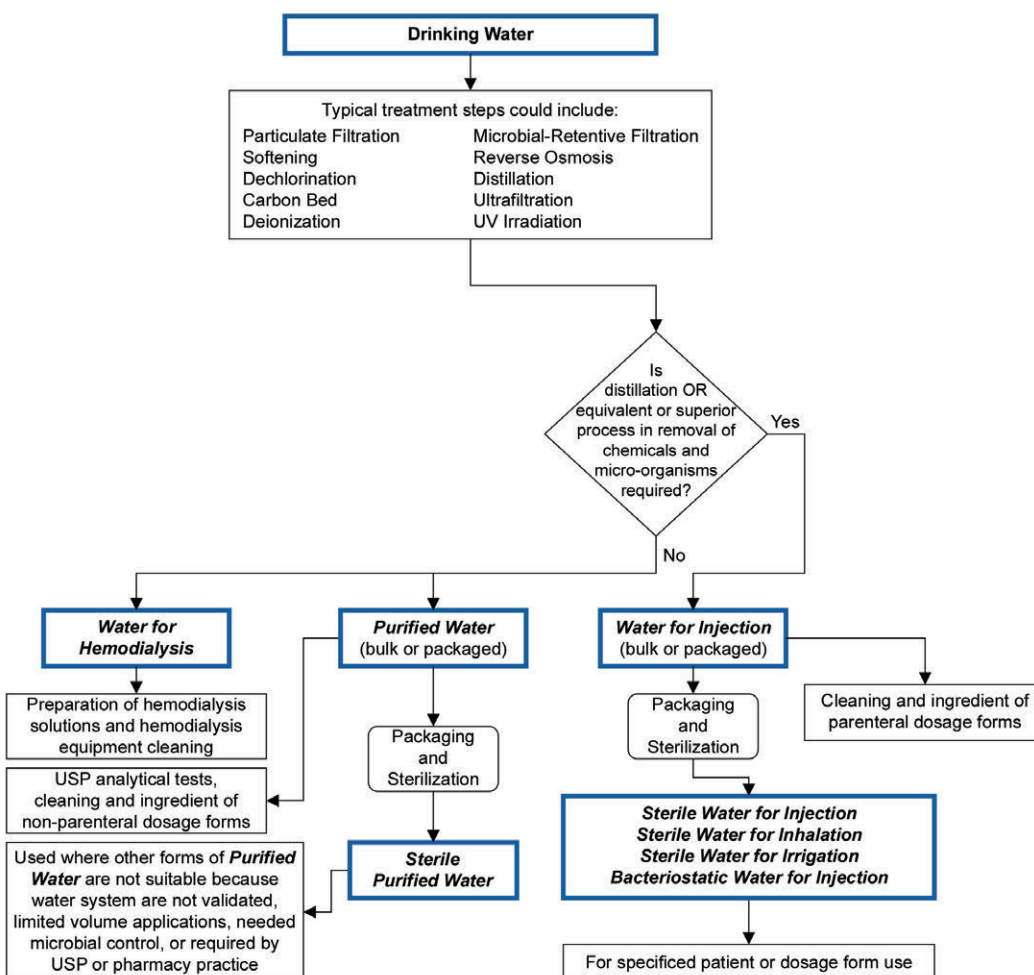


Table 6.2: Water Grades and Testing Requirements from the USP [19]

Water Grade	Testing Requirements
Purified Water	USP <643> Total Organic Carbon USP <645> Water Conductivity, Bulk Water
Water for Hemodialysis	USP <643> Total Organic Carbon USP <645> Water Conductivity, Bulk Water USP <61> Microbial Enumeration Tests USP <62> Tests for Specified Microorganisms USP <85> Bacterial Endotoxins Test
Water for Injection	USP <643> Total Organic Carbon USP <645> Water Conductivity, Bulk Water USP <85> Bacterial Endotoxins Test
Sterile Purified Water	USP <643> Total Organic Carbon USP <645> Water Conductivity, Bulk Water USP <71> Sterility Test
Sterile Water for Injection	USP <643> Total Organic Carbon USP <645> Water Conductivity, Bulk Water USP <788> Particulate Matter in Injections USP <71> Sterility Test USP <85> Bacterial Endotoxins Test

For in-house water generation systems, water sampling should be done on a scheduled basis to show that the water meets the grade requirements. Water systems need to be qualified/validated. Alternatively, different grades of water can be purchased for use in compounding.

FDA has issued a compliance-enforcement inspection guide on High Purity Water System [52]. Although not binding to FDA, it gives insight into their expectations for system design and validation, and provides information on microbial limits for water. According to this inspection guide, there are no pass/fail limits for water, all are action limits. *“When action limits are exceeded the firm must investigate the cause of the problem, take action to correct the problem and assess the impact of the microbial contamination on products manufactured with the water and document the results of their investigation.”* Action limits are listed in the FDA guide for different grades of water.

6.3.2.2 Establishment of Critical Process Parameters

One of the most crucial aspects of compounding (as well as commercial scale manufacturing) is to establish Critical Process Parameters (CPPs) for the operation that are based upon prior knowledge, scientific principles, and quality risk management. Two of the most significant CPPs related to sterile compounding are prefiltration bioburden and bulk holding time.

Prefiltration bioburden measurement involves obtaining a sample from the bulk holding tank and analyzing it for microbial (bioburden) content. Industry best practices establish this value at <100 CFUs (Colony Forming Units) per 100 mL of bulk product, but this should be determined for the specific product. Prefiltration bioburden is equally important for both aseptically filled products and terminally sterilized products since it indicates the microbiological quality of the starting material and helps determine the size sterile filter needed. Aseptic processing and Terminal Sterilization (TS) are discussed in Section 6.5.

Bulk Hold Time (BHT), the second significant CPP, ensures that the compounded product remains stable, both chemically and microbiologically, from the time it is compounded to the time filling is completed. Establishment of BHT involves holding the compounded bulk for a specified time according to required storage conditions, such as temperature (e.g., 48 hours at 20–25°C). Testing for key quality attributes, such as bioburden, assay, and pH, at defined intervals, including the final time point prior to filling should be performed to using established validated analytical methods and compared to established specifications.

In addition to BHT, time limits must also be established for each phase of aseptic processing (§ 211.111 [21]). These phases include the period between the start of bulk product compounding to the start of when the aseptic connections are made, the sterile filtration time period, the filling time period (including time from after filtration to stoppering) as well as for the storage of sterilized equipment, containers, and closures. Time limits should be established for the various production phases and should be supported by robust scientific data and media fills. Bioburden and endotoxin load (for sterile products) should also be assessed when establishing time limits for stages such as the formulation processing stage.

CPPs are essential for all segments of manufacturing and should be established during development or as part of the validation process. CPPs should be developed in alignment with the concepts in ICH Q8 [53], ICH Q9 [54], FDA GFI – Process Validation [44], and FDA Guidance for Industry Sterile Drug Products Produced by Aseptic Processing – Current Good Manufacturing Practice¹⁰ [55] based on the specific compounded product. Example CPPs for manufacturing processes include, but are not limited to:

- Compounding
 - Calculation of active quantity based on potency and water content/Loss On Drying (LOD)
 - Verification of ingredients (dispensing log)
 - Mixing speed during compounding
 - Verification of quantities of ingredients added
 - Verification of complete dissolution
 - Mixing time
 - In-process pH
 - Final weight of bulk after q.s.
 - Prefiltration bioburden In-Process Control (IPC)
 - BHT
- Sterilizing Filter Post-Use Integrity Testing
 - Bubble point after filtration
- Autoclave Sterilization (e.g., for stoppers, utensils, and final product)
 - Temperature (e.g., 121°C ±1°C)
 - Time (e.g., 30 minutes)

¹⁰ Referred to as GFI – Aseptic Processing CGMP throughout this ISPE Guide.

- Pressure (e.g., 15–18 psi (pounds per square inch))
- Filling and Capping
 - Filling speed
 - Fill weight
 - Container-closure check
 - Mixing speed
 - Number of samples for testing
 - Calculation of percent yield
 - Visual inspection
- Testing
 - Correct number of samples, e.g., for endotoxin, sterility, pH, viscosity, and assay
 - Investigation of all microbiological isolates found in ISO 5 areas

6.4 Control of Raw Materials and Container-Closure Systems

6.4.1 Raw Materials

As discussed in other chapters, materials used for compounding include RMs (excipients, sterile drug products, APIs, etc.), containers, and closures. The draft GFI – CGMPs for 503B Facilities [9] states that drug products compounded in an outsourcing facility may qualify for an exemption to label drug products with adequate directions for use (as required under Section 502(f)(1) of the FD&C Act [4]) if these requirements for RMs and components are met:

- The excipient or API complies with the applicable monograph, if one exists, in USP-NF [19] or another compendium or formulary.
- Each lot of API or excipient is accompanied by a valid COA. (The COA may also be used to minimize testing of incoming components (§ 211.84 [21].))
- The bulk API or excipient is manufactured in an FDA-registered establishment.

In addition, each shipment of each lot of RM and components must be tested to verify identity (at a minimum) and evaluated for conformity with appropriate specifications before use in manufacturing. RMs should not be used beyond the supplier's labeled expiration (or retest) date; if there is no expiration date, the supplier should be contacted to provide a date, or a conservative date should be assigned, not to exceed three years after receipt. [9]

At times, the RMs used in compounding are manufactured drug products. In this case, the drug products must have been manufactured in an FDA-registered facility and labeled with a batch control number and expiration date. [9]

Per the draft GFI – CGMPs for 503B Facilities [9], excipients must be tested to verify identity and evaluated for conformity with appropriate specifications and, depending on intended use, tested for endotoxin level and bioburden before use in compounding. As described in 21 CFR 211.84(d)(2) [21], once a supplier has been qualified (see

Chapter 4), the supplier's COA can be evaluated and accepted in lieu of testing each shipment of each RM to determine whether the lot can be used, provided the reliability of the supplier's analyses has been established, if the following is met:

- Confirmation that the ingredient meets the applicable USP or NF monograph [19], if one exists
- The supplier's test results for those tests listed in the specifications show that they conform to the specifications for the RM used for that established compounded drug product
- At minimum, one specific identity test is performed to confirm that the RM is the correct one specified in the purchase order. If additional tests are critical, they may be required as well.

Confirmatory testing and remote audit of the supplier's procedures should also be conducted at least annually for API suppliers and every 2 years for excipient suppliers. As described in the draft GFI – CGMPs for 503B Facilities [9], results from the supplier COA may be used in lieu of testing, depending on whether or not the supplier is verified by the outsourcing facility. See Table 6.3.

Table 6.3: Using Supplier COA Results

When	What the Facility Must Do
If the supplier has not been verified	• If the applicable USP or NF monograph exists, test each lot against the monograph • If a monograph does not exist, test each lot against the product-specific specification
If the supplier has been verified	The following should be performed for all lots: • Confirm the supplier's test results meet the product specifications • Confirm that the component meets the applicable USP or NF monograph, if one exists For each lot and for each shipment of each lot: • Conduct at least one specific identity test before use to confirm that the component is as specified in the purchase order For each shipment of materials: • Verify that the label matches the contents of the shipment for each container • Verify the package integrity of each container or grouping of containers (e.g., whether the box is intact and whether the seal is showing that the package has not been opened) Full testing is typically conducted annually for APIs and every 2 years for other components (excipients and CCS).
Depending on intended use	Components that are not approved finished drug products should be evaluated for requirement to test for endotoxin level and bioburden before use in compounding (21 CFR 211.84 [21])

6.4.2 Container-Closure Systems

As with RMs, appropriate and scientifically sound specifications should be established for the containers and closures to ensure that they are suitable for each drug product for which they will be used. [9] Each lot must conform to its written specifications (§ 211.160(b) [21]). See Table 6.3 for accepting results from the supplier COA depending on whether or not the supplier is verified by the outsourcing facility.

6.5 Sterility Assurance and Aseptic Technique

6.5.1 Introduction

Sterility assurance and aseptic techniques apply to the manufacture of compounded drug products labeled as sterile, although basic sterile manufacturing principles and concepts detailed in CGMPs may also be applied to all 503B drug compounding. These include facility design, EM, personnel monitoring, and finished drug product testing. Unless otherwise stated, the descriptions in this section apply to products labeled as sterile.

In accordance with 21 CFR 211.113 [21]:

“Appropriate written procedures, designed to prevent microbiological contamination of drug products purporting to be sterile, shall be established and followed. Such procedures shall include validation of all aseptic and sterilization processes.”

Sterilization is defined as a “validated process used to render a product free of viable organisms. Note: In a sterilization process, the nature of microbiological death or reduction is described by an exponential function. Therefore, the number of microorganisms which survive a sterilization process can be expressed in terms of probability. While the probability may be reduced to a very low number, they can never be reduced to zero.” (PDA 22 [56])

Depyrogenation is the removal of pyrogens by one of four methods: dry heat, chemical, filtration, or physical removal. The choice of which method to use is determined at least partially by the potential effects of the depyrogenation treatment on the materials themselves. For pharmaceuticals, depyrogenation refers to the destruction or removal of bacterial endotoxins, the most prevalent and quantifiable pyrogen in parenteral preparations, as discussed in USP <1228> Depyrogenation [19].

Depyrogenation procedures are applicable to product streams, equipment, and drug product containers and closures. This section focuses on the CCS. USP <85> Bacterial Endotoxins Test [19] is the primary procedure used to measure bacterial endotoxins. Dry Heat Depyrogenation (USP <1228.1>) and Depyrogenation by Rinsing (USP <1228.4>) are the two most common processes used for CCS. Moist heat sterilization processes are not able to remove pyrogens because they do not reach a temperature high enough to kill pyrogens, whereas dry heat depyrogenation reaches temperatures of 170°C–400°C. The depyrogenation process used must be validated for each container and closure size/material of construction/load. It should be noted that most container-closure components (vials, stoppers, syringes, etc.) can be purchased depyrogenated or Ready-to-Use (RTU); however, the supplier should issue a certificate for each lot certifying that it was depyrogenated and under what conditions.

Sterile products may be manufactured using aseptic processes or TS. Aseptic processing is used when a drug product is not stable to be terminally sterilized as a drug product, often because the API degrades under the necessary heat conditions.

In an aseptic process the drug product, container, and closure all must be sterilized separately before they are put together in a sterile environment. For example, glass containers are subjected to dry heat, rubber closures are subjected to moist heat sterilization, and liquid dosage forms are subjected to sterile filtration. Each of these processes requires validation and control. Because there is no process to sterilize the product in its final container, it is critical that containers be filled and sealed in an extremely clean and high-quality environment. Products that are heat or radiation labile must be manufactured by aseptic processing.

TS involves filling and sealing product containers under appropriate environmental conditions to minimize the microbial and particulate content of the in-process product and to help ensure that the subsequent sterilization process is successful. It is important that the environment is controlled in TS as well, as the TS process is designed and validated to sterilize the product based on an expected incoming microbial load. The product in its final container and closure is then subjected to a sterilization process such as heat or irradiation. (GFI – Aseptic Processing CGMP [55])

Manual or mechanical interventions during the filling processes may be necessary, for example, removal of vials that have fallen over or adjustment of a filling line part. These interventions pose a higher risk of loss of product integrity during aseptic processing due to the inability to resterilize components that are being filled, as opposed to TS in which the final product is sterilized at the end of the process. Therefore, additional measures must be implemented during aseptic processing to ensure the safety and quality of the products.

EM is required in all sterile manufacturing environments, both TS and aseptic; this is discussed further in Section 6.5.2.

Additional references that are helpful when performing Sterile compounding include:

- PDA
 - Technical Report 13, Fundamentals of an Environmental Monitoring Program [57]
 - Technical Report 13-2, Fundamentals of an Environmental Monitoring Program, Annex 1: Environmental Monitoring of Facilities Manufacturing Low Bioburden Products [58]
 - Technical Report, Points to consider for Aseptic Processing- Part 1 [59]
 - Technical Report, Points to consider for Aseptic Processing- Part 2 [60]
- USP <1211> Sterility Assurance [19]

6.5.2 **Environmental Monitoring**

Regardless of the type of sterilization used, controls must be established for the facility design and EM. Room environments are most commonly referred to by ISO standards, which are discussed in Chapter 5. Examples of the types of operations to be carried out in different ISO environments based on type of processing are found in the *ISPE Baseline® Guide: Sterile Product Manufacturing Facilities (Third Edition)*, Table 2.3 [61].

EM ensures that the ISO environment remains in control during setup, washing of components, weighing of RMs, compounding, filtration, and filling through stoppering/closure of the final product. The draft GFI – CGMPs for 503B Facilities [9] discusses some of FDA's recommendations for outsourcing facilities EM. For example, air sampling is expected to be performed under dynamic conditions (see Chapter 5), and in critical areas (ISO 5), nonviable monitoring should be performed during dynamic conditions. A risk assessment should be conducted to determine the appropriate sampling and frequency of sampling for an area. The following items should be considered:

- Viable monitoring
 - Active air sampling
 - Passive air sampling
 - Gown/personnel sampling
 - Swabbing of surfaces
- Nonviable monitoring
 - Performed during static conditions (not manufacturing)
 - Performed during dynamic conditions (manufacturing)

- Activities/unit operations and locations to be monitored (see Table 2 of USP <1116> [19] for examples)
- Intervals for the monitoring

The EM approach needs to be predefined and is used to show that the process was in control. Any excursions or trends in EM need to be documented, investigated, and the impact to the compounded drug product evaluated. The following USP [19] general chapters give additional information on microbiological evaluation of cleanrooms:

- USP <1116> Microbiological Control and Monitoring of Aseptic Processing Environments
- USP <61> Microbial Examination of Nonsterile Products: Microbial Enumeration Tests
- USP <71> Sterility Tests
- USP <1227> Validation of Microbial Recovery from Pharmacopeial Articles

6.5.3 Terminal Sterilization

TS is the process in which the finished compounded drug product (i.e., bulk product, APIs, and RMs that have been compounded, filled into containers, closed with an appropriate closure, and sealed if appropriate) is sterilized (usually) either thermally (by heat and pressure) or by radiation. The CCS (stoppers and vials) must be either purchased as nonpyrogenic (depyrogenated) or depyrogenated before use, as sterilization will not remove pyrogens (see Section 6.5.1).

The amount of sterilization (heat, pressure, and time) to be applied is dependent on the bioburden of the presterilization sample. The bioburden is the microorganisms inherently present in the product due to the environment, RMs, CCS, etc. TS is a balance of using enough heat, pressure, and time (for thermal TS) or radiation to achieve the appropriate bioburden reduction, but not using too much that would cause stability issues in the product (including discoloration or impurity formation).

Although not required, sterilization filtration (normally using a pore size of 0.2 µm) is often performed on TS products to reduce bioburden except where this is not possible, for example when the product is a suspension. If filtration is performed as part of the TS process, it must be shown that the filter is compatible with the material and sized appropriately to handle the microbial load going through it. A post-filtration bubble test must be done to demonstrate that the filter remained intact throughout the process. The filter must be compliant with 21 CFR 211.72 [21] and validated as part of the process. Often two filters are used in sequence in sterile filtration in case of a breakthrough; sometimes the first filter is of a larger pore size (e.g., 0.45 µm) to catch larger material that might clog the sterile filter and slow down filtration; however, often the same pore size filter is used for both.

For drug products that are terminally sterilized, at least a 10⁻⁶ Sterility Assurance Level (SAL) should be demonstrated in validation studies during process development using an appropriate sterilization load monitor, such as Biological Indicators (BIs) and Thermocouples (TCs). [9] This means that the TS cycle should show a reduction in bioburden of greater than 10⁶. Most TS cycles are designed as an overkill cycle, i.e., the standard bioburden for that product is determined by testing and the cycle is designed to exceed the treatment required to achieve greater than 10⁶ reduction, thus accounting for variances in the presterilization microorganism bioburden. For terminally sterilized drug products not subjected to an overkill TS cycle, presterilization bioburden limits should be established and measured before sterilization. The selected sterilization method should both sterilize and maintain the strength, purity, quality, and package integrity of the sterile product (GFI – 503B Facility Definition [25]).

Validation studies must be performed for each load size (container closure, number of vials/bags/syringes/containers, and placement in the chamber) intended for sterilization. [9] This is necessary because all chambers have hot and cold spots, and it must be demonstrated that all the locations achieve adequate bioburden reduction while maintaining product stability. A summary of TS from compounding through sealing, including example room classification for each step, is presented in Table 6.4. This table should be only used for general engineering guidance; it is not intended to be used as a CGMP requirement. A formal risk assessment process should be used to determine and substantiate the requirement for the required cleanliness levels. Additional information can be found in the *ISPE Baseline Guide: Sterile Product Manufacturing Facilities (Third Edition)* [61].

Table 6.4: Example Room Classifications for Terminal Sterilization Processing Steps

Adapted from ISPE Baseline Guide: Sterile Product Manufacturing Facilities (Third Edition) [61]

Process Step	Room Environment	Product/Container/Closure Exposure
Dispensing of RMs (API and excipients)	ISO 8	LP/GAAS ¹
Loading and unloading – sterilization/depyrogenation of components ²	ISO 8	LP/GAAS ¹
Compounding	ISO 8	LP/GAAS ¹
Sterile filtration	ISO 8	LP/GAAS ¹
Filling and stoppering of containers	ISO 8	LP/GAAS ¹
Terminal sterilization	CNC ³ /Operational Grade D	N/A
Inspection	CNC/Operational Grade D	N/A
1. LP = Local Protection; GAAS = Grade A Air Supply – the terms are considered as equivalent and are used to indicate any instance where a localized HEPA filtered air supply is used to reduce the risk of total or viable particulate contamination of critical surfaces within a specified working zone. 2. Required unless components are purchased RTU (ready-to-use), which means they have been sterilized and depyrogenated by a validated process. Documentation should be provided with them. 3. Controlled, Not Classified		

6.5.4 Types of Aseptic Processing Primarily Used in Compounding

There are many types of aseptic processing. The primary type of aseptic processing used in compounding is where components are compounded, filter sterilized (typically through one or two presterilized 0.2 µm filters), filled into depyrogenated and presterilized vials or containers, capped with depyrogenated and presterilized stoppers or coverings, and sealed with aluminum seals. When utilizing filtration as a final sterilization step, validation of the filter sterilization processes must be performed.

Since the product cannot be sterilized at the end of the process (as in TS), a higher classification manufacturing environment is required for aseptic processing than for TS products. In general, the better the air quality in the manufacturing suites, the less opportunity and risk for microbiological and particulate ingress into the final product. To minimize risk, compounding operations may occur in an ISO 7 (Grade C) area and continue by transferring the bulk product through supply lines to the sterilizing filter(s). As discussed in Section 6.5.3, two filters are often used in sequence (known as redundant filters) so that if one has a breakthrough, there is still adequate filtration for the product to meet requirements to be sterile.

For products that rely on sterile filtration for bioburden reduction (TS and/or aseptic), prefiltration bioburden limits should be established and measured before sterile filtration. Filter physicochemical compatibility must be shown for the sterilizing-grade filter and filter integrity testing (usually a bubble point test) should be done after each filtration or production run to show that the filter was functioning appropriately throughout the sterile filtration process.

The remaining processes – filtering of the bulk, filling, and capping – must be conducted in appropriate ISO-classified areas. At this step, the drug product should be protected from environmental ingress and the next step of the process, sealing, may be conducted in an ISO 7 or ISO 8 area, depending on whether the aseptic process is in a closed (e.g., isolator) or open aseptic system. Table 6.5 provides a summary of aseptic processing, from compounding through sealing, including the recommended room classification for each step, as in the *ISPE Baseline Guide: Sterile Product Manufacturing Facilities (Third Edition)* [61]. This table should be only used for general engineering guidance; it is not intended to be used as a CGMP requirement. A formal risk assessment process should be used to determine and substantiate the requirement for the required cleanliness levels.

Table 6.5: Example Room Classification for Aseptic Processing Steps

Adapted from ISPE Baseline Guide: Sterile Product Manufacturing Facilities (Third Edition) [61]

Process Step	Room Environment	Product/Container/Closure Exposure
Dispensing of non-sterile RMs (API and excipients)	ISO 8	LP/GAAS ¹
Sterile raw material dispensing	Open – ISO 7 Closed – ISO 8	ISO 5
Depyrogenation of components ¹	Loading – ISO 8 Unloading – ISO 7	ISO 5 (within isolator)
Non-sterile compounding and sterile filtration	ISO 8	LP/GAAS ¹
Aseptic compounding and formulation of sterile materials not later filtered or sterilized (open)	ISO 7	ISO 5
Aseptic compounding and formulation of sterile materials later filtered, sterilized, or isolated (closed)	ISO 8	Sterile vessel/lines, post-sterilized-in-place/sterile single-use components, LP/GAAS ¹
Sterile filtration (open)	ISO 7	ISO 5 (Assemblies, aseptic connections)
Sterile filtration (closed)	ISO 8	Sterile vessel/lines, post-sterilized-in-place/sterile single-use components, aseptic connectors
Filling and stoppering	Open Aseptic Processing – ISO 7 Closed Aseptic Processing Equipment – ISO 8	Open Aseptic Processing – ISO 5 Closed Aseptic Processing equipment – ISO 5 (within isolator)
Transfer into/out of lyophilizer	Open Aseptic Processing – ISO 7 Closed Aseptic Processing Equipment – ISO 8	Open Aseptic Processing – ISO 5 Closed Aseptic Processing equipment – ISO 5 (within isolator)
Inspection	CNC ² /Optional Grade D	N/A
1. LP = Local Protection; GAAS = Grade A Air Supply – the terms are considered as equivalent and are used to indicate any instance where a localized HEPA filtered air supply is used to reduce the risk of total or viable particulate contamination of critical surfaces within a specified working zone 2. Controlled, Not Classified		

6.5.5 Aseptic Technique and Media Fills

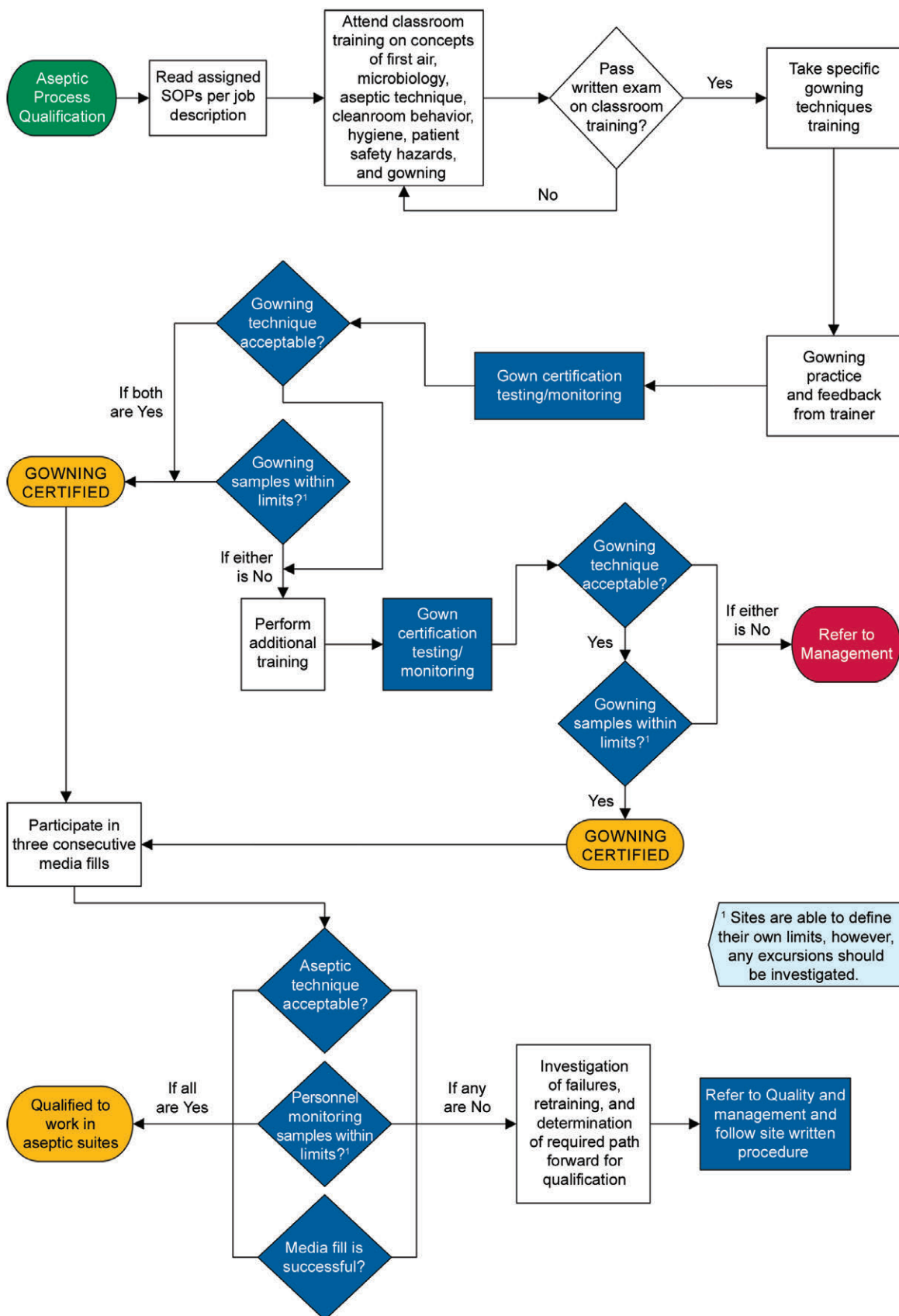
6.5.5.1 Personnel Training – Aseptic Technique

Before an individual is permitted to enter the aseptic manufacturing area or conduct operations in a Laminar Air Flow Hood (LAFH), introductory training on the concepts of first air, microbiology, aseptic technique, cleanroom behavior, hygiene, gowning, patient safety hazards posed by a non-sterile drug product, and procedures covering aseptic manufacturing area operations must be conducted (see 21 CFR 211.25(a) [21]). A process flow showing a typical methodology for aseptic process training is shown in Figure 6.2.

Once introductory training is completed, further training based on department-specific requirements and individual job descriptions should be conducted (personnel gowning techniques are discussed in Section 6.6). After initial training, ongoing training programs should be part of the training curriculum for personnel who work in the sterile processing area. The components of a training program for personnel working in a sterile processing area should include:

1. Personnel to complete all prerequisite training assignments (SOPs, policies, procedures).
2. Personnel to complete classroom training sessions, as discussed in the previous paragraph, followed by a written exam to verify the trainee's knowledge on the concepts presented during the classroom session.
3. Trainer to demonstrate gowning techniques for each personnel to be certified and discuss the risk to the product from personnel not following proper aseptic technique, e.g., interruption of first air and HEPA-filter air flow, walking or moving too quickly, and being improperly gowned.
4. Following the demonstration, personnel to practice gowning once at a minimum or additional times as needed to feel confident on the proper steps in the aseptic gowning process. The additional number of times for practicing is determined by the trainer and the personnel being certified; however, the minimum practice criteria is one time.
5. Following the practice session(s), personnel dons the gown per current gowning procedure and is monitored for microbial enumeration three consecutive times over a minimum of two calendar days. Samples should be obtained from each individual at the end of each training session. Examples of common sample sites on gowns are:
 - Top of the hood
 - Left Glove (fingertips)
 - Right Glove (fingertips)
 - Left Forearm (top of the glove)
 - Right Forearm (top of the glove)
 - Chest under chin
 - Waist (optional)
6. Trainer to observe and document the trainee during the assessment.

Figure 6.2: An Example of Initial Aseptic Certification Process for Personnel



Individuals are considered qualified to conduct aseptic operations after passing at least three successful, successive media fill simulations based on a scientifically sound protocol designed to verify the adequacy of their technique and behavior. Production simulations should be conducted in the same area and on the same equipment where production occurs. As stated in Figure 6.2, supervisors should routinely evaluate each individual's conformance to written procedures during actual operation and provide coaching on any areas of improvement to ensure that best practices are maintained; this should be documented as part of the individual's continued ability to work in the aseptic processing area. Procedures should be in place to state what is considered a nonconformance that is not easily correctable and require removal from the aseptic processing area before retraining is done, and how many times retraining can be done before an operator should be considered for transfer to another area, as the risk to product quality is too great.

Essential aseptic techniques include:

- Personnel, material, and process flow should be optimized to prevent unnecessary activities that increase the potential for introducing contaminants to exposed product, containers or closures, or the surrounding environment.
- After initial gowning, sterile gloves should be regularly sanitized (e.g., using sterile 70% isopropyl alcohol) during production or, when needed, changed. Within the ISO 5 area, operators should not engage in activities that pose an unreasonable contamination risk to the gown.
- Torn, contaminated, or defective gowns or gloves should be changed immediately. Gowning procedures should ensure that there is no exposed skin on personnel involved in any production activities in the ISO 5 areas.
- Sterile products, product-contact surfaces of containers or closures or other critical surfaces should not directly touch any part of the gown or gloves.
- Sterile materials should be contacted only with sterile instruments (e.g., sterilized tongs) and never with hands. Sterile instruments should be held under ISO 5 conditions and changed out if they (even potentially) have become contaminated.
- Personnel should move slowly and deliberately within the cleanroom or LAFH and should keep their bodies and objects out of the path of unidirectional flow above open containers and products being filled.
- Necessary manipulations (addition of stoppers, etc.) should be done in a manner that does not compromise the sterility of the product. This is why routine and non-routine interventions are done in media fills, and the technique to be used documented in an SOP. The same technique used in the media fill should be used during production, as that is what has been shown to maintain an aseptic environment.

6.5.5.2 Media Fills/Process Simulations

Validation of an aseptic process is done by conducting media fills, which are process simulations that mimic the production process using a microbial growth medium in place of the product. The process simulation/media fill includes exposing the microbial growth medium to the entire process and product-contact surfaces of the equipment, CCS, environment, and handling to mimic the normal process. It should incorporate, as appropriate, worst-case activities and conditions challenging to the integrity of the aseptic operations to simulate exposure that could occur during processing. Media fills are also used to qualify personnel to work in the aseptic environment, as discussed in Section 6.5.5.1.

Details of how to conduct media fills are outlined in the FDA GFI – Aseptic Processing CGMP, Section IX [55] and in the draft GFI – CGMPs for 503B Facilities [9].

As described in the GFI – Aseptic Processing CGMP [55], there are multiple steps to design and execute a media fill. These are discussed here briefly.

1. **Study Design** – A media fill program is designed to incorporate risk factors specific to the facility, product, and materials, so that it can accurately assess the state of control of the process. Media fill runs use a BR, with documentation justifying the conditions and activities to be simulated, including the number of times something should be simulated.

A media fill is used to simulate practices that are expected to occur during normal runs, such as those listed below. There are both routine interventions (occur on every run) and non-routine interventions (do not necessarily occur on every run, but likely to occur at some time). These interventions set up the allowances for the production runs. The technique used for these interventions should be included in an SOP so that operators are properly trained on the procedure and all perform it in the same manner, both during media fills and during production.

Routine interventions:

- Equipment setup
- Manufacturing shift change(s)
- Loading compounding vessel
- Line speed adjustment (minimum and maximum)
- Filling the stopper bowl
- Filling the seal bowl
- Removing or standing back up a fallen container
- Removing a broken container
- Technician rotation
- Technician break – minimum 30 minutes
- Spill cleanup and glove change
- Adjustment of filling tubing set
- Missing stopper/seal/closure on container/vial
- Stopper jam in the stopper rail
- End of filling
- Maximum run time

Non-routine Interventions:

- Maximum number of technicians in cleanroom
- Maximum time a technician can be in the cleanroom
- Change of the filling line tubing set

- Change of filling needle
 - Mechanical intervention 1 (minimum 30 min)
 - Mechanical intervention 2
 - Power Failure – minimum 2 minutes
2. **Frequency and Number of Runs** – When initially qualifying a processing line, individual media fills *“should be repeated enough times to ensure that results are consistent and meaningful,”* (GFI – Aseptic Processing CGMP [55]), with at least three consecutive separate successful runs. Semiannually, media fill qualifications should be run for each processing line to evaluate the state of control of the aseptic processing at the facility. The activities and interventions to be done semiannually should be representative of each shift and shift changeover.
- The draft GFI – CGMPs for 503B Facilities [9] states that *“the media fill programs should address applicable issues such as”...“factors associated with the longest permitted run of the aseptic process operation that can pose contamination risk (e.g., operator fatigue, quality of processing environment).”*
 - Each change to a process line should be evaluated using a written change control system and should include a determination whether a media fill is required to assess the ability of the aseptic process to exclude contamination from the sterilized product with this change.
 - If there is data from a media fill showing that the process may not be in control, an investigation should be conducted to determine the origin of the problem. CAPAs should be put in place (see Section 3.7) and additional media fill(s) run to confirm that the preventive actions are working, and the process has returned to a state of aseptic control. If no conclusion can be found for the origin of the failures, it is recommended that after all investigative work is exhausted, a thorough cleaning is conducted followed by three consecutive, successful media fills, with increased scrutiny to show due diligence to future investigators.
3. **Duration of Runs** – The length of the aseptic media fills should be a major consideration when designing the studies. The most accurate simulations would be a full batch size and duration, however, other models can be justified. The time of the media fill needs to be sufficient to allow the manipulations and interventions specified in the media fill BR, in order to demonstrate those activities can be done while maintaining an aseptic environment, as well as taking into consideration the length of the actual manufacturing process/production being simulated. For lyophilization operations, FDA recommends that unsealed containers be exposed to partial evacuation of the chamber to simulate the process, but the freezing portion of the process should not be done, as this potentially inhibits the growth of microorganisms.
4. **Size of Run** – In accordance with the GFI – Aseptic Processing CGMP [55], the size of the run should be adequate to mimic production conditions and accurately assess the potential for contamination of compounded drug product batches. When production is a more manual process design, FDA expects to see a larger number of units similar to actual production size. However, if the process is done primarily in an isolator, justification can be made that direct human intervention is minimal, so a lower number of units can be run to simulate the overall process. If a production batch is produced over multiple shifts, the media fill should simulate this and be designed to address this additional risk. See the GFI – Aseptic Processing CGMP for additional information on run size.
5. **Line Speed** – The media fill needs to address the range of line speeds used in production. The media fill should be run at low and high speeds to simulate different worst-case situations, as discussed in the GFI – Aseptic Processing CGMP [55].
6. **Environmental Conditions** – Media fills need to be conducted in environments representative of production environments. They should neither be excessively cleaned, nor challenged to have poor airflow for a worst-case environment.

7. **Media** – Soybean casein digest medium is a commonly used growth medium. There are situations where an anaerobic growth media or an alternate growth medium may be a better choice for the media fills at a facility. This should be determined by an experienced microbiologist familiar with the facility. Growth promotion testing should also be done to show that the media can support growth of representative microorganisms.
8. **Incubation and Examination of Media-Filled Units** – “Media units should be incubated under conditions adequate to detect microorganisms that might otherwise be difficult to culture.” (GFI – Aseptic Processing CGMP [55]): Incubation conditions should be established by experienced personnel in accordance with the media type, and also with the following general guidelines [55]:
 - “Incubation temperature should be suitable for recovery of bioburden and environmental isolates and should at no time be outside the range of 20-35°C. Incubation temperature should be maintained within + 2.5°C of the target temperature.”
 - “Incubation time should not be less than 14 days. If two temperatures are used for the incubation of the media filled units, the units should be incubated for at least 7 days at each temperature (starting with the lower temperature).”

All of the filled containers during a media fill are then incubated to detect microbial contamination (microbial growth). Unlike when conducting aseptic processing, no container (vial/syringe/bag, etc.) should be excluded from incubation/evaluation unless the defect is related to integrity of the container. All excluded containers need justification, for example, they are cracked/broken/leaking. In a media fill, low fill is not an acceptable reason to exclude a container from growth promotion testing as it could still show issues with aseptic technique. Results are interpreted to determine the likelihood of a compounded drug product becoming contaminated during aseptic processing (start-up, sterile ingredient addition(s), aseptic connections, filling, closing).

After incubation, the compounding containers are evaluated for turbidity by personnel with appropriate education, training, and experience in inspecting media fill units for microbiological contamination; this should be done by QC, or under QC oversight. If any units are found to be turbid, these are brought to the attention of the microbiologist and post-incubation growth promotion testing should be done to determine the type of microbial contamination. If containers cannot be examined visually because they are not clear (amber or opaque, for example), then other techniques can be used. It is good practice to do a formal growth promotion on “good” units to show the media is capable of supporting growth during/after incubation; this demonstrates that there are no false negative results.

EM data should be generated during a media fill and evaluated along with results of the microbial, and if needed, growth promotion testing to determine if a media fill passes or fails.

Successful completion of media fills with appropriate interventions demonstrates the reliability of the aseptic process to manufacture sterile products. Ideally, the media fill process simulation determines the maximum time the process can be run, the operators who are qualified to run the process, how many routine interventions can be done without investigation, and which non-routine interventions are qualified to be done without investigation.

The number of compounded media vials/containers used during the media fill should be representative of the number of compounded media vials/containers produced during normal production. Table 6.6 shows the acceptance criteria per the GFI – Aseptic Processing CGMP [55]:

Table 6.6: Acceptance Criteria for Media Fills

Number of Filled Units	Number of Turbid Units	Action	Recommendation
< 5,000	1	Investigation, Revalidation	Conduct three consecutive runs
5,000–10,000	1	Investigation (including considering a repeat media fill)	Repeat one media fill
	2	Investigation, Revalidation	Conduct three consecutive runs
> 10,000	1	Investigation	Best practice is to repeat one media fill
	2	Investigation, Revalidation	Conduct three consecutive runs

6.6 Gowning and Personnel Monitoring

Operators are the primary source of microbiological and nonviable particulate ingress, and comprehensive practices and procedures must be in place to ensure that the risk they pose is minimized. Risk may first be minimized by creating a strong “good aseptic technique” training program in which individuals are educated on the rationale and taught and evaluated by written and practical exercises as discussed in Section 6.5.5.

In accordance with 21 CFR 211.28(a) [21], appropriate clothing should be worn for the duties being performed. Procedures should be in place for each facility that govern the requirements for the facility. Some examples of industry best practice are:

- Street clothes (specifically external clothing, jewelry) are removed and replaced with company provided scrubs/uniforms
- Personal shoes are removed and replaced with plant-dedicated (i.e., does not leave the facility), cleanroom-designated shoes prior to donning sterile gowning for entry into the cleanroom.
- No street clothing or shoes allowed into the cleanroom to minimize opportunities for microbial organism ingress.

Once personnel have been trained, aseptic techniques should be reinforced daily through surveillance of activities and adherence to the techniques learned during the training process. Maintenance of aseptic practices must be a continual coordinated effort overseen by supervisors and the QU; it is the most basic and critical activity in the manufacturing process of aseptically produced products.

Aside from training, two additional critical practices are mandatory to ensure that personnel do not contribute to microbiological and nonviable particulate ingress into the process: gowning and personnel monitoring.

6.6.1 Gowning (Gowns/Garments)

Garments provide a barrier from particles and microorganisms by offering a protective layer of gowning components to protect the exposed sterilized materials from particles generated by, and microorganisms shed from, the body. In alignment with GFI – Aseptic Processing CGMP [55], FDA recommends gowns that are non-shedding and cover the skin and hair of the individual (e.g., face masks, hoods, beard/mustache covers, protective goggles, and elastic gloves).

In order to not contaminate the garments while putting them on (donning them), a gowning qualification program must be established to demonstrate the capability of an individual to don the complete set of sterile garments in an aseptic manner for ISO 5 areas. Gowning requirements for lower classification areas (e.g., ISO 7 or 8) are not as stringent and may only require a hair net, face mask, outer gown, booties, and gloves (as specified by the site SOP). The garments should be non-shedding. Non-sterile garments are often used in these areas, but the risk of extraneous ingress is naturally reduced by using sterile garments.

Written procedures should detail the methods used to don each gown component in an aseptic manner. Personnel should be regularly requalified or audited to ensure their conformance to aseptic manufacturing requirements. The assessment should include *“microbiological surface sampling of several locations on a gown (e.g., glove fingers, facemask, forearm, chest). Sampling sites should be justified.”* [55] Annual requalification ensures continued compliance with aseptic gowning techniques.

Sterile gloves should be regularly sanitized or changed, as appropriate, to minimize the risk of contamination. Personnel should not directly contact sterile products, containers, closures, or critical surfaces with any part of their gown or gloves. [55]

Mastering the gowning procedures and completing certification enables technicians to advance to the next stage of aseptic training, which is the media fill as discussed in Section 6.5.5.

6.6.2 Personnel Monitoring

Personnel pose the greatest risk to the sterile processing process; therefore, a comprehensive personnel monitoring program should be established and implemented. Monitoring can be accomplished by obtaining surface samples of the operators. Sampling locations and frequency should be determined and justified in writing. The QA unit should establish a more comprehensive monitoring program for operators involved in operations that are particularly labor intensive (i.e., those requiring repeated or complex aseptic manipulations). Additionally, personnel monitoring should always be done on the operators involved in a non-routine excursion.

Samples should be obtained from an individual every time the person exits the aseptic area during routine manufacturing or media fill operations in accordance with the site SOP. The following are examples of common sample sites on gowns, but location and frequency of sampling should be determined by the facility based on risk and documented:

- Top of the hood
- Left Glove (fingertips)
- Right Glove (fingertips)
- Left Forearm (top of the glove)
- Right Forearm (top of the glove)
- Chest under chin
- Waist (optional)

Each individual is resampled upon exiting the area and must don a new set of gloves prior to reentering the area. Care should be taken that gloves are not sanitized prior to conducting personnel sampling.

Sampling should be done by a qualified individual. Samples are most often taken using a microbiological media (RODAC¹¹) contact plate with an appropriate media as determined by the microbiologist. After sampling is completed, the lid of the RODAC should be replaced and the bottom of each plate (samples are read by looking at the top) labeled with:

- Sample date
- Sampled person's initials

¹¹ RODAC = Replicate Organism Detection and Counting

- Product name
- Product lot number (if applicable)
- Time of sampling

The plates should be incubated at the temperature specified by the site procedure and any growth should be confirmed by testing with known organisms, ideally organisms isolated from the facility.

USP <1116> Microbiological Control and Monitoring of Aseptic Processing Environments [19] discusses sampling techniques and how to determine appropriate alert and action limits for personnel monitoring.

Interpretation of personnel monitoring results:

1. Alert and action-level excursions during initial gown certification and recertification are used to assess the person's aseptic technique and their eligibility to be certified or recertified for aseptic gowning. An investigation is not required for initial gown certification and recertification samples obtained when product is not being produced as the potential for product impact is minimal.
2. An investigation is necessary if any of the following microorganisms are recovered:
 - Water borne organisms (e.g., *Pseudomonas*)
 - Enteric microorganisms (e.g., *E. coli*)
 - *Staphylococcus aureus*
 - *Bacillus* sp.
 - Yeast (e.g., *Candida* sp.) or mold (e.g., *Aspergillus* sp., *Cladosporium* sp.)
3. Action-level excursions on the hood, chest, gloves, or forearm sites require an impact assessment of any lots in which the employee had participated in during the day of the excursion. The impact assessment is performed as part of the investigation.
4. Any employee with an adverse trend and/or atypical garment recoveries is subject to decertification.
5. Document recertification of all personnel

7 Product Storage, Packaging, and Labeling

7.1 Introduction

This chapter provides an overview of storage, packaging, and labeling considerations that should be evaluated for the drug product, the RMs that make up the compounded drug product (including API, excipients, and processing aides), and the CCS.

The draft GFI – CGMPs for 503B Facilities [9] and USP-NF [19] provide general and specific requirements for the materials used to package compounded drug products.

CGMPs require 503B compounding facilities to have a QU with oversight to approve or reject all components, drug product containers, closures, in-process materials, packaging materials, labeling, and drug products (§ 211.22 [21]). These requirements apply to all topics in this chapter. The QU is also required to issue labels and oversee that the appropriate storage conditions are maintained, among other activities covered in other chapters. See Section 3.2 of this ISPE Guide.

7.2 Storage

It is important that both the RMs and CCS are appropriately stored prior to use in order to preserve the SQIPP and safety of the compounded drug product. The protection of operators needs to be considered, as does the integrity (protection and maintenance of SQIPP) of the materials that go into the drug product. RMs and packaging materials should be stored according to the manufacturer's specifications. The requirements for storage of excipients and API include, at minimum, controlled temperature and humidity, and product isolation and segregation (discussed in Section 4.3.1).

The FDA expects appropriate storage conditions to be maintained for all RMs and CCS. Temperature and/or humidity conditions specific to a product must be clearly documented in the BRs, analytical methods, and other controlled documentation so that the instructions related to these conditions are followed by those handling the material and executing the batches (§ 211.80 [21]). As these are considered critical to product quality, the appropriate personnel must sign off on the steps showing that these conditions are followed.

7.2.1 Temperature and Humidity

USP <659> [19] defines storage conditions for APIs and excipients. The RM storage and shipping requirements stated within specific monographs for each component must be followed. Storage and shipping requirements specified in the USP apply except where the label or its Prescribing Information (PI) specify differently. Where no specific storage, shipping, and distribution instructions or limitations are provided in the product's labeling or PI, products should be protected from moisture (humidity), freezing, excessive heat, and, where applicable, light.

Storage spaces should be labeled with the defined conditions including the temperature and humidity controls. Temperature mapping and validation is an FDA expectation and are likely to be reviewed during regulatory inspections. Logbooks, chart recorders, validated electronic systems, and other appropriate methods for EM are used for documenting environmental conditions. When excursions from specified limits are observed, a deviation is initiated. The deviation procedure should be documented and should explain how excursions are defined (time frame), tracked (alarm systems, who is notified, etc.), and investigated. There should be an assessment conducted for any excursions to determine impact to product SQIPP. See Section 3.6 for handling deviations and investigations.

Table 7.1 lists the conditions as defined in USP <659> [19].

Table 7.1: Defined Conditions [19]

Source: USP <659>. Used with permission.

Defined Condition	Control	Excursions/Exceptions
Freezer	Temperature controlled between –25° and –10°C (–13° and 14°F)	In some instances, articles may have a recommended storage condition below –20°C (–4°F). In such cases, the temperature of the storage location should be controlled to ±10°C (14°F) of the recommended storage condition.
Refrigerator	Temperature controlled between 2° and 8°C (36° and 46°F)	N/A
Cold	Any temperature not exceeding 8°C (46°F)	N/A
Cool	Any temperature between 8° and 15°C (46° and 59°F)	[Note: An article for which storage in a cool place is directed may, alternatively, be stored and shipped as refrigerated, unless otherwise specified by the individual monograph.]
Controlled Cold Temperature	Temperature maintained thermostatically between 2° and 8°C (36° and 46°F)	Temperature excursions between 2° and 15°C (36° and 59°F) that may be experienced during storage, shipping, and distribution, but not to exceed 24 hours, such that the allowable calculated mean kinetic temperature (MKT) is NMT 8°C (46°F) with no excursions below 2°C (36°F) or above 15°C (59°F). These limits (time and temperature) and the calculated MKT must be documented (see <i>Mean Kinetic Temperature in the Evaluation of Temperature Excursions During Storage and Transportation of Drug</i> <1079.2>)
Room Temperature (also referred to as ambient temperature)	The temperature prevailing in a working environment	N/A
Controlled Room Temperature (CRT)	Temperature maintained thermostatically that encompasses the usual and customary working environment of 20°–25°C (68°–77°F)	MKT can be used during an excursion provided: • MKT does not exceed 25°C (77°F). • Excursions between 15°C and 30°C (59°F and 86°F). • Transient excursions are NMT 40°C (104°F). • Excursion time is NMT 24 hours.
Warm	Any temperature between 30° and 40°C (86° and 104°F)	
Excessive Heat	Any temperature above 40°C (104°F)	
Dry Place	A place that does not exceed 40% average relative humidity at 20°C (68°F) or the equivalent water vapor pressure at other temperatures	The determination may be made by direct measurement at the place. Determination is based on NLT 12 equally spaced measurements that encompass either a season, a year, or, where recorded data demonstrate, the storage period of the article. There may be values of up to 45% relative humidity provided that the average value does not exceed 40% relative humidity. Storage in a <i>Container</i> validated to protect the article from moisture vapor, including storage in bulk, is considered a <i>Dry Place</i> .

21 CFR 211.142 [21] discusses warehousing procedures for drug products. Written procedures must be in place and followed that define how the drug product will be warehoused, including use of an off-site warehouse for storage. The procedures should include information on the storage conditions and controls specific to temperature, humidity, and light so that the SQIPP of the drug product are maintained.

Drug product storage is often required to be in a tighter range than RM or CCS. Storage conditions for the drug product, defined in the monograph, product label, or specifications, need to be appropriately maintained during warehousing and shipping. Storage conditions are determined by stability data. For details on stability, refer to Chapter 9.

7.2.2 Segregation/Isolation Requirements

The RM and drug product storage areas are expected to be clean and have appropriate environmental controls based on the requirements stated in the USP-NF monographs [19] or on product labeling (as discussed in Section 7.2.1). Segregation requirements for materials used in compounding should be established based on CGMP requirements. Materials should be separated by status (received, sampled, quarantined, rejected, released) and from each other within a status area. Also, materials should be isolated based on specific requirements established for each material, ensuring there is adequate spatial segregation to prevent cross-contamination and mix-ups during processing.

Additionally, requirements need to be in place for the storage, handling, and sampling of potent compounds or DEA-controlled substances. These aspects are especially important when considering personnel safety and legal requirements. References include:

- Pharmacist's Manual, Drug Enforcement Agency [63]
- Chemical Handler's Manual, Drug Enforcement Agency [63]
- Controlled Substances Act, Title 21 United States Code, Chapter 13 [64]

Specific requirements for facilities, equipment, process, and sterile processes are detailed within this ISPE Guide in Chapters 3, 5, and 6.

7.3 Packaging, Labeling, Storage, and Transport

The primary packaging materials (CCS) chosen for the drug product are critical, as they protect the product from the environment, help maintain chemical stability, and contain and/or deliver the drug product to the patient. It must be shown that the CCS chosen is suitable for the intended use.

Labeling is also critical to the drug product, as it identifies the product and safety information for the patient. Packaging and labeling operations need to be recorded on BRs and overseen by the QU. See Section 3.2.1.

The FDA expects that packaging and labeling requirements are documented in the BRs, procedures, and other controlled documentation so that the instructions regarding these requirements are followed during batch production. (§ 211.80 [21]). As these are considered critical to product quality, the appropriate personnel must sign off on the steps showing that these conditions are followed.

7.3.1 Packaging Requirements

The key requirements to be followed are [21, 9]:

- Packaging materials must not interact physically or chemically with any of the packaging components in a way that would cause a product to be compromised or fail to conform to its established specifications.

- The container, closure, and overall packaging systems must provide adequate protection against external factors that could cause contamination or deterioration to the drug product.
- Packaging materials must meet USP [19] requirements and their respective monographs, including storage conditions.
- There should be a packaging BR that includes steps for examination of the CCS components before use, confirmation that the correct materials are used, and checks for the correct dosage fill.
- Packaging and labeling lines for different products, or different strengths of the same product, need to have adequate separation to prevent mix-ups.
- SOPs need to be in place to ensure proper identification of any filled containers of drug product that are to be stored unlabeled (sometimes known as bright-stock) for any time period.
- Packaged product must comply with the Poison Prevention Packaging Act [65].
- Contract packagers and labelers must also comply with these requirements. Appropriate SOPs should be in place for handling, storage, and transport between the compounding site and the contract facility. If non-compounders are fulfilling these roles, they should be trained on these SOPs to ensure that product SQIPP and sterility are maintained throughout distribution.

In addition, for compounders of sterile products:

- Packaging must be selected that preserves the sterility of the product and maintains the strength of the product until the BUD (see Chapter 9 for information on BUD).

7.3.2 Labeling Requirements

Section 503B(a)(10) [5] discusses the requirements for labeling drugs at a 503B outsourcing facility. It states that the label of the drug must include:

- The statement "This is a compounded drug." or a comparable alternative statement that prominently identifies the drug as a compounded drug;
- The name, address, and phone number of the outsourcing facility; and
- With respect to the drug:
 - Lot/batch number
 - Established name of the drug
 - Dosage form and strength
 - The statement of quantity or volume (as appropriate)
 - Date that the drug was compounded
 - Expiration date
 - Storage and handling instructions
 - National Drug Code number (if available)

- The statement "Not for resale" and, if the drug is dispensed or distributed other than pursuant to a prescription for an individual identified patient, the statement "Office Use Only"
- List of active and inactive ingredients, identified by established name and the quantity or proportion of each ingredient

It also states that the container from which the individual units of the drug are removed for dispensing or for administration (such as a plastic bag containing individual product syringes) shall include:

- A list of active and inactive ingredients, identified by established name and the quantity or proportion of each ingredient, if there is not space on the label for such information
- The following information to facilitate adverse event reporting: www.fda.gov/medwatch and 1-800-FDA-1088 (or any successor internet website or phone number)
- Directions for use including, as appropriate, dosage and administration

"The label and labeling of the drug shall include any other information as determined necessary and specified in regulations promulgated by the Secretary." [5]

Labeling of the finished compounded drug product is often included in the production BR or performed under a separate BR. As labeling is considered a critical process, adequate separation between labeling operations is needed in order to prevent mix-ups. Written procedures that should be in place for labeling requirements in a 503B facility include the following [9, 21]:

- Labels must contain the required information as detailed above, including a lot or control number, that permits determination of the history of the manufacture and control of the batch.
- Labels must be controlled by the QU, with restricted access, and the QU should have a process for:
 - Maintaining the master label for use in label inspection/release, where applicable, and controlling revisions to the master label
 - Regulatory approval, when required
 - Examining labels to make sure they contain the required information
 - Issuing labels to the manufacturing unit, reconciliation of labels received back from manufacturing, and verification of label reconciliation
- The manufacturing unit must be able to account for all labels, even if they are damaged, during manufacturing. Label reconciliation is performed as part of the BR or other appropriate document.
- Line clearance must be performed to verify all labels from the prior lot have been removed prior to introduction of a new lot on the labeling line. This should occur even if running the exact same product in a row.
- A representative label is placed in the BR.
- Examination of a representative number of samples of labeled drug product to ensure label accuracy prior to drug product release.

For drug products containing DEA-controlled substances, special packaging with tamper evidence should be used. Controlled substance products usually require special storage, depending on the DEA schedule it belongs to (see Section 4.5 for more information). Labeling of the controlled substance must reflect the DEA schedule that the drug product is under, which may be different than the API DEA schedule. [66]

8 Laboratory and Testing (Microbiology and Chemistry)

8.1 Introduction

Critical aspects of laboratory controls to ensure the quality of non-sterile and sterile drug products are further defined in 21 CFR 211.160 Subpart I: Laboratory Controls, General Requirements [21]. The FDA's current thinking (once the draft guidance is finalized) on how these regulations can be implemented regarding laboratory controls for outsourcing facilities are provided in Section J of the draft GFI – CGMPs for 503B Facilities [9]. In addition, Section I of the draft guidance prescribes ways to implement the regulations regarding specifications and sterility and endotoxin limits related to laboratory controls. The guidance should be consulted for additional information regarding the interpretations of the regulations for all testing. Table 8.1 presents where specific laboratory controls information can be found in § 211.160 and the draft GFI – CGMPs for 503B Facilities [9]. In addition, if a USP-NF monograph exists, the drug product must comply with the applicable USP-NF monograph.

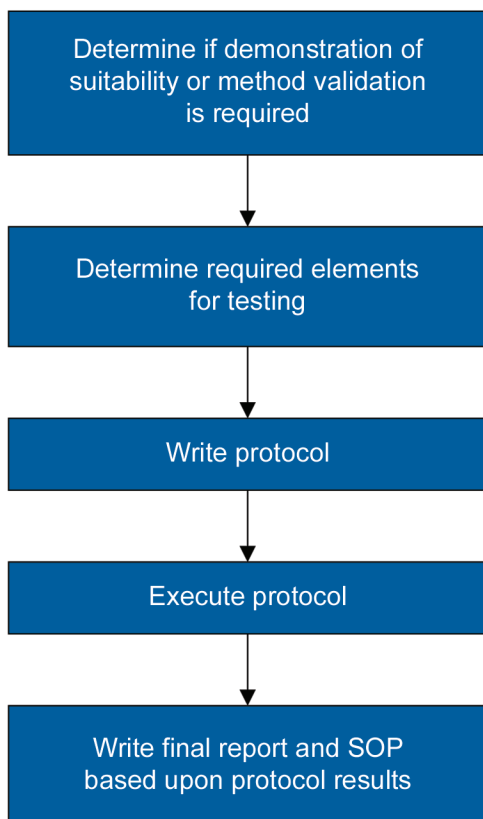
Table 8.1: Laboratory Controls Information in 21 CFR 211.160 [21] and the draft GFI – CGMPs for 503B Facilities [9]

§ 211.160 Subpart I	Sections I and J of the Draft GFI – CGMPs for 503B Facilities
Specifications, standards, sampling plans, test procedures, or other laboratory control mechanisms shall be drafted by the appropriate organizational unit and reviewed and approved by the quality control unit.	Follow appropriate written procedures for the conduct of each test and document the results.
Any deviation from the written specifications, standards, sampling plans, test procedures, or other laboratory control mechanisms shall be recorded and justified.	
Establish scientifically sound and appropriate specifications, standards, sampling plans, and test procedures designed to assure that components, drug product containers, closures, in-process materials, labeling, and drug products conform to appropriate standards of identity, strength, quality, and purity. Conformance to applicable written specifications for the acceptance of each lot within each shipment of components, drug product containers, closures, and labeling used in the manufacture, processing, packing, or holding of drug products. Such procedures shall also require appropriate retesting of any component, drug product container, or closure that is subject to deterioration. Conformance to written specifications and a description of sampling and testing procedures for in-process materials. Such samples shall be representative and properly identified. Conformance to written descriptions of sampling procedures and appropriate specifications for drug products. Such samples shall be representative and properly identified.	Design sampling and testing procedures to ensure that components, in-process materials, and drug products conform to the specifications set for the drug product. If using a validated or an established compendial test procedure in a specification, the test has been verified and documented to work under the conditions of actual use. Appropriate specifications must be established for each drug product. The product should also meet any other specifications included in an applicable USP monograph (see, e.g., section 501(b) of the FD&C Act). Other appropriate specifications for generally recognized attributes for the dosage form, such as those described in ICH Q6A [42], should also be considered. Finished product sterility testing provides additional verification of sterility, even for those products compounded from sterile starting materials, because an unexpected event posing a risk to sterility may have occurred but may not have been detected. For finished products purporting to be nonpyrogenic, the product must meet endotoxin limits before release. Keep complete records of all tests performed to ensure compliance with established specifications and standards, including examinations and assays.
Calibration of instruments, apparatus, gauges, and recording devices at suitable intervals in accordance with an established written program containing specific directions, schedules, limits for accuracy and precision, and provisions for remedial action in the event accuracy and/or precision limits are not met. Instruments, apparatus, gauges, and recording devices not meeting established specifications shall not be used.	Use analytical methods and equipment that are suitable for their intended use and are capable of producing valid results.
Note: Requirements for use of contract laboratories are not addressed in § 211.160.	When a contract facility is used to perform all or part of the testing of a drug, the outsourcing facility's quality control unit is responsible for approving and rejecting drugs tested by the contractor. The FDA recommends that contract facilities performing testing of a drug be ISO 17025 [37] accredited.

8.2 Microbiology Testing and Methods

The schematic in Figure 8.1 summarizes the components of a protocol plan that should be used for all microbiology testing conducted.

Figure 8.1: Schematic of Protocol Plan for Microbiology Testing



8.2.1 Method Suitability versus Validation

Most microbiological methods are compendial (e.g., USP [19]) and require suitability testing, i.e., demonstration that they can detect the microbiological attributes being tested. For example, the suitability testing for USP <71> requires that low numbers of microorganisms introduced into the product can be recovered by the methodology. Non-compendial methods may be used but need to be validated to demonstrate equivalency. An example is a rapid test for sterility, which must be at least equivalent to the compendial USP <71> method. In either instance, whether a method requires suitability testing or validation, a protocol should be generated and testing conducted at the intended testing laboratory. Elements of the protocol include:

- Signature of the preparer, review and approval by QC management and QA management
- Detailed operational parameters (e.g., reagents, instrumentation, preparation of standard and sample solutions, etc.)
- Clear, step-by-step instructions for execution
- Acceptance criteria, as specified in the associated USP chapter

- Use of well-characterized reference materials, reagents and solvents of suitable grade, and microbiological samples representative of samples to which the method will be applied.
- Where there are compendial requirements for verification associated with a particular testing technique/ technology, these requirements must be included in a suitability or validation protocol

8.2.2 Non-compendial Methods

Validation of non-compendial test methods must demonstrate that the methods are at least equivalent to USP methods, if there is a compendial method, as detailed in USP <1223> Validation of Alternative Microbiological Methods [19]. Table 8.2 presents the four options available to establish the equivalence of an alternative method.

Table 8.2: Alternative Methods – Options to Establish Equivalence [19]

Adapted from USP <1223> Table 2. Used with permission.

Option	Demonstration	Comparison to Official Compendial Method	Based on Numerical Results or Conclusion	Number of Characteristics
1. Acceptable Procedures	Acceptable: A reference material with known properties may be used, such as a standard inoculum of a specific microorganism, a quantity of highly purified bacterial genome, an Adenosine Triphosphate (ATP) level or another appropriate signal.	No	Results	Multiple
2. Performance Equivalence	Equivalent: Demonstration of equivalent or better results with respect to validation criteria such as accuracy, precision, specificity, limit of detection, limit of qualification, robustness, and ruggedness.	Yes	Results	Multiple
3. Results Equivalence	Equivalent: The alternative and compendial test methods give equivalent numerical results.	Yes	Results	Single
4. Decision Equivalence	Equivalent: A numerical result is not generated, instead a pass/fail result is obtained. The frequency of positive and negative results generated should be no worse than with the compendial method.	Yes	Conclusions	Single

Method suitability may be demonstrated using three independent tests. Only the accuracy and precision validation parameters are required for quantitative methods. For qualitative methods, recovery of challenge organisms as indicated in USP <62>, <71>, and <1227> [19] is sufficient.

After an alternative method has been shown to be equivalent to the compendial test with one product, it is not necessary to repeat the equivalency parameters for every new product; it is merely necessary to verify the method suitability for each additional product. For example, when employing a nucleic-acid-based method, with each new product, one should demonstrate that residual product does not interfere with the concentration, extraction, purification, and recovery of the target nucleic acid, or the Polymerase Chain Reaction (PCR) amplification and chemical probe detection of the target Ribosomal Ribonucleic Acid (rRNA) gene sequence.

8.2.3 Compendial Methods

Detailed requirements for the suitability of compendial test methods are prescribed in specific USP chapters [19]. For verification of compendial test methods, the methods must be categorized per USP chapters listed in Table 8.3 and/or USP <1227>, Validation of Microbial Recovery from Pharmacopeial Articles, in which antimicrobial properties are neutralized using neutralizing reagents, dilution, or membrane filtration.

Table 8.3: Compendial Test Method Verification by USP Chapter

USP Test [19]	Title	Description
USP <51>	Antimicrobial Effectiveness Testing	Demonstration of the effectiveness of antimicrobial protection.
USP <60 >	Microbiological Examination of Nonsterile Products Tests for <i>Burkholderia cepacia</i> Complex	Determination of the absence of <i>Burkholderia cepacia</i> complex.
USP <61>	Microbiological Examination of Nonsterile Products: Microbial Enumeration Tests	The tests allow quantitative enumeration of mesophilic bacteria and fungi that may grow under aerobic conditions.
USP <62>	Microbiological Examination of Nonsterile Products: Tests for Specified Microorganisms	The tests allow determination of the absence of, or limited occurrence of, specified microorganisms that may be detected.
USP <63>	Mycoplasma Tests	Testing for mycoplasma is a necessary QC requirement to ensure reliably pure biotechnological products and allied materials used to generate these products. The agar and broth media procedure or the indicator cell culture procedure may be used.
USP <71>	Sterility Tests	The test is applied to substances, preparations, or articles which, according to the Pharmacopeia, are required to be sterile.
USP <81>	Antibiotics – Microbial Assays	Procedures for the antibiotics recognized in the USP for which the microbiological assay is the standard analytical method. The cylinder-plate (or plate) assay and the turbidimetric (or tube) assay may be used.
USP <85>	Bacterial Endotoxins Test	This is a test to detect or quantify endotoxins from Gram-negative bacteria using amoebocyte lysate from the horseshoe crab (<i>Limulus polyphemus</i> or <i>Tachypleus tridentatus</i>). Three techniques are acceptable: the gel-clot technique, based on gel formation; the turbidimetric technique, based on the development of turbidity after cleavage of an endogenous substrate; and the chromogenic technique, based on the development of color after cleavage of a synthetic peptide-chromogen complex.

Each USP test has criteria to demonstrate suitability; see the corresponding USP chapter [19] for this information.

8.2.4 Final Report

A final report should be generated documenting the execution of the method validation or verification protocol, results obtained, and conclusion as to whether the method was successfully validated or verified for its intended purpose. The report should contain:

- Cover Page – The cover page should be in the current format for official reports and should include the table of approvers.
- Creator of the document
- QC director or designee
- QA director or designee
- Table of contents
- Abbreviations – All abbreviations used in the method validation/verification report should be listed and defined in this section.
- Purpose – The purpose of the method validation/verification in the context of the intended use of the method should be stated.
- Scope – The scope of the method validation/verification should be presented. The following components should be included:
 - General statement regarding the microbiological approach and technology of the method.
 - Elements that were validated.
- Summary of results:
 - Acceptance criteria as stated in the method validation/verification protocol.
 - Brief presentation of results obtained.
- Discussion of any deviations that occurred during the execution of the method validation/verification; the discussion should include assessment of impact to the validation/verification and the suitability of the method for its intended purpose.
- Conclusion – Assessment of results against respective acceptance criteria.
- Microbiological method in its final form. This will be the basis for writing the microbiological test method for the specific test that was conducted.

8.3 Analytical Testing, Methods, and Validation

Analytical testing follows the same pathway used for microbiological testing as detailed in Section 8.2. The following aspects are similar for both microbiological and analytical testing:

- Determine if validation or verification is required
- Determine required elements for testing
- Write protocol
- Execute protocol
- Generate final report and analytical test method based upon protocol results

Sections 8.3.1 to 8.3.3 present aspects of analytical testing that are different from microbiological testing (discussed in Section 8.2).

8.3.1 Non-compendial Methods

Non-compendial analytical methods must be validated as suitable for their intended purpose. Where the material under test is a compendial article, the validated non-compendial analytical method must also be demonstrated to be equivalent or superior to the compendial method. Consistent with the requirements of USP <1225> Validation of Compendial Procedures [19], the following hierarchy is recommended for categorization of non-compendial test methods:

1. **Category I:** Methods for quantitation of major components of APIs or active ingredients (including preservatives) in drug products
2. **Category II:** Methods for determination of impurities in APIs or degradation compounds in drug products
3. **Category III:** Methods for determination of performance characteristics (e.g., dissolution, drug release)
4. **Category IV:** Methods that are identification tests, i.e., tests that are intended to assure the identity of an analyte in a sample

Additional elements to those indicated in Table 2 of USP <1225> Validation of Compendial Procedures [19] may need to be demonstrated. Where a required validation element will not be demonstrated, the rationale should be justified in the approved validation protocol.

Validation of a non-compendial method must include the following analytical characteristics (refer to USP <1225>, Validation of Compendial Procedures [19] and ICH Q2(R1) Validation of Analytical Procedures: Text and Methodology [67]):

- **Accuracy:** The closeness of test results obtained by that procedure to the true value. The accuracy of an analytical procedure should be established across its range.
- **Precision:** The degree of agreement among individual test results when the procedure is applied repeatedly to multiple samplings of a homogeneous sample. Intermediate precision expresses within-laboratory variation, as on different days, or with different analysts or equipment within the same laboratory.
- **Specificity:** The ability to assess the analyte in the presence of impurities, degradation products, and matrix components.

- **Detection Limit:** The lowest amount of analyte in a sample that can be detected, but not necessarily quantitated.
- **Quantitation Limit:** The lowest amount of analyte in a sample that can be determined with acceptable precision and accuracy.
- **Linearity and Range:** The ability of an analytical procedure to elicit test results that are proportional to the concentration of analyte in samples within a given range.
- **Robustness:** The ability of a procedure to remain unaffected by small, deliberate variations in procedural parameters during normal use.
- **System Suitability:** Testing for variations in equipment, electronics, analytical operations, and samples to assure the variations do not affect the validity of the procedure. This characteristic may be performed during method validation, however it must be performed every time a piece of equipment or instrument is used to conduct a test, such as an HPLC¹² to conduct an assay.

8.3.2 Compendial Methods

Consistent with the requirements of USP <1226> Verification of Compendial Procedures [19], compendial methods must be verified as suitable for use. Verification is not required for basic methods such as residue on ignition, loss on drying, various wet chemical procedures such as acid value, and simple instrumental determinations such as pH measurement. Other than basic methods, the same categories used for non-compendial methods are applicable to compendial methods (i.e., Categories I - IV). Verification in most cases consists of demonstration of a subset of the elements demonstrated for a full validation. Where a required verification element will not be demonstrated, documented justification should be included in the approved verification protocol.

Analytical tests are detailed in the specifications or COA for each component, CCS, or product. Example specifications for an API are presented in Table 8.4.

Table 8.4: Example of API Specifications

Test	USP Reference [19]
Identification	USP <197K>
Residue on ignition	USP <281>
Assay	Current USP monograph
Impurities	Current USP monograph
Acidity or alkalinity	Current USP monograph
Clarity and Color of Solution	Current USP monograph
Related Compounds	USP <621>
pH	USP <791>

For either method validation or method verification, any result that does not conform to acceptance criteria should be investigated as described in the protocol.

¹² HPLC = High Pressure Liquid Chromatography

8.3.3 Method Transfer

Where a method has been validated by one laboratory (originating laboratory) and testing is to be executed by a different laboratory (receiving laboratory), documented support must be provided that the receiving laboratory is qualified to execute testing using the method for its intended purpose. Such support is provided by a documented method transfer.

The originating laboratory should provide the receiving laboratory with a copy of the test method and its corresponding development report and method validation report. The originating laboratory is also responsible for providing any special training necessary to ensure that personnel from the receiving laboratory can execute the test procedure.

The receiving laboratory should assess the method and validation for any gaps that require remediation prior to execution of the method transfer. The receiving laboratory should confirm that it has instrumentation qualified as suitable for use for the method to be transferred.

The transfer is conducted under an approved protocol. Consistent with the approaches presented in USP <1224> Transfer of Analytical Procedures [19], the *ISPE Good Practice Guide: Technology Transfer (Third Edition)* [68], and the WHO guidelines on transfer of technology in pharmaceutical manufacturing, TRS 961, Annex 7 [69], one of the following approaches should be applied. (Depending upon the intended purpose of the analytical method, other approaches may be supported.)

- **Comparative Testing** – Comparative testing requires the analysis of a predetermined number of samples of the same lot by both the originating and receiving laboratory.
- **Co-Validation** – Transfer may be supported by including the receiving laboratory in an interlaboratory co-validation. (This is most directly done by having the receiving laboratory function as the “second analyst” in the validation of intermediate precision, thereby validating the reproducibility of the method.)
- **Revalidation** – Where the analytical procedure to be transferred does not include complexity or technology that is new to the receiving laboratory, transfer may be supported by the receiving laboratory performing a partial revalidation of the method.
- **Waiver of Transfer** – The waiver of a method transfer may be justified. Typically, a waiver of transfer is applied in the following situations:
 - A new product's composition is comparable to that of an existing product and is analyzed by procedures with which the receiving laboratory has experience.
 - The analytical procedure is compendial and is to be used without modification. (Method verification is applicable in this case.)
 - The analytical procedure to be transferred is the same as or very similar to a procedure in use by the receiving laboratory.
 - Personnel who validated the analytical procedure or who performed the analytical procedure routinely at the originating laboratory have moved to the receiving laboratory and will be performing the testing using the analytical procedure to be transferred.

Upon successful completion of the method transfer and closure of any applicable deviation investigations, a final report will be prepared by the receiving laboratory. The report should be approved by laboratory management and QA of both the originating and the receiving laboratory.

8.4 Sample Collection

Components, in-process materials, and finished drug products need to be sampled to ensure they meet their required specifications. They may be sampled based on ISO 2895, Sampling procedures for inspection by attributes – Part 1: Sampling schemes indexed by Acceptance Quality Limit (AQL) for lot-by-lot inspection [70]. This document provides a comprehensive analysis of sampling, beginning with normal sampling and leading either to reduced sampling after 30 lots have passed their assigned specifications or to increased sampling when 2 of 5 consecutive lots do not pass their assigned specifications. Tightened sampling may be reduced after 5 consecutive lots pass their specifications. Assignment of normal, reduced, or tightened inspection is a fluid process, dependent on the results generated; diligent suppliers are more likely to produce better quality products that may result in less (reduced) testing. In essence, a strong sampling plan can facilitate better quality throughout the various materials purchased and produced.

Sampling plans require input from multiple disciplines including QA, Materials Management, QC (laboratory), R&D (Research and Development) and statisticians to achieve a robust process. ISO 2895 [70] is an excellent reference source for establishing sampling plans and should be consulted to develop a “user-friendly” plan for an outsourcing facility.

8.4.1 Components

Sampling of components is best accomplished by using as many aseptic techniques as possible to avoid ingress of microbiological and chemical contaminants. For example, a hairnet, beard cover, partial gown, and gloves help minimize the ingress of microbiological and chemical contaminants; sterile garments provide additional reductions to microbiological and chemical risk. Sterile sampling equipment, such as a glove box, ladles, spatulas, scoops, sampling thieves, and containers provide added assurance that the equipment will not interfere with the testing results. Clear instructions should be available to facilitate sampling and include the following:

- Each container of sample (e.g., excipients and APIs) should be vacuumed or dusted and transferred to a secured sampling room.
- Ideally the sampling room should be negatively pressurized to the outer environment to prevent cross-contamination once a container is opened.
- The sampler(s) should don the sampling garments prior to entering the sampling room and disinfect the top of the container with isopropyl alcohol or a similar disinfecting agent.
- The sampler(s) should carefully remove the top, open the inner layer of plastic (if used), unwrap the sampling instrument, obtain samples from the top, middle, and bottom of the container and transfer them to a sampling bag.
- A two-person sampling team is an industry best practice in which one person performs the operations on the outside of the container and the second person obtains the samples.
- Once sampled, the sampling bag should be closed, or sealed, and labeled with the date, time, samplers' names, material information, and testing to be performed.
- If both microbiological and analytical samples are required, the microbiological samples should be obtained before the analytical samples.

Sampling of containers and closures requires similar gowning and aseptic practices. If the containers and closures are received as sterile from the manufacturer, the samples must be obtained in an ISO 5 LAFH or isolator. The samplers must be attired in sterile garments and the sampling container or bag must be presterilized.

The number of samples to be obtained varies according to the number of units (for containers and closures) or the number of bulk containers of APIs and excipients. The tables in ISO 2895 [70] may be used to determine the number of samples to obtain; however, typically $\sqrt{N}+1$ units are taken. For example, if 1,000 vials are received, 33 vials should be sampled. For powders, the amount of material to be sampled should be sufficient to allow triplicate testing to be performed, if required (i.e., in the event of a deviation associated with the testing).

The FDA draft GFI – CGMPs for 503B Facilities [9] recommends designing sampling and testing procedures to ensure that components, in-process materials, and drug products conform to the specifications set for the drug product, and to keep complete records of all tests performed to ensure compliance with established specifications and standards, including examinations and assays.

8.4.2 In-Process Samples

In-process sampling is dictated by the CPPs established during validation of the process. Based upon the CPPs, samples should be obtained during the manufacturing process, starting from compounding through filtering, filling, capping, sealing, and sterilizing (where applicable). The number of samples at each step and the time of sampling could be determined during validation of the process. For example, fill volume samples could be taken hourly; pH samples could be taken once a shift; endotoxin samples could be taken every fourth hour. Adjustments to the process could be made during processing, if required, based upon the results of the in-process samples.

8.4.3 Product Samples

Non-sterile and aseptically filled presentations of the product in their final container-closure configurations, that is, the CCS used by patients, need to be sampled from the beginning, middle, and end of manufacturing.

For non-sterile and aseptically filled products, the samples should be obtained during final sealing at the three time points. For terminally sterilized products, the beginning, middle, and end samples should be obtained during final sealing; the samples should be set aside and terminally sterilized with the remaining product in the sterilizer.

Product testing is determined based upon the specifications of each product. Although testing is unique for each product, certain tests remain constant for many products, such as assay, pH, and for sterile products, endotoxin and sterility. Terminally sterilized products do not necessarily need to be tested for sterility but can be released based upon the validation of the sterilization process, if they meet the conditions described in the draft GFI – CGMPs for 503B Facilities, Appendix A, Section B [9]. Annual requalification of the sterilizer verifies the validated state of the process, obviating the need for sterility testing.

8.5 Laboratory Sample Receipt and Handling

Each laboratory should have a designated area or section for sample receipt. The area should be clean and well organized to prevent mix-ups and extraneous ingress of environmental contaminants. Samples may be received either externally (e.g., RMs received from a vendor) or internally (e.g., in-process or drug product samples). Upon receipt, the sample information should be documented in a dedicated logbook. Information should include the sample name, lot number, date of receipt or manufacture, day and time of receipt, testing to be performed, and the storage condition prior to testing, e.g., 2–8°C. Receipt of each sample must be initialed by the person who received and entered the information into the logbook.

Each sample should be assigned a laboratory testing number and the number written on the sample, or a label created and attached to the sample. For facilities in which electronic records are used for sample disposition, such as a Laboratory Information Management System (LIMS), a physical logbook is not required and a laboratory label is automatically generated for each sample.

Once the samples are logged and labeled, they should be transferred to a holding or staging area to await testing. It is imperative to maintain the samples at the appropriate temperature; therefore, a designated refrigerator should be used for holding refrigerated samples. Typically, a test sheet is prepared for each sample that accompanies the sample as it is retrieved from the staging area and tested. Once testing is completed, the results may be documented on the test sheet for review by laboratory management.

8.6 Analytical Instrument Qualification

Instrument validation (qualification) is a well-defined set of processes that verify instruments are properly installed, work correctly, and provide accurate and reliable results. Formal validation consists of the following steps:

1. Installation Qualification (IQ): The equipment, as installed or modified, complies with the approved design and manufacturer's recommendations.
2. Operational Qualification (OQ): The equipment installed or modified performs as intended in the specified operating ranges.
3. Performance Qualification (PQ): The equipment can perform effectively with reproducibility based on the approved methods.

USP <1058> Analytical Instrument Qualification [19] divides instruments in the laboratory into three categories, designated as groups. The groups are presented below:

- **“Group A:** Includes the least complex, standard instruments that are used without measurement capability or user requirement for calibration, such as a magnetic stirrer or vortex mixer. Proper function is ensured by observation, and no further qualification activities are needed for this group.
- **Group B:** Includes instruments that may provide a measurement or an experimental condition that can affect a measurement. Examples include a pH meter or an oven. Proper function of instruments in this group may require only routine calibration, maintenance, or performance checks. The extent of activities may depend on the criticality of the application. Generally, these instruments may have firmware but not software that is updated by the user.
- **Group C:** Comprises analytical instruments with a significant degree of computerization and complexity, such as high-pressure liquid chromatographs and mass spectrometers. All elements of qualification, including software validation, must be considered to ensure proper functioning of instruments in this group.”

The following fundamentals of IQ, OQ, and PQ are presented in Table 8.5 and were discussed in the context of manufacturing equipment in Chapter 5.

Table 8.5: Fundamentals of IQ/OQ/PQ

Validation Activity	Description
Installation Qualification (IQ)	Formal checks that confirm or verify that the equipment's software and hardware were placed and installed properly according to the manufacturer's requirements. The IQ should include the following checks: • System components • Environmental conditions • Utility requirements • Instructions for operation and backup mechanism • LIMS communication • Configuration access • Documents and records
Operational Qualification (OQ)	OQ confirms/verifies that the equipment can perform as expected. Examples of operational attributes include: • Calibration • QC testing • Process control limits, including monitoring and alarms • Data transfer across electronic interfaces
Performance Qualification (PQ)	PQ confirms that the equipment performs as expected under normal operating conditions by testing both the device and the process's ability to manage workflow in the anticipated time frame. PQ components include: • Accuracy (to include the effects of interferences, analytical sensitivity, and carryover) • Precision • Linearity • Method comparison to previous methodology or equipment • Defining, establishing, and verifying reference intervals • Completeness and correctness of operating procedures • Staff training and competency checklists
Performance Qualification Revalidation	For instrumentation that is moved, serviced, or modified, IQ and OQ need to be revalidated. If the instrument is moved to a different laboratory site, a modified PQ needs to be conducted to include precision.

IQ, OQ, and PQ need to be completed one time, prior to commissioning the instrument or piece of equipment. PQ revalidation, or requalification, is normally repeated annually for each instrument or piece of equipment to ensure the parameters established during initial PQ have not changed and therefore do not affect the results that are generated.

Additional information regarding equipment, instrument, and computer validation may be found in *ISPE GAMP®* series [71]. Briefly, there are four categories applicable to computerized systems (i.e., systems controlling equipment and instruments):

- Infrastructure: Platforms on which computer applications or elements necessary to operate and manage information technology environments run, such as operating systems, firewalls, and antivirus programs.
- Non-configurable software: Software that is marketed freely or is integrated into hardware to allow their operation, such as tools for statistical calculation, software for data acquisition, control panel viewers, and spreadsheets used as databases.

- Configurable software: Software that can be configured for specific purposes and validated, such as Enterprise Resource Planning (ERP), LIMS, and production equipment control systems associated with the process. Spreadsheets may be used as long as they are validated for the intended use.
- Custom software: Software tailored to meet specific needs, such software add-ons for non-configurable and configurable software, spreadsheets, dedicated systems, and ERP systems.

The *ISPE GAMP® 5: A Risk-Based Approach to Compliant GxP Computerized Systems (Second Edition)* [72] and the *ISPE GAMP® Good Practice Guide: A Risk-Based Approach to GxP Compliant Laboratory Computerized Systems (Second Edition)* [73] provide updates on supplier involvement, software development, current FDA activity on Computer Software Assurance (CSA) and links to *ISPE GAMP® Data Integrity Series* [74]. The reader is encouraged to use the GAMP approach to instrument, equipment, and computer system validations.

8.7 Analyst Training

Each facility should have a training program in which all employees receive basic CGMP training. Analysts must also be trained to perform the specific tasks for which they are required. A good analyst training program should start with a job description:

- Position title
- Position description
- Specific job duties
- Experience, qualifications, and education required

Based on the job description, a general training plan should be created for each analyst containing the following elements:

- General training required
- CGMP training required
- Company policies or standards training required
- Procedures, forms, logs required to execute job responsibilities
- Certifications required; timing for certifications
- Professional development training required
- OJT required

Subsequently, the analyst should be trained for each specific task as follows:

- Classroom lectures/presentations
- Read and understand the method or SOP (self-taught)
- Written test confirming the analyst understands the method
- OJT demonstrations by the trainer

- Hands on practice by the analyst
- Evaluation of results

OJT is a continuous program whereby analysts are trained on basic laboratory methods and receive training on more advanced activities as their proficiency increases. Table 8.6 is a partial list of training for analysts who perform laboratory testing.

Table 8.6: Example/Partial List of Training for Laboratory Analysts

Laboratory Analyst	Specific Training	Evaluation
Analytical Analyst	Laboratory housekeeping	OJT
	Glassware washing	OJT
	Preparation of solutions and suspensions	Skills Test
	Description, Color, and Clarity	Skills Test
	pH	Skills Test
	Acidity or Alkalinity	Skills Test
	Identification	Skills Test
	Residue on Ignition	Skills Test
	Assay	Skills Test
	Impurities	Skills Test
	Related Compounds	Skills Test
	Particulate Matter in Injections	Skills Test
	Loss on Drying	Skills Test
	Specific Rotation	Skills Test
	Chromatographic Purity	Skills Test
	Water Determination	Skills Test
Microbiology Analyst	Laboratory housekeeping	OJT
	Glassware washing	OJT
	Preparation of media	Skills Test
	Plate counting	Skills Test
	Culture transfer	Skills Test
	Growth promotion	Skills Test
	Microorganism ID	Skills Test
	Antimicrobial Effectiveness Testing (AET)	Skills Test
	Microbiological Examination of Nonsterile Products Tests for <i>Burkholderia cepacia</i> Complex	Skills Test
	Microbiological Examination of Nonsterile Products: Microbial Enumeration	Skills Test

Table 8.6: Example/Partial List of Training for Laboratory Analysts (continued)

Laboratory Analyst	Specific Training	Evaluation
Microbiology Analyst (continued)	Microbiological Examination of Nonsterile Products: Tests for Specified Microorganisms	Skills Test
	Mycoplasma Tests	Skills Test
	Sterility Tests	Skills Test
	Antibiotics - Microbial Assays	Skills Test
	Bacterial Endotoxins Test (BET)	Skills Test

A major expectation of a good training program is demonstrating that an analyst can repeat the same test as the trainer and obtain comparable results. Many laboratories develop a rotation in which the analyst performs one or two tests for a week or two – up to a month – followed by rotation to conduct another test for two weeks or a month. For example, in an analytical laboratory, an analyst might perform assays and impurities testing for one month and perform loss on drying and specific rotation the following month. In a microbiology laboratory, an analyst might conduct media preparation and growth promotion for one month, and sterility testing the following month.

Rotation of responsibilities ensures that the analysts maintain their training and skills for each task; rotation also alleviates boredom from repeating the same test day after day and enables them to develop expertise across all skills sets in the laboratory. As analysts develop proficiency in each test method, more experience is attained that over time allow them to become an expert. As an expert, the analyst is able to recognize problems before they happen, and conduct investigations that find and correct the root cause of a problem. Training is one of the most important tools for a company and it is the most important activity to maximize human accuracy and precision.

9 Stability and Beyond-Use Date

9.1 Introduction

The regulatory expectations for establishing a BUD or an expiration date for a compounded drug product manufactured by an outsourcing facility are prescribed in Section K and the corresponding Appendix B of the draft GFI – CGMPs for 503B Facilities [9]. This FDA guidance also presents the criteria under which the FDA generally does not intend to take regulatory action regarding stability testing for a 503B manufacturer that meets the conditions outlined in Section K. These conditions may be considered as two groups, each with subgroups. The two primary groupings are:

- 503B products for which a default BUD¹³ is applied
- 503B products for which limited stability testing is required

Note 1: In all cases, before any batch of **sterile** 503B product can be released, Container-Closure Integrity Testing (CCIT) must have been conducted on samples aged to or beyond the desired BUD to ensure that sterility is maintained over that time period. The FDA has indicated that CCIT (see USP <1207> Package Integrity Evaluation – Sterile Products [19]) is required only once for each formulation/container-closure configuration. Where there are multiple strengths, concentrations, and/or container-closure configurations, bracketing or matrixing may be applied to minimize the amount of testing (see Section 9.3.3).

Note 2: In all cases, before any batch of 503B product labeled or intended to be **multidose** can be released, Antimicrobial Effectiveness Testing (AET, see USP <51> Antimicrobial Effectiveness Testing [19]) must be conducted on samples aged to the desired BUD. As with CCIT for sterile products, AET is required for one lot of each formulation/container-closure configuration, and bracketing or matrixing may be applied to minimize the amount of testing.

For some 503B products, there is overlap between the regulatory expectations supporting BUDs (normally associated with compounded products) and those supporting expiration dates (normally associated with stability programs for drug products to be filed for regulatory approval). Accordingly, understanding the overall regulatory expectations for establishing BUDs and/or expiration dates is facilitated by prior familiarity with and understanding of the following compendial general chapters and regulatory guidances:

- USP-NF chapters [19]
 - <7> Labeling
 - <51> Antimicrobial Effectiveness Testing
 - <1207> Package Integrity Evaluation – Sterile Products
- FDA Guidances for Industry
 - ICH Q1A(R2) Stability Testing of New Drug Substances and Products [75]
 - ICH Q1E Evaluation for Stability Data [76]
 - ANDAs: Stability Testing of Drug Substances and Products [77]

¹³ The phrase “default BUD” refers to a BUD supported, as described in the draft FDA guidance, without stability testing.

- ANDAs: Stability Testing of Drug Substances and Products – Questions and Answers [78]
- Investigating Out-of-Specification (OOS) Test Results for Pharmaceutical Production [39]

9.2 Default BUDs

The subgroups under which a default BUD may be applied are presented in the draft GFI – CGMPs for 503B Facilities [9] and summarized below. Note that it is possible for a product to be in more than one subgroup. Where use of a default BUD is supported, no additional testing is required beyond that described in Section 9.1 of this ISPE Guide. The FDA draft guidance [9] provides the following requirements for a default BUD to be used:

- 503B product compounded solely by combining two or more drug products approved under Section 505 of the FD&C Act [4]:
 - At least one of the drug product components has labeling specifying the assignment of an in-use time.
 - The 503B product (compounded drug product) has been prepared and labeled with an in-use time.
 - The in-use time for the 503B product is also applied as the expiration date and that expiration date is the shortest of the in-use times of any of the component drug products.
- Non-sterile drug product with aggregate batch size $\leq 5,000$ units produced within a six-month period:
 - Water activity determined for products (not solid dosage forms) is expected to be ≤ 0.6 .
 - The default BUD is assigned as prescribed in Table C of Appendix B of the guidance.
 - Literature or other available scientific information does not indicate physicochemical instability over the BUD time period.
 - The default BUD is used as the expiration date.
- Sterile drug product with aggregate batch size $\leq 1,000$ units produced within a six-month period:
 - The default BUD is assigned as prescribed in Table D of Appendix B of the guidance.
 - Literature or other available scientific information does not indicate physicochemical instability over the BUD time period.
 - The default BUD is used as the expiration date.

9.3 Limited Stability Testing

9.3.1 Applicability of Limited Stability Testing

Where 503B facility wishes to establish an expiration date and/or approved storage conditions beyond those based upon component in-use times or default BUDs (Chapter K and Appendix B, respectively, of the draft GFI – CGMPs for 503B Facilities [9]), stability testing within a compliant stability program must be performed. The requirement for a stability program with limited stability testing applies to the following:

- To support a BUD/expiration date greater than the default for products with aggregate batch sizes $\leq 5,000$ units for non-sterile or $\leq 1,000$ units for sterile products.
- In all cases when aggregate batch sizes are $> 5,000$ units for non-sterile or $> 1,000$ units for sterile products.

The ICH Q1A [75] and FDA Guidance for Industry for A/NDAs or BLAs¹⁴ require multiple stability batches to establish expiration dating. For 503B products, where a submission for regulatory approval is not required, FDA indicates that under certain conditions it does not intend to take regulatory action against an outsourcing facility for a limited stability testing program (applied to one lot) used in support of expiration dates. The conditions apply to sterile products or non-sterile preserved products and are summarized as follows: [9]

- The stability-testing regimen must include reliable, meaningful, and specific stability-indicating methods.
- The CCS (including labeling and label adhesive) for the stability samples must be representative of that used for the marketed product.
- Stability samples must be representative of the batch from which they were obtained and stored under suitable conditions.
- For products labeled or intended to be multidose, testing to evaluate antimicrobial effectiveness is included; where antimicrobial effectiveness has been previously established for the formulation/continuer-closure configuration, a test for preservative content may be used instead.
- Samples are evaluated under long-term storage conditions that are consistent with those prescribed in ICH Q1A(R2) [75] and with labeled storage conditions.
- Results obtained at each time point are evaluated against the specifications for the compounded drug product.
- The BUD (expiration date) is not longer than 12 months.
- If the result(s) for any test falls outside of the established specification limits, the BUD must be restricted to either:
 - The last time point within which all results remained within specifications
 - The default BUD as described in Section 9.2 of this ISPE Guide to the last time point at which the data remained within specifications, or the default BUD as prescribed in Table C and Table D of Appendix B of the draft GFI is used.

The draft GFI includes a recommendation that the stability testing schedule include at least one interim time point before the BUD. In this way, if an OOS result is obtained for testing at the BUD, but all results at the interim time point are within specification, a BUD of the interim time point may be supported. It should be noted that an OOS investigation is expected to be conducted for any testing result that falls outside of the established specification limits during stability testing as well.

¹⁴ BLA = Biologics License Application

The guidance also prescribes the minimum set of tests to be conducted; the sets differ for sterile versus non-sterile products. It is noted that for sterile drug products, CCIT or sterility testing is conducted to document that sterility is maintained over the BUD.

9.3.2 Justification of In-Use Time

Where 503B products require reconstitution, re-dilution, or other manipulation prior to administration, an in-use time must be indicated. In the draft GFI – CGMPs for 503B facilities [9], the FDA has indicated that under certain conditions, it does not intend to take regulatory action regarding the requirement to have stability studies to support the in-use time. These conditions are summarized below:

- **For Sterile Products:**
 - When the product has directions for use that include an in-use time less than 4 hours at room temperature or less than 24 hours at refrigerated conditions.
- **For Multidose Products:** up to a 28-day in-use period is supported, provided that limited stability testing is conducted as follows:
 - The testing includes appropriate laboratory testing demonstrating that the product conforms to its specifications and maintains the product SQIPP. Additionally, AET is required.
 - The testing is conducted on samples that have been:
 - > Aged to at least 2/3 of the labeled BUD
 - > Manipulated (diluted/reconstituted) as described in the labeling and held for the in-use time not to exceed 28 days.

9.3.3 Use of Bracketing Strategy

The draft GFI – CGMPs for 503B Facilities [9] indicates conditions under which the stability of intermediate strengths or fill volumes may be supported by demonstration of the stability of the extremes (bracketing). The conditions are summarized below:

- The formulations should vary only in strength (or concentration) or fill volume.
- The same excipients must be in all formulations, although proportional formulations are not required.
- The same CCS, where possible, must be used for the samples placed on stability. The stability CCS should be as close as possible to the compounded drug CCS, including same material of composition.

The draft GFI [9] presents scenarios for bracketing based upon the number of different strengths, concentrations, or volume presentations. These are presented in Table 9.1.

Table 9.1: Scenarios for Testing for Bracketing Studies [9]

Number of Strengths, Concentrations, or Volume Presentations	Bracketing Strategy
3 or 4	Test the high and low extreme cases
5–10	Test the high and low extreme cases and 1 intermediate case
> 10	Test the high and low extreme cases and 2 intermediate cases

The draft GFI cautions that the decision on extremes may not always be meaningful if based upon fill volumes alone, as the ratio of head space to fluid volume may be a more relevant consideration for determining extremes.

The application of bracketing is not considered appropriate for sterility, endotoxins, and bioburden tests. Bracketing may be appropriate for water activity testing and AET when used in conjunction with a preservative content testing strategy.

9.3.4 Stability Protocols

Stability studies are expected to be executed under an approved protocol. There should be a governing procedure that describes the requirements for creation, review, approval, and issuance of the stability protocol. The governing procedure should also specify the protocol content, which should minimally include:

- Protocol identification number – the number should be unique to the protocol, with the format and assignment of the number prescribed by procedure
- Name of the drug product under study – this should match the formal name as found on the product label
- Container and closure information
- Purpose of the study (e.g., developmental, first commercial batch, annual commercial batch)
- Orientation (upright, on side, inverted) of the container in the chamber
- Storage conditions
- Testing schedule
- Number of units/amount of material required – in addition to the number of samples required to complete all testing, it is recommended that there be an appropriate overage in case of laboratory error or investigations
- Creator of the protocol
- Approver of the protocol

Note: For sample pulls, the governing procedure should indicate the acceptable tolerance around the scheduled pull date for removal of samples from the stability chamber, delivery to the laboratory for testing, and for completion of testing. Stability pulls that occur outside of these tolerance windows should generate an incident within the site deviation system and a corresponding investigation and impact assessment.

9.4 Recommended Practices

9.4.1 Time Zero (T₀) Results

Manufacturers of 503B products should establish a maximum time between execution of release testing and the date of placement of samples into a stability chamber for initiation of a study. Where the release testing is executed within the acceptable time frame, the results of release testing may be used for the T₀ results for the stability study. Where the maximum time is exceeded, separate testing apart from release testing should be conducted to provide the results for the T₀ time point.

9.4.2 Sample Pull at the Expiration Date

For 503B products, the date of manufacture is considered the date of compounding. The expiration date is set based upon the date of manufacture.

“Because of the possibility that a sample may not meet specifications at the final time point, FDA strongly recommends the inclusion of testing at least one interim time point. If the data at the final time point do not confirm the stability of the product at the desired BUD (e.g., some measurements fall outside of the established specifications), but the data at the interim time point are acceptable (i.e., measurements meet the established specifications), a BUD equal to the interim time point meets the second condition above.” (Draft GFI – CGMPs for 503B Facilities [9])

9.4.3 Use of Studies under Accelerated and Freeze-Thaw Conditions

Although the FDA draft guidance [9] does not speak to stability studies under accelerated and/or freeze-thaw conditions, manufacturers of 503B products should consider the inclusion of such studies during product development. Data obtained from studies under accelerated conditions may support planning for studies to extend established BUDs. Data from both types of studies assist in evaluating the impact to the product in cases of exposure to temperature excursions outside of labeled storage.

10 What to Expect in an Inspection

503B facilities are inspected by the FDA and SBOPs. The following lists were developed to aid 503B outsourcing facilities in preparing for these inspections. The lists are not all-inclusive but represent key areas on which regulatory officials typically focus. Having these documents readily available during an inspection may help facilitate inspections that are efficiently conducted, making good use of the limited resources of regulators.

10.1 Quality System (Chapter 3)

During an inspection, a regulator expects to see an established QU, independent of manufacturing in reporting structure and daily activities, that is responsible for ensuring the finished drug product meets the approved specifications for strength, identity, quality, and purity as well as sterility for sterile products and microbial quality for non-sterile products. The following items may be requested for review during an inspection:

- Organizational chart for the QU
- Document control procedures
- Index of all QS documents, including unique identifying number, title, and version
- List of all deviations, CAPAs, customer complaints, and adverse events for the last 24 months
- Batch production records
- Annual quality reviews
- Identified trends and trend investigations/outcomes
- Training records for employees
- Procedures related to the following:
 - Production and process controls
 - Cross-contamination and line clearance procedures
 - Established specifications for drug products, APIs, components, labeling, and packaging materials
 - Warehouse procedures for production materials and finished product as well as related sampling procedures, tools, process, and number of samples to be pulled/tested
 - Process and computer validation
 - EM and personnel monitoring
 - Release of finished drug product
 - Training program

10.2 Supply Chain and Materials (Chapter 4)

During an inspection, regulators are likely to request:

- Supply chain department (Purchasing, Planning, Warehousing, Logistics functions) organization chart and procedures, including Quality oversight
- Procedures for material receipt, inspection, identification, testing, material status (quarantine, rejected, release, for return/destruction), and control of manual and automated warehousing systems
- Sampling procedures for RMs and components with assurance that sampling is representative of incoming lot (see Chapter 8 for additional information on sampling procedures)
- Evidence that the bulk API was manufactured in an FDA-registered facility
- Procedures to maintain sterility of sterile drugs/APIs, components, and packaging materials
- Procedures for control of materials and labels used in compounding
- Procedures in place for controlled substances and compliance with DEA requirements, if appropriate
- Evidence of established specifications and comparison of materials purchased against specifications for components used in compounding
- Storage conditions and controls for components
- Controls and procedures for quarantine, reject, and destruction of rejected products/components/packaging/labeling
- Supplier/vendor qualification and management program
- Signed quality agreements with approved suppliers/vendors
- Approved vendors list and controls in place to ensure purchase/receipt of materials from approved vendors
- Evidence of supplier audits

Note: The outsourcing facility company is not required to show actual audit reports. The company instead presents evidence of when the audit was done, such as an audit agenda, and whether or not the supplier is acceptable.

- Shipping procedures, with consideration given to BUD and storage requirements (temperature, humidity, and light)
- Shipping studies, preferably for all four seasons, where applicable
- Procedures for material segregation and tracking
- Computerized inventory management system, where applicable
- Environmental controls in the warehouse or storage areas
- Temperature-mapping studies and/or validation of the warehousing areas

- ISO 5 and ISO 7 smoke studies
- EM data and validation of the EM system, when applicable
- Handling of temperature/humidity excursions, such as investigation, root cause analysis, and actions taken to address the issue/prevent recurrence
- Investigations related to warehousing or components/packaging/labeling to ensure the events are appropriately investigated and resolved with Quality oversight

10.3 Facilities and Equipment (Chapter 5)

During an inspection, regulators are likely to request (as appropriate):

- Facility diagrams with the following layouts, at minimum:
 - Room layouts with ISO classifications (when appropriate)
 - Equipment layouts
 - Personnel flow
 - Material flow
 - Equipment flow
 - Air pressure differentials
 - For ISO 5 rooms especially, EM sampling locations
- Paperwork on recent lots including personnel monitoring and aseptic logs
- Room logs, equipment logs, maintenance logs, and calibration logs
- Air handling system and air flow records
- Filter data
- Written procedures on any topics mentioned Chapter 5
- IQ, OQ, or PQ records or requalification records of equipment
- Risk assessments
- Cleaning schedules and cleaning agent rotation

They may request to see information on the material of construction of the walls/floors/ceilings.

Be prepared to answer questions from the insanitary conditions list of items (Section 5.4).

10.4 Manufacturing Controls in Compounding (Chapter 6)

During an inspection, regulators are likely to request:

- Risk assessments used to develop cleaning validations as well as the validations of the processes and the methods
- Water system, sampling, which water system is used for what areas, how they keep controls, etc.
- CPPs and how they have been determined
- Systems for verifying that the RM, API, and CCS are released for use prior to using them in the production BR
- Smoke study videos
- Aseptic procedures and practices as well as a listing of aseptic related deviation investigations
- Records for depyrogenated materials or how they are depyrogenated
- Information about the type of manufacturing operations performed
- Information regarding validation of the loads if TS or lyophilization is conducted
- Information about media fills, including:
 - Most recent media fill results and the justification for the bracketing scheme
 - Investigations conducted and, if any failures were dismissed, these will be closely evaluated by the inspector
 - Any growth promotion testing that was done for media fills
 - Any EM results over the past 6–12 months
- BRs to make sure they provide enough information to release the batch and determine if the information recorded is enough to give confidence that the product is of sufficient quality and sterility

Inspectors may also ask technicians/operators:

- About their training for aseptic manufacturing, including specific questions about concepts stated on the training curriculum and/or on the test
- To perform a gowning demonstration

10.5 Product Storage, Packaging, and Labeling (Chapter 7)

The FDA may ask for proof of adequate setup of the engineering controls for maintaining storage and isolation requirements as well as proof of continued monitoring to demonstrate maintenance of these requirements. They may ask to follow a specific product through its manufacturing process storage. For example, they may ask to review:

- Receiving records for RMs
- Logbook or e-recording system for RM storage including temperature and humidity information and location

- Records showing quarantine, sampling, release of RMs as well as logbooks showing movement to the appropriate locations when they move through these statuses
- Investigations of any storage excursions

FDA inspectors expect that any product-specific isolation requirements and instructions related to these requirements are clearly documented in the BRs, methods, and other controlled documentation.

For packaging and labeling, inspectors are likely to request:

- Records of what products were run on which lines and sign-off for line clearance, where applicable
- Confirmation that compounding areas are adequately separated to prevent mix-ups and cross-contamination
- Training records and proof that the personnel are trained for the job they are doing
- QU label storage including restricted access and controls
- Written procedures for, at minimum:
 - Examination of packaging components
 - Storage of packaged but unlabeled product
 - Setup of the line and line clearance
 - QU policies for label issuance, examination, control, and reconciliation
 - Information required on a label, including revision of labels

10.6 Laboratory and Testing (Microbiology and Chemistry) (Chapter 8)

The laboratory is the place where the quality of the compounded product as well as the starting materials is measured. While “testing into compliance” almost always results in a regulatory citation, testing under controlled conditions is the final arbiter to determine the quality and safety of a product or ingredient.

As in all manufacturing facilities, the FDA expects the following when inspecting laboratories:

- Clearly written and comprehensive SOPs and test methods to conduct testing
- Each method to be validated or verified
- Each analyst to be properly trained and continuously monitored
- Each piece of equipment or instrument to be properly validated and revalidated
- Samples to be properly obtained without allowing microbiological or chemical ingress
- Samples to be properly delivered to and recorded by the appropriate laboratory personnel

10.7 Stability and Beyond-Use Date (Chapter 9)

The FDA expects the process for supporting a BUD or expiration date to be proceduralized. This process should be consistent with regulatory recommendations described in the draft GFI – CGMPs for 503B Facilities [9]. For each product, the support for the BUD/expiration date should be documented and available in an approved report(s). Where a stability program is required in order to support an expiration date, the FDA expects that there are written procedures for the following:

- Governing the operation of the program
- Qualification, control/monitoring of storage chambers and access to them
- How stability samples are obtained, placed into storage, and retrieved at required time points
- How stability protocols are created, approved, and executed
- How temperature excursions in storage chambers are addressed
- Studies to support freeze/thaw of products where applicable

Testing time points should be consistent with ICH requirements [75].

Where a stability program is in place, there should be prescribed tolerances around pull dates. The FDA expects a procedure that includes requirements for testing turnaround (usually 30 days); such turnaround should be appropriate to allow for timely action in case of OOS stability results. The occurrence of late or missed testing should be rare. When testing is late or missed, the expectation is that the incident is investigated through the company's formal deviation process. Such an investigation should always include an assessment of the impact to the marketed product.

The FDA is likely to evaluate the test methods in place for a stability program. Where a test method was transferred to a testing laboratory, documentation supporting the transfer and the corresponding qualification of the testing laboratory should be available.

The results of stability testing along with any statistical trending of stability data are likely to be reviewed by the FDA for assurance that the data supports the claimed BUDs. Where an OOS result is generated during stability testing of a 503B product, the corresponding investigation is likely to be reviewed to determine if the claimed BUD remained supported. The FDA is likely to evaluate if the outsourcing facility conducted an appropriate assessment of impact to commercial product and whether required regulatory notifications occurred. Where an in-use time is indicated for a product, the inspector is likely to evaluate if appropriate testing of in-use time was included in the stability program and whether the in-use time proposed for the product is supported by such testing.

The control of records (electronic and/or paper) associated with the stability program and stability testing should meet current regulatory expectations for data governance and data integrity. Where software is used to manage a stability program (stability database) and/or to retain records of stability testing (stability database or a LIMS), such software is expected to be validated with validation documentation available for review.

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12 Appendix 2 – Glossary (Acronyms and Abbreviations)

ADE	Adverse Drug Experience
AET	Antimicrobial Effectiveness Testing
ANDA	Abbreviated New Drug Application
APIC	Active Pharmaceutical Ingredients Committee
AQL	Acceptance Quality Limit
ATP	Adenosine Triphosphate
BET	Bacterial Endotoxins Test
BHT	Bulk Hold Time
BI	Biological Indicator
BLA	Biologics License Application
BR	Batch Record
BUD	Beyond-Use Date
CAPA	Corrective Action and Preventive Action
CCIT	Container-Closure Integrity Testing
CCS	Container-Closure System
CDC	Centers for Disease Control and Prevention (US)
CGMP	Current Good Manufacturing Practice
CHT	Clean Hold Time
CMMS	Computerized Maintenance Management System
COA	Certificate of Analysis
CPP	Critical Process Parameter
CSA	Computer Software Assurance
CV	Curriculum Vitae
DEA	Drug Enforcement Administration (US)
DHT	Dirty Hold Time
DQSA	Drug Quality and Security Act
EM	Environmental Monitoring
ERP	Enterprise Resource Planning
FD&C Act	Food, Drug, and Cosmetic Act (FDA)
FDA	Food and Drug Administration (US)
FIFO	First In, First Out
GAAS	Grade A Air Supply
GFI	Guidance for Industry
GxP	Good “x” Practice
HBEL	Health Based Exposure Limits
HEPA	High-Efficiency Particulate Air
HPLC	High Pressure Liquid Chromatography
HVAC	Heating, Ventilation, and Air Conditioning
ICH	International Council for Harmonisation

IQ	Installation Qualification
ISO	International Organization for Standardization
LAFH	Laminar Air Flow Hood
LIMS	Laboratory Information Management System
MACO	Maximum Allowable Carryover
MBR	Master Batch Record
MFR	Master Formulation Record
MKT	Mean Kinetic Temperature
NABP	National Association of Boards of Pharmacy (US)
NDA	New Drug Application
NF	National Formulary
OCQC	Office of Compounding Quality and Compliance (FDA)
OJT	On-the-Job Training
OOS	Out of Specification
OQ	Operational Qualification
OSHA	Occupational Safety and Health Administration (US)
PCR	Polymerase Chain Reaction
PDA	Parenteral Drug Association
PEC	Primary Engineering Controls
PI	Prescribing Information
PPA	Pharmacy Practice Act
PPE	Personal Protective Equipment
PQ	Performance Qualification
q.s.	<i>quantum sufficit</i>
QA	Quality Assurance
QC	Quality Control
QS	Quality System
QU	Quality Unit
REMS	Risk Evaluation and Mitigation Strategy
RM	Raw Material
RODAC	Replicate Organism Detection and Counting
RTU	Ready-to-Use
SAL	Sterility Assurance Level
SBOP	State Board of Pharmacy
SDS	Safety Data Sheets
SOP	Standard Operating Procedures
SQIPP	Strength, Quality, Identity, Purity, and Potency
TC	Thermocouple
TS	Terminal Sterilization
US	United States
USP	United States Pharmacopeia
VMP	Validation Master Plan
WHO	World Health Organization



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